VRIJE UNIVERSITEIT AMSTERDAM

Master Thesis

Modelling Patient Functioning Over Time: Automated ICF Categorisation and Its Representation in an Event-Centred Knowledge Graph

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a thesis submitted in fulfillment of the requirements for the degree of

MA Linguistics
(Language and AI)

Vrije Universiteit Amsterdam

Computational Linguistics and Text-Mining Lab
Department of Language and Communication
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Date July 24, 2026
Student number 2866274
Word count 26,731

Abstract

Automatically extracting structured health information from clinical conversations could help healthcare professionals track patient functioning more efficiently. This thesis investigates whether language models can assign labels of the International Classification of Functioning, Disability and Health (ICF) (categories, impairment levels, and relative time) from spoken Dutch interRAI interviews. Two models are evaluated: mBERT, chosen for its multilingual and conversational flexibility, and MedRoBERTa.nl, chosen for its medical domain knowledge.

Four experimental settings are compared: a baseline trained only on Electronic Health Records (EHR) data, two fine-tuned settings using synthetically annotated conversational data, and a prompt-based GPT model. Each model is evaluated under four conditions combining strict and lenient filtering with strict and lenient evaluation.

The results show that fine-tuning on both EHR and conversational data works best for ICF category prediction. For level regression, the EHR-only baseline performs best. Relative time is predicted best by the conversational data model. mBERT generally outperforms MedRoBERTa in fine-tuned settings, while MedRoBERTa has an advantage on the baseline. The prompt-based model performs worst across the tasks. Inter-annotator agreement is low, confirming that the task is genuinely difficult. This may be due to transcription errors, differences in conversation segmentation, and the complexity of conversational discourse. Annotators also often disagreed on which conversational turn should be annotated.

An error analysis investigates agreement at the segment level, disjointness between category and level annotations, over-assignment of level 0 in the synthetic training data, and category confusion. The predictions of the best-performing model are used to populate an event-centred knowledge representation, illustrating how the extracted information can be organised into a knowledge graph. Together, these findings show that domain adaptation of language models to conversational clinical data is feasible, but that data quality and evaluation design remain serious challenges.

Declaration of Authorship

I, Marit Veerle Rozendaal, declare that this thesis, titled *Modelling Patient Functioning Over Time: Automated ICF Categorisation and Its Representation in an Event-Centred Knowledge Graph* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: July 24, 2026

Signed:



Acknowledgments

First, I want to thank my supervisors Piek Vossen and Luis Morgado da Costa. Piek, thank you for always having a solution ready whenever I walked in with yet another problem. Luis, thank you for your critical questions and helping me dig deeper into the material.

I also want to thank the A-PROOF group, a Dutch interdisciplinary research consortium consisting of Amsterdam UMC, Vrije Universiteit Amsterdam, Leiden University, and Amsterdam University of applied science: Edwin Geleijn, Marike van der Leeden, Sabina van der Veen, Jesse Aarden, Carel Meskers, and Maarten van Egmond, and Aminou Kadiri. Thank you for your advice, for thinking along with me, and for taking the time to annotate the test set. A special thanks goes to Edwin, who annotated the entire test set, which was no small task. Thank you also to Hein van Hout for providing the conversational dataset that made this work possible and also for annotating.

Thank you to the Robot Play group of the Computational Linguistics and Text-Mining Lab (Faculty of Humanities, Vrije Universiteit Amsterdam), Piek, Luis, Bram and Thomas, for letting me sit in on your discussions. It was very interesting to join your discussions.

Lastly, I want to thank Marja Jakobs for proofreading my thesis. And to the rest of my family and friends: thank you for the support, the kind words, and above all, the patience. Yes, I am finally done.

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Chapter 1

Introduction

In recent years, there have been significant advancements in healthcare, particularly in the care of elderly fragile people. Keeping track of a patient's health and wellbeing across different care settings is a real challenge, and there is a growing need for tools that can structure and organise this information in a meaningful way. Two well-established frameworks that help with this are the International Classification of Functioning, Disability and Health (ICF) and the Inter Resident Assessment Instrument (interRAI). These frameworks provide a standardised language for describing a patient's health, functioning, and care needs, making it easier to share and compare information across different facilities and departments.

The ICF, introduced by the World Health Organisation, provides a standard language and framework for describing health and health-related states (World Health Organization et al., 2002). InterRAI, first introduced by Hirdes et al. (1999), is an assessment tool that generalises patient information to ensure it can be used across different care facilities and departments.

Existing classifiers have been trained on Electronic Health Records (EHR) to predict seventeen ICF categories (ICF-17) (Chen and Vossen, 2026). However, not all relevant data comes in the form of written text. The dataset used in this project consists of spoken conversations, specifically conversations in Dutch between healthcare professionals and elderly fragile patients. During these conversations, the healthcare professional follows an interRAI questionnaire. This differs considerably from the EHR data that existing classifiers were trained on. While EHR data tends to be more formal, structured, and written by medical professionals, spoken conversations are informal, unstructured, and often contain discrepancies such as incomplete sentences or interruptions. This mismatch between the source domain (EHR data) and the target domain (spoken conversations) means that existing models cannot be applied directly, and adaptation is necessary.

Furthermore, while existing models do produce structured output, this output can be difficult to interpret and navigate in practice. As the number of patient records grows, there is an increasing need to organise clinical data in a way that allows healthcare professionals to retrieve and interpret relevant information efficiently.

One promising way to bring all of this information together is through knowledge graphs (KG). Knowledge graphs are structured representations of facts, organised as a graph rather than a traditional knowledge base. A knowledge graph allows healthcare professionals to query patient information flexibly and get a quick overview of a pa-

tient's functioning. In the longer term, this work could serve as a first step towards a system that can conduct patient assessments autonomously and provide professionals access to structured patient data over time.

Healthcare systems are facing a growing shortage of healthcare workers. As a result, reducing the workload of medical staff through automation is becoming increasingly important. The goal of this thesis is therefore to adapt a multilingual classifier (mBERT)(Devlin et al., 2019) and a medical classifier (MedRoBERTa.nl)(Verkijk and Vossen, 2021) to this target domain so that it can reliably predict ICF labels and levels from these conversations. This could help healthcare professionals by lessening the time and effort required to analyse medical conversations. It could potentially also support more consistent and efficient evaluations than humans can provide. Patients could benefit indirectly from improved clinical methods and, in the future, from AI systems that assist in medical conversations. Such systems could for example provide support outside regular healthcare hours, making assistance more readily available.

This thesis builds on the thesis by Jakubauskaitė (2025), and papers by Chen and Vossen (2026) and Kim et al. (2022).

The results of this work can be used to populate an event-centred knowledge graph, structuring patient functioning in a way that captures how they change over time. This allows healthcare professionals to get a organised overview of what was discussed in a medical conversation, and to query the data to reveal meaningful patterns across patients and time points. This thesis includes an example of such a graph based on the time classification of the level interpretations.

The central research question of this thesis is:

> *Can language models automatically assign ICF labels from interRAI interviews?*

To answer this, the following sub-questions will be addressed:

a. How well can fine-tuned language models extract ICF labels from conversational healthcare data?

b. How well do language models with different pretraining domains generalise to ICF classification in conversational medical settings?

These questions lead to the following hypotheses.

For subquestion (a), fine-tuned models are expected to perform well at extracting ICF labels from conversational data. If a model has been exposed to similar language structures, it should be able to adapt to the conversational domain. Training on both EHR data and conversational data is expected to work best, as it gives the model medical knowledge first and then exposes it to conversational patterns.

For subquestion (b), mBERT is expected to outperform MedRoBERTa in the fine-tuned settings, due to its stronger general linguistic flexibility. MedRoBERTa may have an advantage in the baseline setting, where its medical pretraining is most directly relevant. For the prompt-based models, few-shot prompting is expected to outperform zero-shot, as the examples help the model identify relevant patterns. Overall, the LLM-prompted models (Large Language Model) are expected to perform best.

For the full research question, the expectation is that language models can assign ICF labels from interRAI conversations, and that fine-tuning with conversational data will meaningfully improve performance over the baseline.

In summary, this thesis adapts two language models (mBERT and medRoBERTa.nl) to a conversational healthcare setting, training them to predict ICF categories, impairment levels, and relative time from spoken Dutch patient-provider conversations. The predictions are then used to populate an event-centred knowledge graph, providing a structured overview of patient functioning over time.

Chapter 2

Background

This chapter provides the background needed to understand the work in this thesis. It starts with an overview of interRAI and ICF, which form the foundation of this thesis. It then introduces transformer models, with a focus on mBERT and MedRoBERTa.nl. Furthermore, this chapter describes the A-PROOF project, which forms the basis of this thesis. Finally, it covers knowledge graphs, including how they are structured and how they can be used to model events and perspectives.

2.1 InterRAI and ICF

Elderly people face a wide range of physical, social, and psychological challenges that must be systematically captured to support clinical decision-making (Abdi et al., 2019). To track how people of age are doing across different care settings, countries and cultures, standardised assessment instruments are needed. Ideally, these tools capture a patient's health and wellbeing in a format that is interpretable and adaptable across medical departments, hospitals, languages, and countries. Two such instruments are the interRAI Home Care (interRAI HC) and the International Classification of Functioning, Disability and Health (ICF). While interRAI HC provides a structured patient assessment framework, ICF offers a universal classification system for human functioning. Together, they form the basis for organising patient information on a scale that generalises across settings.

This section will discuss each instrument in turn, while considering what they provide in combination.

2.1.1 InterRAI HC

InterRAI is a collaborative research network that develops standardised assessment instruments for use across care facilities and departments (InterRAI network). The Resident Assessment Instrument (RAI) was first introduced by Hirdes et al. (1999) as the RAI Home Care instrument, and has since been revised to improve compatibility with other assessment systems. By 2007, it was renamed the interRAI HC suite (Mello et al., 2015). It continued to be developed and validated to meet the constantly evolving needs of elderly people (Hirdes et al., 2004, 2008b; Leung et al., 2012; Morris et al., 2000).

Each RAI instrument involves a person-specific assessment recorded on a Minimum Data Set (MDS) form, which can be interpreted using clinical algorithms. The MDS

data aids in care planning, quality monitoring, outcome measurement, and resource allocation (Hirdes et al., 1999). Assessments include questions on cognitive performance, communication, mood and behaviour, social functioning, physical functioning, and more. There are several instruments within the RAI series, each created for its own domain. They include versions for Home Care (RAI-HC), Acute Care (RAI-AC), Mental Health (RAI-MH), and Post-Acute Care (RAI-PAC) (Hirdes et al., 1999). This work uses the interRAI HC 9.1, the instrument developed for nursing home settings worldwide.

InterRAI HC has been used across a wide range of applications, from diverse assessment implementations to nurse care coordination and integrated care model development (Mello et al., 2015; Salahudeen and Nishtala, 2019). Crucially, several studies have also confirmed its reliability. Hirdes et al. (2008a) tested the instrument across nearly 800 individuals in 12 countries, calculating inter-annotator agreement using weighted kappa coefficients (Fleiss et al., 2013; Landis and Koch, 1977). The results showed substantial agreement, and held across multiple languages and countries. This is further supported by Dendere et al. (2023), who evaluated a range of internationally used assessment systems for potential use in Australian geriatric care. The authors found interRAI HC to be among the most thoroughly validated. The instrument is characterised by its clinical assessment protocols (CAPs), risk screeners, outcome scales, quality indicators, and assessor certification programmes.

2.1.2 ICF

The International Classification of Functioning, Disability and Health (ICF) was introduced by the World Health Organisation (World Health Organization et al., 2002) to describe changes in body function and structure, as well as a patient's capacity and performance levels. Rather than viewing health through a purely biomedical lens, ICF adopts a "biological-psycho-social medical model", treating human functioning as the multimodal sum of health conditions and contextual factors (Zhang et al., 2024). This results in four core domains: body functions, body structures, activities and participation, and environmental factors. Each of these contain numerous specific categories and codes.

ICF can be used across nearly every healthcare domain. Unlike interRAI HC, it does not require every code to be filled in for every patient. This flexibility makes it well suited for limiting classification to only the categories relevant to a specific domain. A number of studies have developed such "core sets", containing only domain-specific categories. For example, for geriatric patients in early post-acute rehabilitation (Grill et al., 2005), for adults with vision loss (Billiet et al., 2024), and for people with interstitial lung diseases (Saketkoo et al., 2022), among others. However, on its own, ICF does not have immediate clinical utility for care planning, outcome monitoring, and resource allocation (Berg et al., 2009). Linking it to a structured assessment instrument such as interRAI HC thus becomes valuable.

Berg et al. (2009) demonstrated a conceptual overlap between interRAI HC and ICF. The authors argued that linking the two can open the door to implementing ICF in clinical use. Using interRAI together with common ICF elements across settings ensures standardised reporting at an organisational, regional, and national level. This connection is particularly relevant for the present work, since it forms the basis for mapping interRAI questions to ICF categories.

For home care geriatric patients (the focus of this thesis), two core sets are relevant.

WHO has endorsed an ICF Geriatric Core Set (ICF-GS) for the functional evaluation of older adults (Spoorenberg et al., 2019; Zhang et al., 2024), comprising 38 categories. However, this set lacks sufficient assessment guidelines, which limits its direct applicability. A more suitable option is the core set proposed by Chen and Vossen (2026) (ICF-17), which was specifically developed to match the interRAI HC system. In this set, 58.8% of categories belong to the Body Functions domain and 41.2% to Activities and Participation.

2.2 Transformers

Transformer-based models have become the main architecture in natural language processing since their introduction by Vaswani et al. (2017), consistently outperforming earlier approaches across a wide range of tasks (Bayat and Isik, 2023; Kehinde et al., 2023; Ma et al., 2025; Yildirim, 2024). They have the ability to model contextual relationships across sequences, which makes them well suited for tasks such as text classification and sequence labelling.

However, applying pre-trained transformers to a specific target domain requires careful consideration of what a model was trained on and what it is asked to do. In this thesis, the target data consists of spoken Dutch conversations between patients and healthcare professionals. This data is informal, unstructured, and domain-specific. Neither of the two models used here, mBERT and MedRoBERTa.nl, directly matches this profile. mBERT was trained on encyclopaedic data across many languages, giving it broad linguistic coverage but little medical conversational knowledge. MedRoBERTa.nl was trained on Dutch medical notes, meaning it has a solid clinical vocabulary, but also lacks exposure to conversational language. Furthermore, both models were pretrained on written text rather than spoken language, which likely limits their ability to handle the informal and fragmented nature of conversational data.

Domain adaptation is therefore necessary for both. This is achieved by fine-tuning. By adding task-specific layers on top of the pre-trained model and training end-to-end on (a small amount of) target data (Amatriain et al., 2023), the models are expected to adjust to the target data: medical conversations.

The following subsections describe the transformer architecture underlying these models, followed by a closer look at each model and its suitability for this task. Additionally, the challenges in medical conversational data are discussed.

2.2.1 Transformer Architecture

The core of the transformer architecture is the self-attention mechanism, paired with a fully connected feed-forward layer (Vaswani et al., 2017). Together these process the input tokens. Self-attention works by mapping each input token to three representations: a query, a key, and a value. The output representation of each token is then computed as a weighted sum over the values of all tokens, where the weights are determined by the compatibility between that token's query and every other token's key. This allows the model to capture relationships between tokens regardless of their distance in the sequence. Through multi-head attention, this process runs in parallel across multiple attention heads, each learning different aspects of token relationships. The resulting representations are then combined into a single output representation for each token (Amatriain et al., 2023).

A practical advantage of this design is that all tokens are processed in parallel rather than sequentially, which enables efficient computation (Amatriain et al., 2023). The trade-off is that transformers tend to be computationally expensive and have limited interpretability (Zhu, 2023).

The models used in this thesis (mBERT and medRoBERTa.nl) are encoder-only transformers, meaning they are designed to produce rich contextual representations of input text rather than generate new text. Both are pre-trained using Masked Language Modelling (MLM), where a certain proportion of input tokens is randomly masked and the model is trained to predict them based on the surrounding context. This forces the model to develop a bidirectional understanding of language. BERT, on which mBERT is based, additionally uses Next Sentence Prediction (NSP) as a second pre-training objective, where the model learns to determine whether two sentences are consecutive in the original text.

2.2.2 mBERT

Multilingual BERT (mBERT) is a BERT-based model trained on Wikipedia pages from 104 different languages using a shared WordPiece vocabulary (Rietberg et al., 2023; Luo et al., 2025). BERT was originally introduced by Devlin et al. (2019) as a model that pre-trains deep bidirectional representations by conditioning on left as well as right context across all layers. mBERT broadens this to a multilingual setting. The model finds lexical overlap between different languages using their shared vocabulary. This allows semantically similar tokens from different languages to share subword representations.

This expansive training gives mBERT strong general language understanding which may make it easier for the model to adapt to the relatively unstructured nature of spoken dialogue, meaning it would be better equipped to handle informal or conversational Dutch (Luo et al., 2025; Arnoult et al., 2021). The key limitation, however, is the lack of any medical domain knowledge. Since it was trained on encyclopaedic text, mBERT has limited knowledge of clinical terminology. It may miss subtle medical distinctions that are important for ICF categorisation. Domain adaptation here therefore includes introducing medical and task-specific knowledge, rather than adapting to conversational style.

2.2.3 MedRoberta.nl

MedRoBERTa.nl (henceforth also known as MedRoBERTa) is a RoBERTa-based encoder model trained from scratch on 13GB of Dutch Electronic Health Record (EHR) data (Verkijk and Vossen, 2021). RoBERTa builds on BERT by taking out the NSP objective and using a more robust training procedure. It also uses Byte Pair Encoding (BPE) for tokenisation. This encodes text into variable-length subword units and guarantees that no input token is ever unknown (Verkijk and Vossen, 2021, 2025).

EHR data consists of clinical notes written by healthcare workers about individual patients, and tends to be dense with medical terminology. Words and sentences are often abbreviated, may omit function words or contain spelling errors (Verkijk and Vossen, 2021). Training on this data gives MedRoBERTa a good comprehension of clinical vocabulary and concepts, and the model has shown promising performance on tasks involving medical terminology and patient-healthcare professional content (Verkijk and Vossen, 2021, 2025; Muizelaar et al., 2024). With only 117 million parameters, it is relatively compact compared to more recent state-of-the-art models (Scheeres, 2025).

For this thesis, MedRoBERTa's medical background could be beneficial, since the conversations involve clinical concepts that are also present in EHR data and directly relevant to ICF categories. However, EHR notes are written, structured, and concise, unlike spoken conversation. The model may therefore struggle with the informal, fragmented nature of patient-healthcare professional dialogue, and domain adaptation to conversational data will be necessary.

2.2.4 Challenges in conversational medical data

Conversational medical data comes with several challenges. It is full of domain-specific terminology (Blee et al., 2021), and suffers from severe class imbalance. Most utterances do not refer to any ICF category at all, making it difficult to locate the relevant information (Song et al., 2020). Additionally, strict privacy regulations limit the amount of data available for research (Conduah et al., 2025).

Apart from this, converting audio to text is not straightforward, and correctly recovering the structure of the dialogue from the transcription adds another layer of difficulty.

2.3 A-PROOF

A-Proof (*Automated Prediction of post-COVID-19 RecOvery Of Functioning*) is a project that applies AI to healthcare (A-PROOF Consortium, 2025). It is a collaboration between Amsterdam UMC, VU University Amsterdam, Leiden University, and Amsterdam University of Applied Sciences. The project develops tools that can automatically monitor the functioning of individuals who require healthcare support.

Several relevant publications have come out of this collaboration. Verkijk and Vossen (2021) introduced MedRoBERTa.nl, a domain-specific language model trained on Dutch EHR data. The model outperforms general-purpose models on a range of clinical text mining tasks. Kim et al. (2022) used MedRoBERTa to classify ICF categories and levels on EHR data using a set of nine categories (ICF-9). Performance was strong for the main categories, with F1-scores above 80%. For level prediction, the average error was below one point on a five-point scale. Chen and Vossen (2026) extended the set to include 17 categories (ICF-17). Their approach combined manual and synthetic annotations to fine-tune a domain-specific encoder, outperforming models trained on either annotation type alone.

This thesis builds on these works in several ways. It predicts ICF categories and levels on the ICF-17 set, but applies the models to conversational data rather than EHR text. Additionally, a multilingual model is introduced alongside MedRoBERTa to assess whether broader linguistic training is better suited to conversational language. Furthermore, relative time prediction is added as a new task, which enables the construction of an event-centred knowledge graph. Together, these extensions adapt the pipeline from written clinical records to spoken patient-healthcare professional interactions.

2.4 Knowledge graphs

Healthcare data, and conversational data in particular, is inherently temporal: Observations, assessments, and patient states all exist relative to a point in time (Hooshafza

et al., 2022). Managing this temporal element well is important for information retrieval, healthcare decision support, and monitoring a patient's trajectory over time. One promising way to organise this kind of interconnected, time-stamped information is through knowledge graphs (KGs).

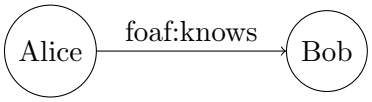
A knowledge graph is a structured representation of facts organised as a graph. Nodes represent entities, either real-world objects or abstract concepts, and edges represent typed relationships between them (Ji et al., 2021; Hogan et al., 2021). Apart from storing information, KGs introduce explainability by making relationships between concepts explicit and enabling reasoning over them (Rajabi and Kafaie, 2022). Their structure is flexible. Builders can extend the graph, and queries can follow paths across multiple connected nodes to find direct as well as indirect answers.

Previous work has demonstrated that KGs are very capable of representing EHRs, particularly via semantic annotation linked to ontologies (Carvalho et al., 2024). This thesis takes a related approach, constructing a KG that links ICF categories to the patient-healthcare professional conversations.

This section goes on to describe the definition and structure of knowledge graphs, as well as how to model events and perspectives.

2.4.1 Definition and Structure

Knowledge graphs store information using RDF-style triples of the form *(subject, predicate, object)*. The subject and predicate are IRIs (Internationalised Resource Identifiers) and the object can additionally be a literal value such as a string or number (Zamini et al., 2022; Ali et al., 2022). Blank nodes can appear in the subject or object position and are interpreted as existential variables. A simple example is the triple `:Alice foaf:knows :Bob`, where the prefix `foaf` expands to `http://xmlns.com/foaf/0.1/`. The full predicate IRI is thus `http://xmlns.com/foaf/0.1/knows`. Graphically, this looks like so:



The semantics of RDF are dictated by RDF Schema (RDFS), which supports class and property hierarchies, domains and ranges, and other constructs (Zamini et al., 2022; Ali et al., 2022).

Knowledge graphs consist of two parts, namely the TBox and the ABox. The TBox defines the general structure of the graph, specifying the types of concepts and relationships that can exist. The ABox contains the actual data, linking specific instances to these concepts and relationships (De Giacomo et al., 1996). This way, the graph has a clear overall framework while still capturing detailed information. This makes knowledge graphs a practical way to organise and explore complex, connected data.

Temporal knowledge graphs extend this structure by adding timestamps to facts. They can be understood as graph snapshots captured at different points in time, or as graphs that record events happening over continuous time (Cai et al., 2024; Hooshafza et al., 2022).

SPARQL is the standard query language used to retrieve information from a knowledge graph (Hogan, 2020). A SPARQL query is made up of six components. First, *base and prefix declarations* allow commonly used IRIs to be abbreviated, just like in

RDF. Second, the *dataset clause* defines which graphs the query operates on, since SPARQL works across datasets that can consist of multiple graphs. Third, the *query type* determines what kind of result is returned: `SELECT` returns a table of results, `ASK` returns a boolean, `CONSTRUCT` returns a new subgraph, and `DESCRIBE` returns a graph describing the matched resources. Fourth, the *where clause* specifies the pattern that needs to be matched against the graph. Fifth, *aggregation* controls how results are grouped. Finally, *solution modifiers* determine how the results are ordered and how many are returned.

2.4.2 Modelling Events and Perspectives

Representing the content of conversations in a KG requires more than storing facts. It requires modelling who said what, when, and from what perspective. Several frameworks have been developed to address this.

Van Hage et al. (2011) introduced the Simple Event Model (SEM), designed to represent events across diverse domains without using domain-specific vocabulary. SEM defines four core classes: `sem:Event` (what happened), `sem:Actor` (who or what participated), `sem:Place` (where), and `sem:Time` (when). It is intentionally minimalistic: it requires no consensus on the characteristics of the event and allows different viewpoints, which makes it broadly interoperable.

Next, Fokkens et al. (2013) proposed the Grounded Annotation Framework (GAF), which introduces predicates such as `denotedBy` and `denotedIn` to connect instances to mentions in text. These instances can be anything from events to entities.

The GRaSP framework introduces a source perspective (Van Son et al., 2016; Fokkens et al., 2017). The framework represents not just what is said, but also the viewpoint a source takes on it. This combines a mention layer, an instance layer, and an attribution layer into a single model. It builds on GAF to relate mentions in text to real-world instances, and adds annotations on perspective, factuality and sentiment.

For conversational data, where a patient and a healthcare professional may express different views on the same situation, perspective modelling is very relevant. These frameworks together give the foundations for the knowledge graph constructed in this thesis.

Chapter 3

Data and Annotation

This thesis uses two datasets, namely EHR data and conversational data. The EHR data serves as the baseline and is described in Chen and Vossen (2026). This chapter covers the conversational data used for fine-tuning, including how it was collected, processed, and annotated for training and evaluation.

3.1 Curation Rationale

There are two main purposes of this dataset. First, it is used to investigate whether MedRoBERTa and mBERT can adapt from their respective training domains to conversational medical data. Second, it allows for a comparison between the two models. The intention is thus to analyse whether a medically trained model like MedRoBERTa outperforms a multilingual model like mBERT in this setting, or if mBERT's broader language coverage give it an advantage in handling conversational input.

The conversations are drawn from real clinical interactions, meaning that if the models perform well here, there is a realistic prospect of utilising them in practice.

3.2 Conversational Data

The data comes from an international study involving home care organisations in Belgium, Finland, Germany, Iceland, Italy, and the Netherlands (Van der Roest et al., 2019). Three Dutch organisations participated to collect the data. This thesis uses only a sample of the Dutch conversations, recorded by Cordaan and Buurtzorg.

The data consists of real conversations between geriatric patients and healthcare professionals following the interRAI-HC questionnaire. The conversations were recorded in Dutch (nl-NL) between January 2014 and August 2016, and took place at the patient's home where possible. In some cases, a proxy responded on behalf of the patient. The conversations are provided by the Amsterdam UMC.

Each conversation is structured as a dialogue with turns assigned to each speaker. The typical pattern is a healthcare professional asking a question and the patient (or proxy) answering, mostly turn by turn. Exceptions arise naturally. For example, conversations are sometimes interrupted by visitors or phones ringing, and patients occasionally do not answer directly, requiring the healthcare professional to rephrase or steer the conversation.

Time	Number of conversations	Number of turns	Average number of turns per conversation
T0	172	412,425	2,398
T1	143	229,326	1,604
T2	40	54,707	1,368

Table 3.1: Statistics conversations.

The data was collected as part of a longitudinal study, with assessments at baseline (T0), six months (T1), and twelve months (T2). Not all patients completed the full study, so some have fewer than three conversations. Most conversations were about 45 to 90 minutes. Follow-up conversations tended to be shorter, as they focus only on changes since the previous assessment.

The study received ethical approval from the Medical Ethics Review Committee of VU University Medical Center (No. 2013.333), in accordance with local regulations. Written informed consent was obtained from all participants.

The final dataset used in this thesis contains 216 patients, resulting in 355 conversations (400 files) and 696,458 turns in total. Each turn is to be considered a data instance. A breakdown of conversation lengths is given in Table 3.1.

3.3 Participant demographic

Patients were eligible if they received home care from one of the enrolled organisations, were at least 65 years old, and were expected to remain in care for at least six months after inclusion. Patients were excluded if they had a terminal illness, received only short-term care, were planned to be institutionalised within six months, or had cognitive impairment (score Cognitive Performance Scale (CPS) ≥ 3) (Morris et al., 1994) without a known informal caregiver or legal representative.

The Dutch home care clients were predominantly female (71.0%) with an average age of 82.5 years ($\pm$7.1). Most lived alone (69.4%), and just over a third were married or with a partner (35.9%). Nearly 79% had at least one informal caregiver. On average, participants received 7.9 ($\pm$14.2) hours of informal care in the last three days, and 4.6 ($\pm$4.7) hours of professional home care per week. In the Dutch data, cognitive decline was low (1.8%), though depressive symptoms were common (23.6%) and daily pain was reported by 31.4% of patients.

Most patients were native Dutch speakers, so language proficiency was generally high. Nevertheless, the age of the participants means that some disordered speech is present; Patients occasionally mumbled or struggled to speak clearly. Also, since conversations took place in patients' homes, interruptions sometimes occurred.

The healthcare professionals were predominantly female and fluent in Dutch. They worked for the home care organisations as trained (research) nurses, using licensed interRAI software (InterRAI network). All assessors completed a standardised training on the interRAI HC questionnaire.

3.4 Annotator demographic

The test data is annotated by medical experts from the A-Proof team. All annotators are fluent in Dutch. Eight of the nine annotators have prior experience annotating datasets using the same guidelines; the ninth practised beforehand. Annotations followed the guidelines provided by World Health Organisation (WHO).

The training data was annotated synthetically using `gpt-4o`. This is an autoregressive multimodal model pre-trained on data up to October 2023, drawn from publicly available datasets, web crawls, and proprietary data partnerships (OpenAI, 2024)

3.5 Linguistic situation and text clarification

The recordings are spontaneous spoken conversations, recorded synchronously during in-person assessments. They were collected for research purposes. The conversations follow the structure of the interRAI-HC questionnaire, meaning the topic is the patient's medical status and functional wellbeing. The non-linguistic context is thus a formal clinical assessment, though the conversations themselves are natural and unscripted.

3.6 Data Processing

The audio recordings are transcribed and diarised as described in Chapter 4, resulting in a JSON file per conversation. Placeholders for the ICF category, ICF level, and relative time are also added. An example is shown below.


```
{
  "filename": "T0 999000 MR",
  "patient": "999000",
  "HP": "MR",
  "T": "T0",
  "conv": {
        "turn_0: {
                "speaker": "SPEAKER_03",
                "text": "Bent u vandaag nog buiten geweest?",
                "relative_time": null,
                "ICF_category": null,
                "ICF_level": null
        },
        "turn_1: {
                "speaker": "SPEAKER_02",
                "text": "Ik heb vandaag een rondje gewandeld.",
                "relative_time": null,
                "ICF_category": null,
                "ICF_level": null
        },
  }
}
```


The resulting documents are used as input for model training and testing.

Time	Number of conversations	Number of turns	Average number of turns per conversation
T0	5	9,622	1,924
T1	5	5,907	1,181
T2	1	941	941

Table 3.2: Statistics test conversations.

3.7 Test set

The test set was taken as a sample from the final dataset. It consists of 11 conversations from 5 patients. Each patient has a T0 and T1 conversation, and one patient also has a T2 conversation. Including all time points for each patient makes it possible to observe changes in a patient's situation over time.

The test set was constructed empirically, guided by a set of diversity criteria. Variation was desired in both the healthcare professional and the patient. Some assessors ask questions directly from the questionnaire, while others let the conversation develop more naturally. Some patients respond concisely and directly, while others elaborate at length. The test set also includes cases where a proxy answered some or most questions. Conversations with inaudible dialogue were excluded. Finally, diversity in interviewee gender and level of independence at home was taken into account.

Based on these criteria, the test set contains 16,470 turns across 11 conversations from 5 patients. See Table 3.2 for a breakdown of the test conversation lengths. The average number of turns in the test set files is lower than in the overall dataset, because the full dataset contains some very long conversations that increase the average. The test set conversations were each around 60 minutes long.

3.8 Data Annotation

All turns are annotated with ICF categories and levels from the ICF-17 set listed in Appendix A. Each category has a code, a name, and a difficulty scale (level) ranging from 0 (complete problem/disability) to 4 (no difficulty). If no category from the ICF-17 set applies, the turn is assigned the label "None". Throughout this thesis, having 'no category assigned' means exactly this.

A turn can accept zero, one, or more ICF category labels. If a single category occurs in multiple turns, each of those turns is labelled accordingly. For turns where the difficulty level is specified, the turn is also annotated with the relative time point it refers to: "past" (something that was the case but no longer is), "present" (something currently the case, even if it began earlier), or "future" (something yet to happen). If the relative time point is not mentioned, then it can be assumed that it is currently the case and "present" is applicable. The annotation of the time point is only applied on turns where a level is explicitly given. An example of the format of the test set given to the annotators is shown in Table 3.3.

The annotation guidelines are based on the ICF browser (World Health Organisation (WHO)) and are included in Appendix B. They specify what each category covers and what is excluded, since several categories overlap.

Training and test data were annotated differently. Manually annotating the full training set was not feasible, so training annotations were generated synthetically lead-

conv_id	turn	speaker	text	cl	past	present	future	B1300	D465	...	D450	None	notes
T0 999000 MR	0	MR	Bent u vandaag nog buiten geweest?	0.95									
T0 999000 MR	1	999000	Ik heb vandaag een rondje gewandeld.	0.95		x					4		
T0 999000 MR	2	MR	U heeft wel het perfecte week gekozen om te gaan wandelen, zeg.	0.90							x		
T0 999000 MR	3	MR	Liep u met uw rollator?	0.90					x		x		

Table 3.3: Example test data for annotators.

ing to bronze labels. The test set required reliable gold-standard labels and was therefore annotated by human experts.

3.8.1 Test set

The test set was annotated by nine medical experts in total. One annotator, referred to as the baseline annotator, annotated all files. The remaining eight annotators each annotated one file and are referred to as additional annotators. Each additional annotator is identified by the file number they annotated. Two additional annotators happened to annotate the same file. These are distinguished as files 1.1 and 1.2, and both were compared individually to the baseline annotator.

The baseline annotator's annotations are used as gold labels, as they are the only annotations consistent across all files. Their reliability is assessed through inter-annotator agreement in Sections 5.1 and 7.1. The identities of the annotators are not disclosed here for privacy reasons, but can be made available upon request

3.8.2 Data filtering and evaluation

The annotated test set is filtered before evaluation in order to remove parts of the conversations that were not annotated due to transcription errors or because they are not relevant to any ICF category. Evaluation is then performed on the filtered annotations only.

The first filtering step removes duplicated sentences. A common transcription error is the repeated occurrence of the same phrase or sentence across multiple consecutive turns. All but the first two occurrences of any repeated turn are removed. This keeps intentional repetitions, such as when a question is repeated because it was not heard, while preventing transcription errors from affecting the evaluation.

Next, the conversations are divided into segments. A segment starts when a category is annotated and closes when the level for that category is annotated. Two filtering conditions are applied. In strict filtering, only the turns within this segment are kept. In lenient filtering, ten turns before and ten turns after the strict segment boundaries are added, providing more conversational context. Both for strict and lenient filtering, there can be turns without a category and without a level annotated. This is illustrated in Figure 3.1. However, every segment is assigned the category and level on which it was filtered. As a result, the gold labels in the filtered data no longer contain the level "None", and this label therefore has no support in the evaluation. The label is retained because the models can still predict it.

The evaluation itself is also split into two conditions. In strict evaluation, predictions are compared to gold labels turn by turn within the filtered segment. In lenient evaluation, a prediction is counted as correct if the gold label appears anywhere in the segment, regardless of the exact turn. This is a first step towards evaluating ICF anno-

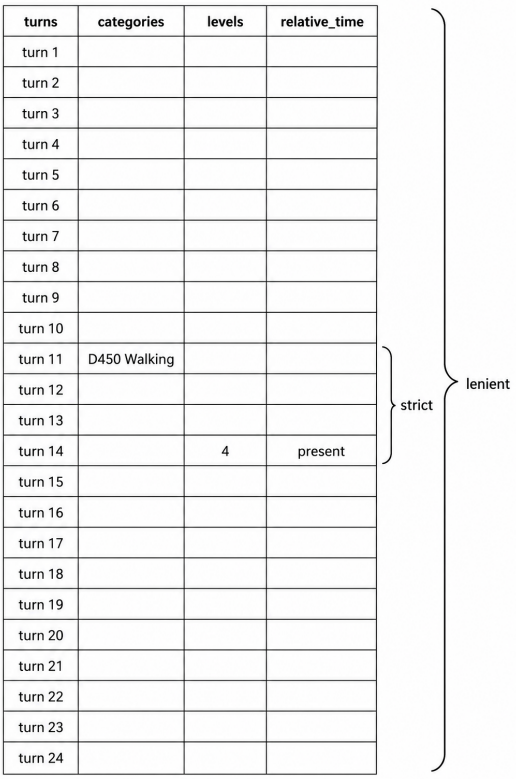


Figure 3.1: Visualisation of the strict/lenient filtering.

tations at the discourse level, though further improvement will be needed to properly assess ICF annotations in conversational data.

Combining the two filtering conditions with the two evaluation conditions gives four evaluation settings in total, summarised in Table 3.4.

		Filtering	
		Strict	Lenient
Evaluation	Strict	Evaluation setting 1	Evaluation setting 2
	Lenient	Evaluation setting 3	Evaluation setting 4

Table 3.4: Overview of the four evaluation settings resulting from the combination of strict and lenient filtering and evaluation criteria.

3.8.3 Annotation density

The mean annotation density per segment is shown in Table 3.5. Annotation density is defined as the number of turns containing at least one annotation divided by the total number of turns in a segment.

The table shows that strict segments are short, averaging less than three turns. Lenient segments are around 20 turns longer, but these extra turns are context turns that mostly carry no annotation. This explains the considerably lower annotation density under lenient filtering compared to strict filtering.

	Strict	Lenient
Total number of turns	2,068	16,023
Average segment length (turns)	2.79	22.78
Average category density	0.936	0.149
Average level density	0.686	0.096
Average time density	0.924	0.225

Table 3.5: Summary statistics of the strict and lenient segments.

Under strict filtering, category and relative time density are both high. Nearly every turn contains at least one annotation. This is expected for the categories, since the filtering was designed to separate densely annotated segments. Level density is lower, which makes sense: levels mark the end of a segment and are discussed less frequently than categories in the conversations.

The relative time density is notable. It is comparable to the category density rather than the level density, even though relative time was meant to be annotated only on turns where a level was also specified. This suggests that annotators did not always follow this instruction consistently, and annotated relative time more broadly than intended.

3.9 Label distribution

3.9.1 Overall task distribution

Table 3.6 shows the number of annotated instances per task in the training and test data. The training data contains over 20 times more instances than the test data. In both datasets, only a small proportion of turns are annotated. Proportionally, the training data contains more category and relative time annotations than the test data, while level annotations are slightly more common in the test data. This suggests that annotators were more conservative when assigning categories and relative time, but more liberal when assigning levels. The lower number of level annotations than category annotations for both the human and synthetic annotators is expected, as an ICF category may be discussed across multiple turns without its level being explicitly stated. However, the difference was larger than anticipated.

Notably, the number of category and relative time annotations in the test data are equal. This suggests that annotators assigned a relative time label to almost every turn with an ICF category, rather than only to turns where a level was also specified, as intended. This also confirms the pattern seen in Section 3.8.3.

3.9.2 ICF categories

The full label distribution for the ICF categories is shown in Figure 3.2. The data is heavily imbalanced, with the majority of turns labelled 'None'. To make the distribution of the remaining categories visible, 'None' has been excluded from the figures.

The category distributions differ between training and test data in several ways. In the training data, "D760 Family relationships" and "B164 Higher-level cognitive functions" both account for 12.9% of instances. In the test data however, "D760 Family relationships" is the most common category at 15.5%, while "B164 Higher-level cogni-

Dataset	Task	Total turns (N)	Annotated instances (n)	% of turns
Train	ICF categories	1,095,655	95,954	8.76
	ICF levels	1,095,655	2,148	0.20
	Relative time	1,095,655	121,043	11.05
Test	ICF categories	52,285	1,104	2.11
	ICF levels	52,285	698	1.33
	Relative time	52,285	1,104	2.11

Table 3.6: Absolute and proportional number of annotated instances per task.

tive functions" drops to only 3.5%. Some categories are substantially more frequent in the test data than in the training data. "B140 Attention functions", for example, makes up only 1.1% of training instances but 9.1% of test instances. The reverse also occurs: "B152 Emotional functions" is the dominant category in the training data at 15.7% and drops to 11.6%. In the test set, "D840-859 Work and employment" rarely occurs (0.9%). These distribution mismatches can cause models to overpredict categories that are common in training but rare in the test set, and to underpredict categories that are rare in training but frequent at test time.

3.9.3 ICF levels

The full label distribution for the ICF levels is shown in Figure 3.3. The data is heavily imbalanced, with the majority of turns labelled 'None'. To make the distribution of the remaining categories visible, 'None' has been excluded from the figure.

The label distributions differ substantially between the training and test sets. In the training set, the lower levels are more common, whereas level 5 is the most frequent label in the test set. This imbalance is expected to bias the models towards predicting lower levels.

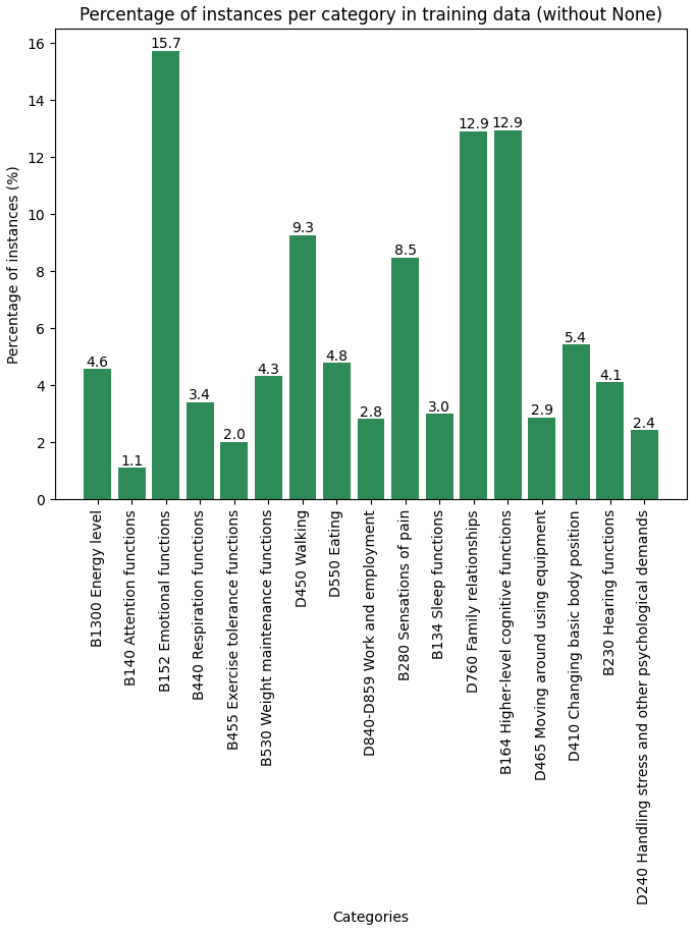
Level 0 accounts for 37.3% of the training annotations but only 1.3% of the test annotations. Such a large difference could strongly influence the models to predict level 0. Therefore, level 0 was removed from the training data. The resulting label distribution is shown in Figure 3.4.

Despite removing level 0, differences between the datasets remain. Level 1 represents more than half of the training annotations but only 6.4% of the test annotations. In contrast, level 5 accounts for 10.8% of the training annotations and nearly one third of the test annotations. These differences are still likely to bias the models towards predicting the lower levels.

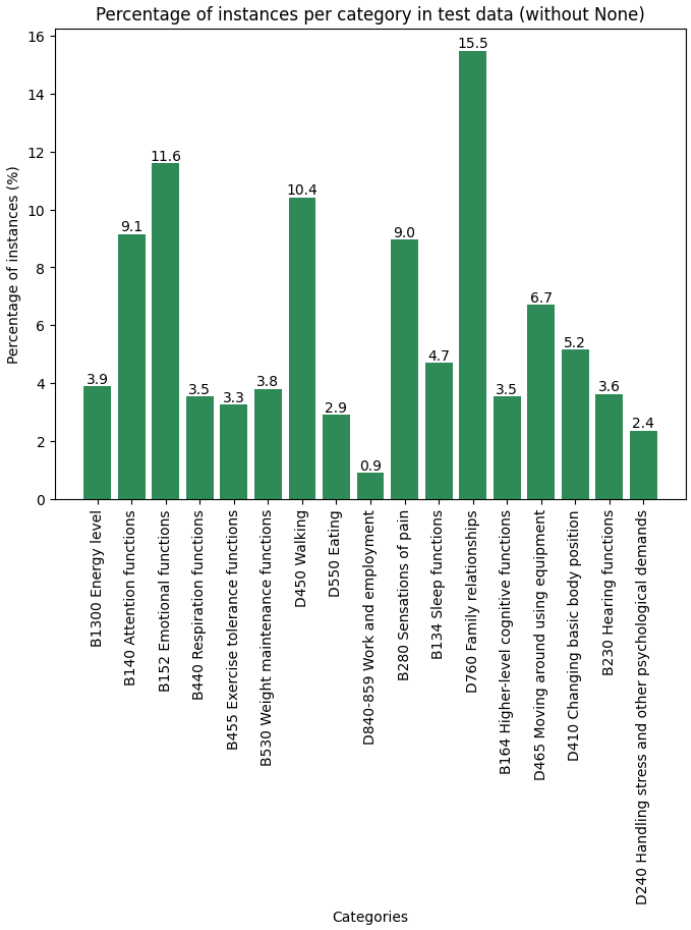
3.9.4 Relative time

The full label distribution for the relative time is shown in Figure 3.5. The data is heavily imbalanced, with the majority of turns labelled 'None'. To make the distribution of the remaining categories visible, 'None' has been excluded from the figure.

The train and test label distributions for relative time are more similar than those for ICF categories and levels. In both datasets, "present" is the most frequent label, whereas "future" occurs only rarely. The main difference is found for the "past" label. It accounts for 24.1% of the training annotations but only 6.3% of the test annotations.

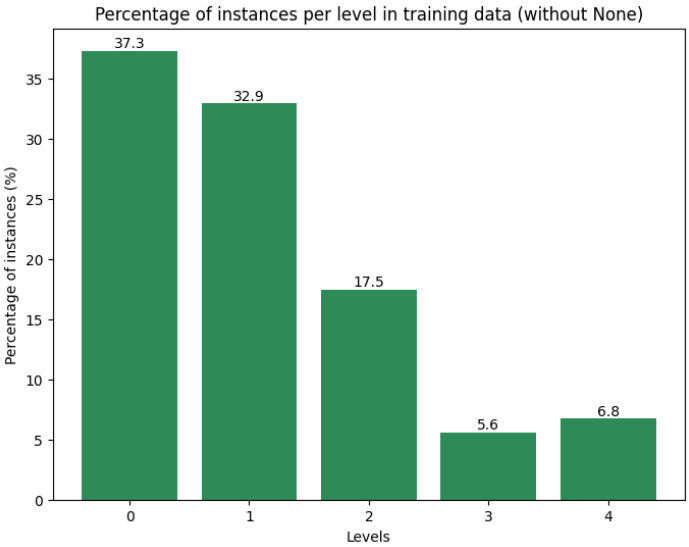


(a) Distribution of categories in the training set. The label "None" was excluded.

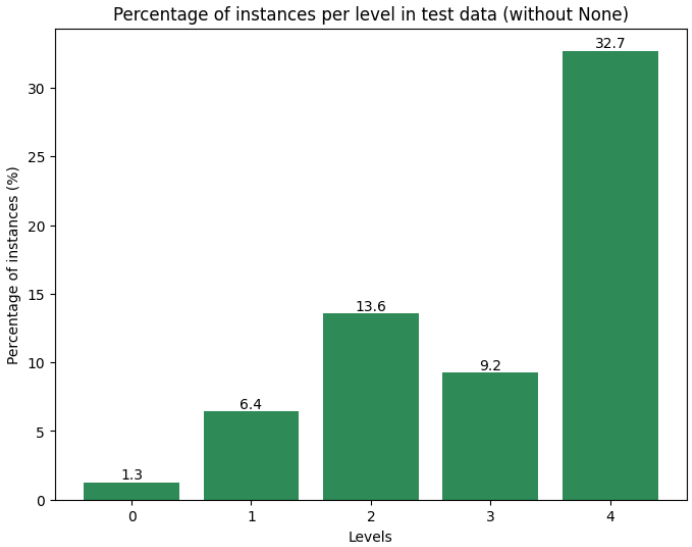


(b) Distribution of categories in the test set. The label "None" was excluded.

Figure 3.2: Category distributions for the training and test sets.



(a) Distribution levels training set. The label 'None' was excluded.



(b) Distribution levels test set. The label 'None' was excluded.

Figure 3.3: Level distributions for the training and test set.

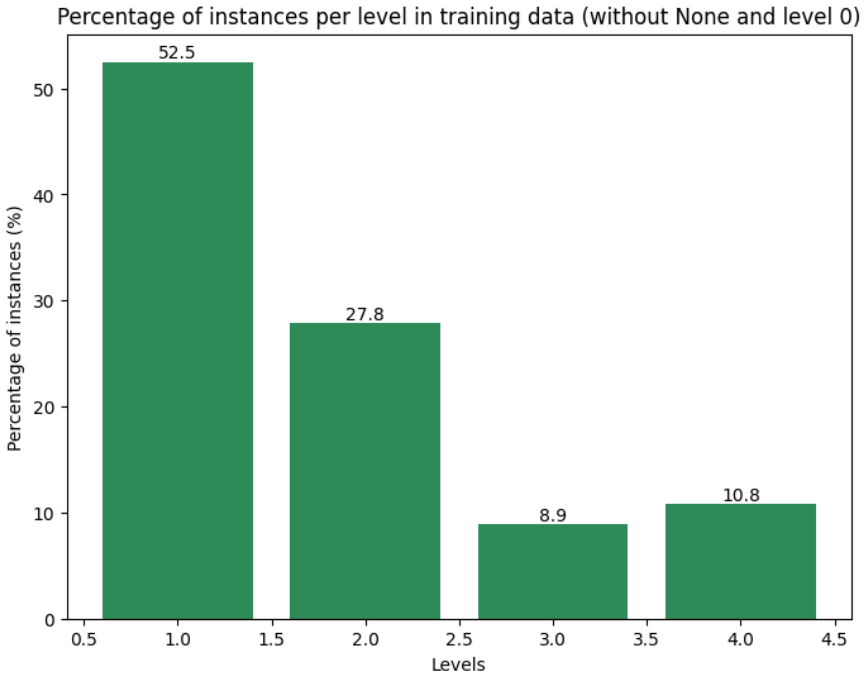
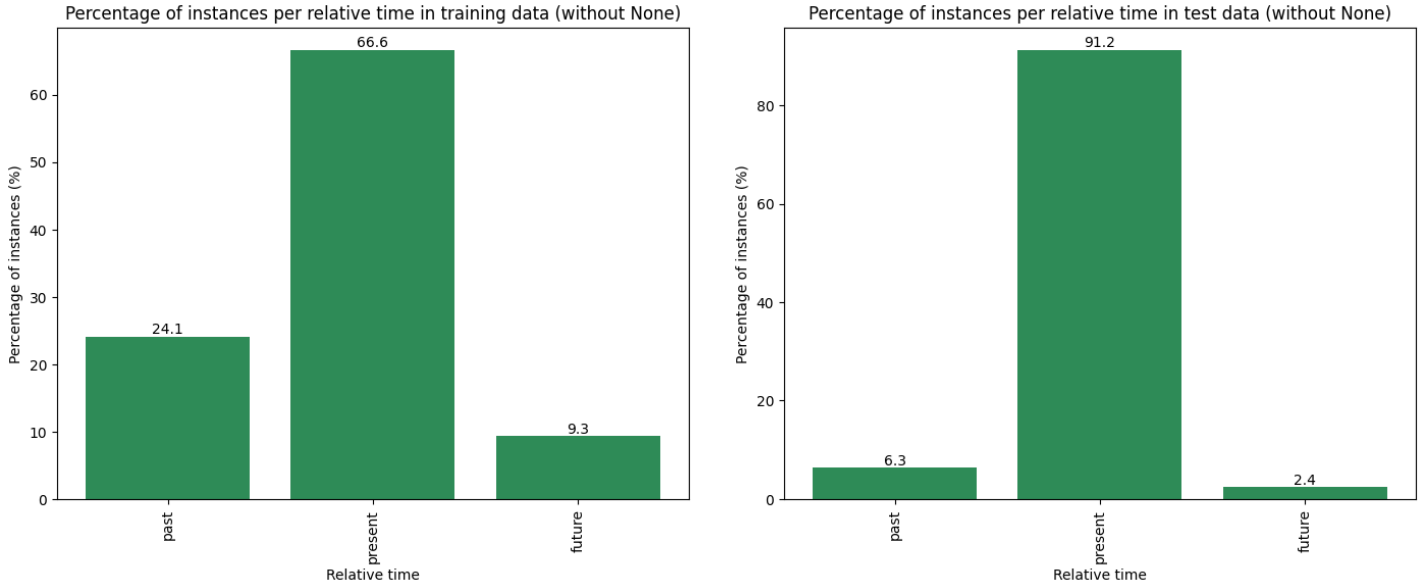


Figure 3.4: Distribution levels training set. The label 'None' The labels 'None' and 0 was excluded.

This difference is likely to bias the models towards predicting "past" more often than it occurs in the test set, while "present" is expected to remain the dominant prediction.



(a) Distribution time training set. The label 'None' was excluded.

(b) Distribution time test set. The label 'None' was excluded.

Figure 3.5: Level distributions for the training and test set.

3.10 Data limitations

There are a few limitations in the data worth noting. Van der Roest et al. (2019) acknowledge that a selection bias may have occurred in the Dutch sample, most notably in the low rate of cognitive impairment. Patients with more advanced cognitive decline are likely underrepresented.

The transcription quality also poses a challenge. The audio recordings are not always transcribed accurately. Some turns are incomprehensible, and repetitions of words or phrases occur. This can in turn affect the quality of the synthetic training annotations, which were generated by an LLM from these transcriptions. Both issues could negatively affect model performance.

Chapter 4

Methodology

This chapter describes the methodology of this thesis step by step. From preprocessing the raw audio data, to classifying ICF categories, to constructing a knowledge graph from the results. The workflow of this thesis can be seen in Figure 4.1.

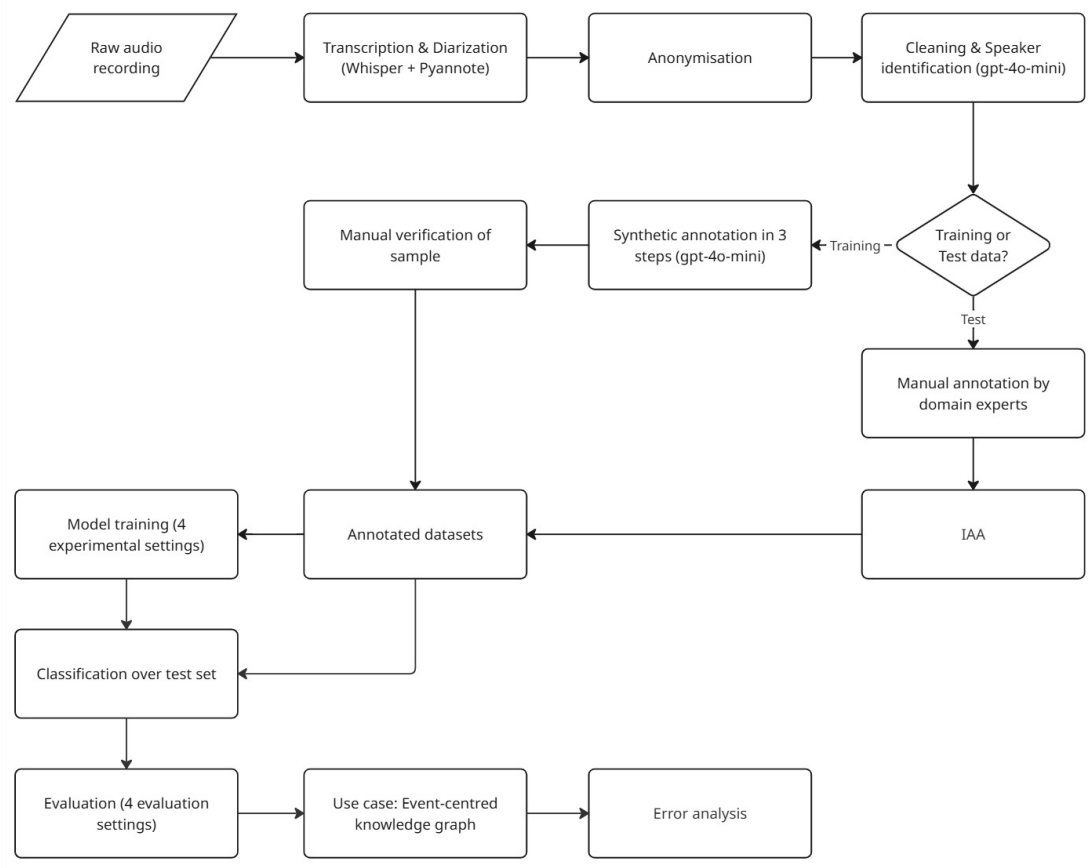


Figure 4.1: Flowchart illustrating the methodological steps taken to answer the research question.

4.1 Preprocessing

The dataset consists of recorded conversations between healthcare professionals and patients. The first step is transcribing and diarising these recordings into written text on a turn-by-turn basis.

Transcription is done using `Whisper` (Radford et al., 2023) on batches of 3 minutes, specifically using the `large` model with 1550 million parameters. Diarisation is handled by `Pyannote` (Plaquet and Bredin, 2023; Bredin, 2023), using the `speaker-diarization-3.1` model. The pipeline runs in parallel on a high-performance computing cluster using `Python 3.9`. Other transcription and diarisation tools were also tried, including `gpt-4o-transcribe-diarize` and `microsoft/VibeVoice-ASR-HF`, but these were either inaccessible or did not produce better results.

Before any further processing, the transcriptions are anonymised following Kim et al. (2022). All person names and geopolitical entities such as countries and cities, are replaced with the general placeholders `PERSON` and `GPE`.

Each conversation is then saved as a JSON file. This file stores the conversation metadata (patient, healthcare professional, recording name) and the diarised dialogue, where each turn contains the speaker label and the turn text. At this stage, speakers are not yet identified by role. They are labelled as `SPEAKER_00`, `SPEAKER_01`, and so on. The fields for relative time, ICF category, and ICF level are initialised as empty for each turn. An example of this has been shown in Section 3.6.

The JSON is then passed to `gpt-4o-mini` for cleaning. The prompt is included in Appendix C. The model assigns speaker roles based on the conversational content. This is alongside the speaker roles defined by the diarisation. It also adds confidence scores for speaker assignment and text coherence, and a notes field for flagging anything unusual such as excessive noise or a likely incorrect assignment. A sample of the cleaned transcriptions is manually checked to verify transcription quality and correct JSON formatting, since errors at this stage could negatively impact the training data. An example of the cleaned transcription is shown below:

```
{
  "filename": "T0 999000 MR",
  "patient": "999000",
  "HP": "MR",
  "T": "T0",
  "conv": {
        "turn_0": {
                "speaker": "SPEAKER_03",
                "text": "Bent u vandaag nog buiten geweest?",
                "relative_time": null,
                "ICF_category": null,
                "ICF_level": null,
                "speaker_llm": "MR",
                "confidence_score_speaker": 1,
                "confidence_score_text": 1
        },
        "turn_1": {
                "speaker": "SPEAKER_02",
                "text": "Ja. Ik ging naar de slager voor een hons
                    worst.",
                "relative_time": null,
```

```
                "ICF_category": null,
                "ICF_level": null,
                "speaker_llm": "999000",
                "confidence_score_speaker": 1,
                "confidence_score_text": 0.75,
                "notes": "Incomprehensible speech."
        },
        "turn_2": {
                "speaker": "SPEAKER_03",
                "text": "Lekker zeg. Voor bij de lunch?.",
                "relative_time": null,
                "ICF_category": null,
                "ICF_level": null,
                "speaker_llm": "MR",
                "confidence_score_speaker": 0.95,
                "confidence_score_text": 0.90
        }
  }
}
```


The training data is filtered using the confidence score of the text assigned to each turn during transcription cleaning. Two thresholds are used, namely a lower bound of 0.70 and an upper bound of 0.80, based on the overall distribution of confidence scores.

Turns with a confidence score above the upper bound are kept. Turns below the lower bound are removed. Turns that fall between the two bounds are kept only if both the preceding and following turn have a confidence score above the upper bound. The boundary check is performed before any turns are removed, so that the decisions are not influenced by already-filtered neighbours.

After filtering at the turn level, the remainders of the conversations are ranked by their average confidence score. Only the top 80% of files are kept, resulting in a final training set of 311 files.

4.2 Annotation

4.2.1 Training data

The training data is annotated synthetically using a generative model (`gpt-4o`). Annotation takes place in three steps, all using batches of 20 turns. The batch size was chosen empirically: annotators indicated that the same ICF category tends to recur within a span of roughly 16 to 18 turns, so 20 was chosen as a safe upper bound.

In the first step, the model is given a batch and asked to identify which ICF categories are relevant given the conversational context. In the second step, the same batches are used to assign an impairment level to each identified category, if one is present in the text. In the third step, the model assigns a relative time (past, present, or future) to each turn where a level was found, again using the surrounding context. If no time point can be determined, the present is used as default.

All three steps are run over a full file before moving to the next. If a batch is not returned in the correct format, for example if the JSON is malformed or the conversation content is missing, that batch is excluded from the final annotations.

The prompts for each step are included in Appendix C. Each prompt uses few-shot examples, which can be found in Appendix D. The examples cover a range of possible combinations of ICF categories, levels, and time points, though not every category or level is represented. They also illustrate common annotation mistakes and the correct alternative. Examples include turns that should not receive a level but previously did, turns from the introduction of the interview that were incorrectly assigned a category, and turns that were given a relative time annotation despite no level being specified. A sample of around 30 batches across multiple conversations was manually reviewed to verify the quality of the synthetic annotations and limit the number of annotation errors.

4.2.2 Test data

The test data is annotated manually by native Dutch medical experts from the A-Proof team. For each turn, they mark the ICF category if one is present. If the impairment level is specified, they mark the relevant turns with the corresponding level. Where a level is marked, the referred time point is also noted. If no clear time point can be determined, the present is used as default.

The quality of the baseline annotator's annotations was assessed by calculating the inter-annotator agreement (IAA) between the baseline annotator and each of the additional annotators. Cohen's Kappa (κ) was used for the categories and relative time, while weighted Cohen's Kappa (weighted κ) with linear weights was used for the levels. Cohen's κ measures the agreement between two observers for categorical values while accounting for agreement that may occur by chance (Dettori and Norvell, 2020). Weighted κ was chosen because the levels are ordinal. Disagreements between levels that are further apart receive a larger penalty than disagreements between adjacent levels.

Four conversations were exclusively annotated by the baseline annotator and were excluded from the IAA analysis.

For the categories and relative time, agreement was calculated based on an exact match between the annotations. For the levels, agreement was also calculated for the same category, but weighted κ accounted for the distance between the assigned levels. A κ score of 1 indicates perfect agreement, while 0 means the annotators agreed no more than would be expected by chance. Scores below 0 indicate that the annotators agreed less than chance would predict (Dettori and Norvell, 2020).

A turn-by-turn analysis is presented in Chapter 5, followed by a more detailed error analysis in Chapter 7. In addition to the agreement scores, the annotation distance was calculated to measure how far apart matching annotations were placed within a conversation. For each annotation made by the additional annotator, the distance in turns to the nearest matching annotation by the baseline annotator was calculated. This distance was then averaged across all matching annotations, as can be seen in Chapter 5.

Filtered test set

Before evaluation, the test set is filtered to remove low-quality turns and to find annotation dense segments. The filtering procedure described in Section 3.8.2 is applied to

obtain both the strict and lenient filtered datasets. This approach ensures that all annotated turns are maintained, keeping the counts of ICF categories, levels, and relative times intact, while reducing the number of 'None' turns.

Table 4.1 shows the number of turns per file after strict and lenient filtering, as well as the resulting number of segments. The lenient turn count can exceed the total number of turns in the original file, since segments may overlap and each segment includes 10 turns of context before and after the strict boundaries. The number of duplicate turns removed are also shown.

File	Original	Strict	Lenient	Duplicates Removed	Segments
1	1,272	348	1,765	536	71
2	1,776	140	1,900	381	88
3	2,189	103	1,123	694	51
4	1,312	456	2,794	114	117
5	1,308	56	836	524	39
6	941	66	686	411	31
7	2,189	170	1,530	694	68
8	1,938	137	1,697	439	78
9	2,447	405	1,645	951	62
10	1,007	60	760	382	35
11	991	127	1,287	212	58
Total	**17,370**	**2,068**	**16,023**	**5,338**	**698**

Table 4.1: Statistics of the filtered test data turns after duplicate removal and segmentation.

4.3 Experimental settings

Four experimental settings are examined, as shown in Table 4.2. Together, these settings help to investigate whether the models can generalise from EHR data to conversational data, and whether additional fine-tuning on conversational data improves performance.

The baseline setting uses the same approach as the A-Proof ICF-17 classifier (Chen and Vossen, 2026), which is based on MedRoBERTa.nl and was fine-tuned on manually and synthetically annotated EHR data. The same EHR data is also used to fine-tune mBERT, giving a second baseline that reflects how a multilingual model performs without any conversational data. Neither baseline predicts relative time. In the next setting, both models are further fine-tuned using the synthetically annotated conversational data in addition to the EHR data. A third setting fine-tunes both base models (mBERT and medRoBERTa) only on conversational data, to see whether EHR data is actually helpful. This model also classifies the relative time. Finally, a prompt-based setting uses `gpt-4o` with zero-shot and few-shot examples directly on the conversations without any fine-tuning.

Experimental Setting	Training Data	Models
Baseline	Manually annotated EHR + synthetically annotated EHR	mBERT, MedRoBERTa
Fine-tuning (All Data)	Manually annotated EHR + synthetically annotated EHR + synthetically annotated conversations	mBERT, MedRoBERTa
Fine-tuning (Conversations Only)	Synthetically annotated conversations	mBERT, MedRoBERTa
Prompt-based Generation	Conversations (zero-/few-shot prompting)	ChatGPT-5

Table 4.2: Experimental settings.

4.4 Model description

All classification and regression models were trained for 3 epochs with a batch size of 16. Performance was checked on a development set after each epoch. For the baseline settings, the existing development set from the EHR data was used. For the conversational data settings, the training conversations were split 80/20 (249/62 conversations) by patient, ensuring all patient files were classified in only one of the datasets.

The mBERT model was initialised from `bert-base-multilingual-cased` and MedRoBERTa from `CLTL/MedRoBERTa.nl`. Training was performed on an HPC with 8 cores, 96GB of memory, and a `gpu 4g.47gb` node. Total training time was approximately 5 hours and 45 minutes for each model.

Hyperparameters were determined through manual tuning and are summarised in Table 4.3. Fine-tuning on existing model weights generally required lower learning rates.

4.4.1 ICF categories

ICF category prediction is constructed as a multi-label classification task. For each turn, the model predicts all applicable ICF categories from the ICF-17 set. The model receives a single turn as input and outputs a list of predicted categories. Performance on the development set is measured using micro, macro, and weighted F1-scores.

4.4.2 ICF levels

ICF level prediction is built as a regression task. The training data is first expanded so that each row corresponds to one ICF category in a turn. Thus, turns with multiple categories produce multiple rows.

The model receives the current turn as input, along with the current ICF category and the five most recently occurring categories from previous turns in the same conversation, passed as special tokens. This five-turn window was chosen based on an analysis of the annotated test data. In most cases, a turn where a level is specified had the corresponding category mentioned within the previous five turns. Since a turn can

Task	Model	Setting	Learning rate (α)
ICF categories	mBERT	Baseline	2e-5
		All data	5e-6
		Conv. only	2e-5
	MedRoBERTa	Baseline	2e-5
		All data	5e-6
		Conv. only	2e-5
ICF levels	mBERT	Baseline	2e-5
		All data	1e-5
		Conv. only	2e-5
	MedRoBERTa	Baseline	2e-5
		All data	1e-5
		Conv. only	2e-5
Relative time	mBERT	Conv. only	2e-5
	MedRoBERTa	Conv. only	2e-5

Table 4.3: Learning rates per task, model, and experimental setting.

carry multiple labels which producing multiple rows, the model also takes these into account across the window. The levels are then predicted for each row (and thus each category).

The baseline model is an exception: it was trained with no previous category context, since the EHR data is non-consecutive and has no meaningful conversational history.

Turns without a level are assigned the value -1 during training, allowing the regression model to handle them. These turns are excluded from evaluation (see Section 4.5.2). Performance is measured using the regression metrics MAE, MSE, and RMSE (further explained in Section 4.5.2).

There are two variants of this model: one that uses gold ICF categories as input for the context, and one that uses predicted categories. This measures how much the level prediction is impacted by the quality of the category prediction.

4.4.3 Relative time

Relative time prediction is also a multi-label classification task. It was trained only on conversational data, since the EHR data does not contain relative time annotations.

The model receives the current turn as input, along with the five most recent categories and their corresponding levels from previous turns. This results in four experimental variants, combining gold or predicted categories with gold or predicted levels. Turns where both the category and level are "None" are excluded from training, as these turns are not expected to carry a relative time.

Performance is measured using micro, macro, and weighted F1-scores.

4.4.4 Prompt-based generation

The prompt-based setting uses the same model and prompts as the synthetic annotation step (see Appendix C). Two variants are evaluated, namely a few-shot variant using the same examples as before (see Appendix D), and a zero-shot variant with no

	Pred. cat	No pred. cat
Gold cat	FP1a	FP1b
No gold cat	FP2a	FP2b

(a) False positives

	Pred. cat	No pred. cat
Gold cat	FN1a	FN1b
No gold cat	FN2a	FN2b

(b) False negatives

Table 4.4: Matrices of level prediction errors. False positives correspond to instances with no gold level but a predicted level. False negatives correspond to instances with a gold level but no predicted level. Rows indicate whether the category is present in the gold annotations, while columns indicate whether the category was predicted.

examples provided.

After training, each model is applied to the test set to predict the ICF category and impairment level for every turn. Relative time is predicted only by the models fine-tuned on only conversational data and the prompt-based generation model.

4.5 Evaluation

Model performance is evaluated on the annotated test set described in Section 4.2.2, consisting of 11 conversations from 5 patients across two or three time points. During evaluation, turns are fed to the models in the same way as during training, including any conversational history or as expanded turns per category if applicable.

ICF categories, levels, and relative time are each evaluated for all four settings described in Section 3.8.2.

4.5.1 ICF Categories

ICF category prediction is evaluated using macro F1-score, with precision and recall reported alongside it. The macro F1-score is chosen because the data is highly imbalanced. Most turns are labelled "None", and the remaining ICF categories are also unevenly distributed. A classification report and confusion matrix are produced for each experimental setting.

4.5.2 ICF levels

ICF level evaluation is split into two parts. The first part addresses turns where either the gold or predicted category is "None", or where either the gold or predicted level is "None". In these cases, a standard regression evaluation is not possible. Instead, these turns contribute to false positives and false negatives. There are eight possible combinations, illustrated in Table 4.4. A false positive occurs when no gold level is assigned but the model predicts a level on the 0-4 scale. A false negative occurs when a gold level is assigned but the model fails to predict one.

The remaining turns, those where both a gold and predicted level are present, are evaluated using three regression metrics: MAE, MSE, and RMSE.

MAE Mean Absolute Error measures the average absolute difference between predicted and gold values (Miller et al., 2024):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

where $\hat{y}_i$ is the predicted value, y_i is the gold value, and n is the number of instances. Because all errors are weighted equally, MAE is relatively robust to outliers. The downside is that it cannot meaningfully compare predictions across datasets with different variance (Miller et al., 2024).

MSE Mean Squared Error averages the squared differences between predicted and gold values (Jaradat et al., 2025):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Squaring the errors means larger mistakes are penalised more heavily, which is useful when large errors are particularly costly. However, this also makes MSE sensitive to outliers, and the squared units make it less intuitive to interpret directly (Jaradat et al., 2025).

RMSE Root Mean Squared Error takes the square root of MSE (Miller et al., 2024):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

This reinstated the original unit of measurement, making RMSE more interpretable than MSE. Like MSE, it is sensitive to outliers due to the squaring of errors (Jaradat et al., 2025).

4.5.3 Relative time

Relative time prediction is evaluated only for turns with an ICF category and level annotation. Turns where both the category and level are labelled "None" are therefore excluded. Relative time prediction is then evaluated using macro F1-score since the data is imbalanced, with the majority of turns labelled "present". Precision and recall are reported alongside it. A classification report and confusion matrix are produced for each experimental setting.

The predictions of all models are saved to CSV files.

4.6 Error analysis

After the results are presented and discussed, an error analysis is conducted. Four aspects are examined. First, inter-annotator agreement is revisited at the segment level, since the turn-by-turn approach used in the Chapter 5 may be too strict for conversational data. Second, cases where a category is annotated but no level is specified are investigated, to see whether this affects model predictions. Third, the synthetic training annotations are examined more closely, focusing on why level 0 was assigned so frequently. The prompt is then adjusted to test whether this can be corrected. Finally, categories that are often confused with one another are analysed to understand why the models may struggle to distinguish them.

4.7 Use-case: event-centred knowledge graph

As part of this thesis, a conceptual example of an event-centred knowledge graph is developed. The graph is constructed for a single patient using both the predicted and gold annotations from the medical conversations. This use case is intended as a proof of concept and does not represent a complete knowledge graph. A full implementation would include all information and relations extracted from the conversations.

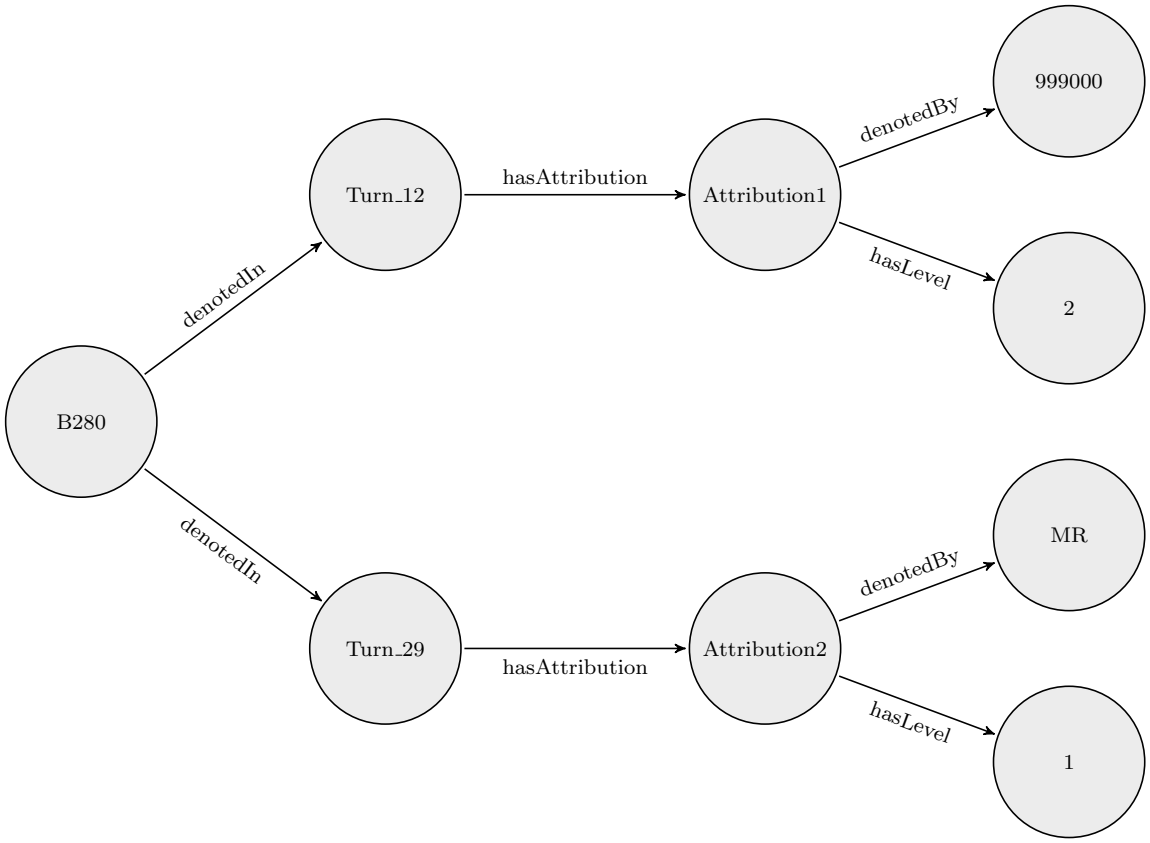
First, the conversations of one patient are selected. For each filtered conversation, the turns containing a category, a level and a relative time annotation are extracted. The relative time annotations are then mapped to a timeline, as shown in Table 4.5. The lower-case t denotes an actual point in time, whereas the upper-case T denotes the conversation in which the information was mentioned.

For this example, it is assumed that the patient participated in three conversations, each taking place six months apart. Conversation $T0$ takes place at time point t_0. Mentions referring to the "past" are therefore assigned to t_{-1}, "present" mentions to t_0, and "future" mentions to t_1. Conversation $T1$ takes place six months later at t_1. Its "past", "present", and "future" mentions are therefore assigned to t_0, t_1, and t_2, respectively. Finally, conversation $T2$ takes place at t_2, with mentions referring to t_1, t_2, and t_3. As a result, multiple conversations may refer to the same point in time. For example, time point t_1 contains the "future" mentions from $T0$, the "present" mentions from $T1$, and the "past" mentions from $T2$.

Time point	**Conversation T_0**	**Conversation T_1**	**Conversation T_2**
t_{-1}	Past_{T_0}		
t_0	Present_{T_0}	Past_{T_1}	
t_1	Future_{T_0}	Present_{T_1}	Past_{T_2}
t_2		Future_{T_1}	Present_{T_2}
t_3			Future_{T_2}

Table 4.5: Mapping of relative time annotations to a shared timeline.

For each time point, the levels assigned to each category are extracted separately for the patient, the healthcare professional, and possibly a 3rd Party. These annotations form the mentions in the knowledge graph. The mentions are then converted into triples for each category. The figure below illustrates this process for the category "B280 Sensations of Pain" (represented as "B280") in the form of a graph, assuming two mentions are available.



The resulting graph provides an overview of the information mentioned for each category at each time point. Because multiple conversations can contribute information to the same time point, statements from different conversations and speakers can be compared. In this use case, triples are constructed for a single patient over time. The mentions are collected from the best-performing model in the best-performing evaluation setting for relative time, since the main focus is to analyse fluctuations in events over time. Summary statistics, including the mean and standard deviation of the levels, are calculated separately for the patient and the healthcare professional for both the gold and the predicted labels. This allows the perspectives of both speakers to be compared for each category in the gold and predicted data. In addition, changes in the reported levels across time points can be explored. The results are presented in a table rather than as a graph, as this provides a clearer overview in the context of this report.

Chapter 5

Results

This chapter describes the results. First, the inter-annotator agreement on the test is presented. Next, the model results are presented for each subtask in turn: ICF category prediction, ICF level prediction, and relative time prediction. Each subtask is evaluated under four conditions, combining strict or lenient test set filtering with strict or lenient evaluation, as described in Section 3.8.2. The results will be further discussed in Chapter 6.

5.1 Inter-annotator agreement

The inter-annotator agreement (IAA) was calculated using (weighted) Cohen's κ, which is shown in Table 5.1 for each additional annotator. As mentioned in Chapter 3, each additional annotator corresponds to the file number they annotated.

Category agreement ranges from poor to fair, with most scores between 0.200 and 0.300 (Dettori and Norvell, 2020). File 7 reached the highest category agreement according to Cohen's κ, and is an exception, reaching moderate agreement ($\kappa = 0.483$). File 9 showed only slight category agreement ($\kappa = 0.131$), indicating that the two annotators rarely assigned the same categories to the same turns. Level agreement is measured using a linearly weighted kappa, considering only turns where both annotators assigned a level. Scores vary significantly across the files, ranging from 0.130 (slight agreement) to 0.616 (substantial agreement). The highest agreement is between the baseline annotator and annotator 7 (weighted $\kappa = 0.616$), while annotator 9 again shows the lowest agreement (weighted $\kappa = 0.146$). For relative time, the highest agreement is between the baseline annotator and annotator 4 ($\kappa = 0.400$), and the lowest is with annotator 1.2 ($\kappa = 0.146$). Annotators 7 and 8 did not annotate relative time, so no score could be calculated for those files.

File	**1.1**	**1.2**	**2**	**4**	**7**	**8**	**9**	**11**
κ categories	0.244	0.255	0.329	0.292	0.483	0.423	0.131	0.281
weighted κ levels	0.130	0.308	0.445	0.374	0.616	0.374	0.146	0.189
κ time	0.293	0.146	0.326	0.400	—	—	0.223	0.273

Table 5.1: Cohen's κ-scores for the test set annotations.

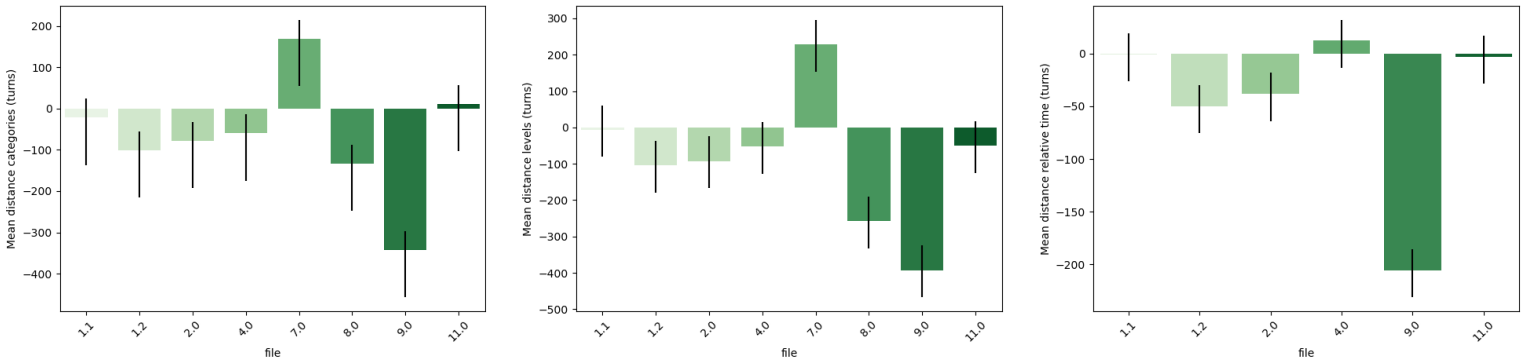
Apart from Cohen's κ, the annotation distance was also calculated and visualised

in graphs.

Figure 5.1a shows the annotation distances for categories. File 9 again stands out with a very large negative mean distance, meaning the additional annotator consistently annotated much earlier in the conversation than the baseline annotator. This can also happen if they annotated different segments of the conversation for each category. Most other files also show negative distances, meaning the additional annotators tended to annotate categories earlier in general. Files 1.1 and 11 have means close to zero, indicating that the annotators placed their annotations at nearly the same turns.

Figure 5.1b shows the distances for levels. File 9 again shows a large negative mean of almost -400 turns. File 1.1 has a mean close to zero, but a relatively large error bar, indicating high variability in where annotations were placed. All distances are negative except for file 7.

Figure 5.1c shows the distances for relative time. Since files 7 and 8 were not annotated for the relative time, they were omitted in the figure. Files 1.1 and 11 again show means close to zero. File 9 shows a mean distance of around -200 turns, though the error bars are relatively small, suggesting the distance was consistent rather than fluctuating. All distances apart from file 4 are negative, meaning the additional annotator consistently annotated earlier than the baseline annotator.



(a) Average annotation distance categories.

(b) Average annotation distance levels.

(c) Average annotation distance relative time.

Figure 5.1: Average annotation distances.

Overall, the annotation results indicate relatively low agreement between annotators. Furthermore, the annotation distances indicate that annotators often did not assign annotations to the same turns and, in some cases, not even within the same segments. These findings suggest that agreement is already difficult for human annotators, which can put the model results in perspective. Lenient evaluation may therefore be more appropriate for this kind of data, and a more thorough analysis will be done in Chapter 7. All model results in the following sections are evaluated against the gold labels of the baseline annotator.

5.2 ICF categories

This section presents the results for ICF category prediction per evaluation setting. For each setting, a classification report is shown. For the best performing model in each setting, the confusion matrix is included in the main text. All remaining confusion

matrices can be found in Appendix E.

5.2.1 Evaluation setting 1 (strict filtering, strict evaluation)

This first section discusses the results obtained using strict filtering in combination with strict evaluation. In this setting, the segments without surrounding context are evaluated turn-by-turn. Given the conversational nature of the data, evaluation setting 1 is expected to punish turn-level mismatches harshly. Results of this setting are shown in Table 5.2.

Macro average F1 ranges from 0.410 to 0.461. For mBERT and MedRoBERTa, precision tends to outperform recall, with precision above 0.500 and recall around 0.390. The LLM prompted models show a more balanced trade-off between precision and recall. The zero-shot prompting approach achieved a macro precision of 0.479 and a recall of 0.437, while the few-shot prompting approach achieved a macro precision of 0.445 and a recall of 0.506.

Overall, mBERT slightly outperforms MedRoBERTa. Among all model settings, few-shot prompting performs best, followed by zero-shot. Both LLM settings slightly outperform the fine-tuned models. Among the fine-tuned models, the All Data setting performs best for both mBERT and MedRoBERTa, and all fine-tuned models outperform their respective baselines.

The best performing categories are "B1300 Energy level" for mBERT (baseline F1=0.623) and "D760 Family relationships" for MedRoBERTa (baseline F1=0.603). In the All Data and Conversational Data settings, "B530 Weight maintenance functions" scores highest for both models (All Data F1=0.706, Conversational Data F1=0.696). The LLM prompted models perform best on "B280 Sensations of pain" (zero-shot F1=0.778, few-shot F1=0.789). "None" scores 0.000 across all models and settings, since the support is 0. "D840-859 Work and employment" also scores 0.000 for all models except the baseline. "B140 Attention functions" scores consistently low, with F1 between 0.120 and 0.130 across all models.

Figure 5.2 shows the confusion matrices for the LLM-prompted models under strict filtering and strict evaluation. In the zero-shot setting (Figure 5.2a), several categories are predicted correctly, including "B1300 Energy level", "B280 Sensations of pain", and "B440 Respiration functions".

The most common source of confusion is "None". Nearly all categories are at some point misclassified as "None". Additionally, "B140 Attention functions" is frequently confused with "B152 Emotional functions" (zero-shot=45 times, few-shot=58 times), and "B455 Exercise tolerance functions" with "D450 Walking" (zero-shot=13 times, few-shot=11 times). Some confusion also exists between "D465 Moving around using equipment", "D410 Changing basic body position", and "D450 Walking".

Additionally, "D840-859 Work and employment" is never predicted. The few-shot setting (Figure 5.2b) shows similar patterns.

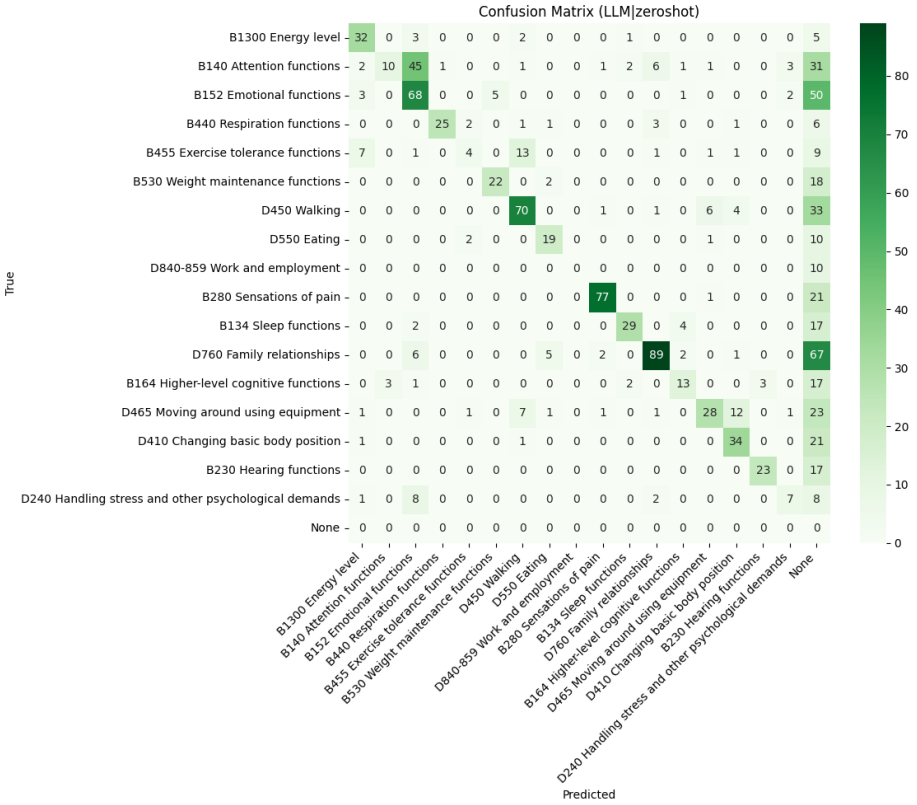
5.2.2 Evaluation setting 2 (strict filtering, lenient evaluation)

In the second evaluation setting, strict filtering is combined with lenient evaluation. Rather than evaluating predictions on a turn-by-turn basis, the evaluation checks whether the predicted category appears anywhere within the segment defined by the

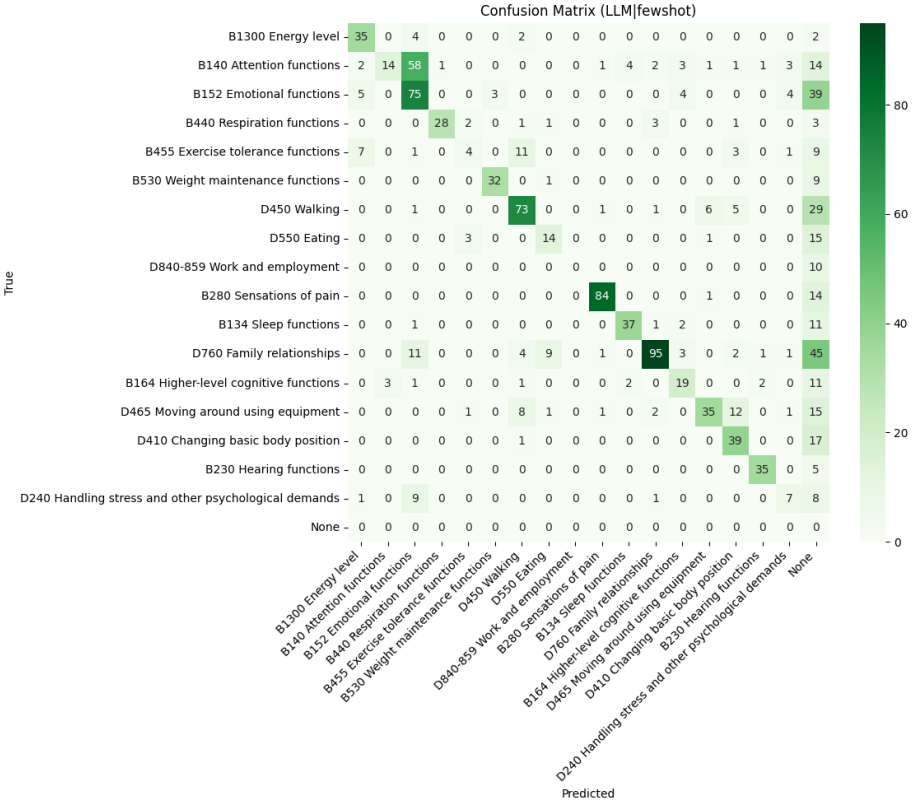
	Baseline						**All data**						**Support**
	mBERT			MedRoBERTa.nl			mBERT			MedRoBERTa.nl			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
B1300 Energy level	0.706	**0.558**	**0.623**	0.632	**0.558**	0.593	0.610	0.581	0.595	0.667	0.558	0.608	43
B140 Attention functions	0.778	0.069	0.127	0.778	0.069	0.127	0.500	0.069	0.122	0.583	0.069	0.124	104
B152 Emotional functions	0.414	0.414	0.414	0.453	0.453	0.453	0.445	0.539	0.488	0.437	0.516	0.473	129
B440 Respiration functions	0.708	0.436	0.540	0.472	0.436	0.453	0.706	0.615	0.658	0.686	**0.615**	0.649	39
D455 Exercise tolerance functions	0.111	0.028	0.044	0.286	0.056	0.093	0.167	0.056	0.083	0.333	0.056	0.095	36
B530 Weight maintenance functions	**1.000**	0.405	0.576	**0.905**	0.452	**0.603**	**0.923**	0.571	**0.706**	**0.923**	0.571	**0.706**	42
D450 Walking	0.600	0.548	0.573	0.602	0.513	0.554	0.517	0.643	0.574	0.481	0.557	0.516	115
B550 Eating	0.565	0.406	0.473	0.517	0.469	0.492	0.516	0.500	0.508	0.556	0.469	0.508	33
D840-859 Work and employment	0.375	0.300	0.333	0.300	0.300	0.300	0.000	0.000	0.000	0.000	0.000	0.000	10
B280 Sensations of pain	0.725	0.505	0.595	0.727	0.485	0.582	0.705	0.626	0.663	0.734	0.586	0.652	99
B134 Sleep functions	0.651	0.538	0.589	0.596	0.538	0.566	0.692	0.519	0.593	0.730	0.519	0.607	52
D760 Family relationships	0.733	0.497	0.592	0.687	0.538	**0.603**	0.752	0.462	0.572	0.772	0.456	0.574	172
B164 Higher-level cognitive functions	0.414	0.308	0.353	0.344	0.282	0.310	0.288	0.385	0.330	0.311	0.359	0.333	39
D465 Moving around using equipment	0.492	0.405	0.444	0.508	0.432	0.467	0.500	0.446	0.471	0.516	0.432	0.471	76
D410 Changing basic body position	0.611	0.386	0.473	0.559	0.333	0.418	0.615	0.421	0.500	0.605	0.456	0.520	57
B230 Hearing functions	0.400	0.500	0.444	0.397	0.575	0.469	0.676	**0.625**	0.649	0.667	0.500	0.571	40
D240 Handling stress and other psychological demands	0.184	0.269	0.219	0.429	0.231	0.300	0.500	0.231	0.316	0.667	0.231	0.343	26
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
micro avg	0.566	0.409	0.475	0.562	0.419	0.480	0.573	0.464	0.513	0.585	0.441	0.503	1112
macro avg	0.526	0.365	0.412	0.511	0.373	0.410	0.506	0.405	0.435	0.537	0.386	0.431	1112
weighted avg	0.603	0.409	0.464	0.588	0.419	0.464	0.580	0.464	0.494	0.603	0.441	0.486	1112

	Conversational data						**LLM prompt generation**						
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
B1300 Energy level	0.581	0.581	0.581	0.632	0.558	0.593	0.604	0.744	0.667	0.583	0.814	0.680	43
B140 Attention functions	0.538	0.069	0.123	0.538	0.069	0.123	0.500	0.099	0.165	0.438	0.139	0.211	104
B152 Emotional functions	0.447	0.523	0.482	0.461	0.555	0.504	0.389	0.531	0.449	0.323	0.586	0.417	129
B440 Respiration functions	0.706	**0.615**	0.658	**0.615**	0.615	0.615	**0.781**	0.641	0.704	0.609	0.718	0.659	39
D455 Exercise tolerance functions	0.091	0.028	0.043	0.250	0.056	0.091	0.235	0.111	0.151	0.138	0.111	0.123	36
B530 Weight maintenance functions	**0.889**	0.571	**0.696**	**0.889**	0.571	**0.696**	0.759	0.524	0.620	**0.800**	0.762	0.780	42
D450 Walking	0.476	0.600	0.531	0.466	0.539	0.500	0.452	0.609	0.519	0.462	0.635	0.535	115
B550 Eating	0.571	0.500	0.533	0.536	0.469	0.500	0.528	0.594	0.559	0.400	0.438	0.418	33
D840-859 Work and employment	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	10
B280 Sensations of pain	0.708	0.636	0.670	0.686	0.596	0.638	0.778	**0.778**	**0.778**	0.737	0.848	0.789	99
B134 Sleep functions	0.718	0.538	0.615	0.750	0.519	0.614	0.725	0.558	0.630	0.712	0.712	0.712	52
D760 Family relationships	0.726	0.497	0.590	0.768	0.444	0.563	0.712	0.520	0.601	0.709	0.556	0.623	172
B164 Higher-level cognitive functions	0.263	0.385	0.312	0.302	0.410	0.348	0.265	0.333	0.295	0.211	0.487	0.295	39
D465 Moving around using equipment	0.500	0.419	0.456	0.469	0.405	0.435	0.406	0.378	0.392	0.449	0.473	0.461	76
D410 Changing basic body position	0.591	0.456	0.515	0.521	0.439	0.476	0.425	0.596	0.496	0.424	0.684	0.523	57
B230 Hearing functions	0.686	0.600	0.640	0.639	0.575	0.605	0.697	0.575	0.630	0.745	**0.875**	**0.805**	40
D240 Handling stress and other psychological demands	0.500	0.231	0.316	0.500	0.231	0.316	0.368	0.269	0.311	0.280	0.269	0.275	26
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
micro avg	0.564	0.463	0.508	0.562	0.445	0.496	0.533	0.498	0.515	0.495	0.567	0.529	1112
macro avg	0.500	0.403	0.431	0.501	0.392	0.423	0.479	0.437	0.443	0.445	0.506	0.461	1112
weighted avg	0.571	0.463	0.491	0.575	0.445	0.479	0.547	0.498	0.502	0.519	0.567	0.523	1112

Table 5.2: Classification report of the category predictions with strict filtering and strict evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing category per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.



(a) Zero-shot



(b) Few-shot

Figure 5.2: Confusion matrices for the categories prediction task of Prompt-Based Generation in the strict filtering and strict evaluation setting.

filtering procedure. This evaluation setting is expected to produce better results than strict evaluation, as predictions are no longer required to match the exact turn of the gold annotation. The results for the second evaluation setting are presented in Table 5.3 and confirm this expectation.

Under lenient evaluation, macro precision ranges from 0.722 to 0.944, while macro recall ranges from 0.515 to 0.620. Both metrics are higher than under strict evaluation, where macro precision was approximately 0.500 and macro recall around 0.410. As under strict evaluation, the fine-tuned models achieve higher precision than recall. In contrast to the strict evaluation setting, the LLM-prompted models also show relatively higher precision than recall under lenient evaluation.

The baseline is the only setting where MedRoBERTa outperforms mBERT (macro F1 MedRoBERTa=0.660, mBERT=0.653). For the models fine-tuned on All Data, mBERT performs better (macro F1 mBERT=0.664, MedRoBERTa=0.646), while both models score similarly when being fine-tuned on Conversations Only (macro F1 mBERT=0.587, MedRoBERTa=0.587). Few-shot prompting performs best overall, slightly ahead of fine-tuned mBERT on All Data (few-shot macro F1=0.697). Zero-shot performs slightly below the All Data mBERT setting (zero-shot macro F1=0.655). Based on macro scores, the baseline outperforms the Conversational Data model.

In the baseline, all categories except "None" achieve precision of 1.000. The best F1 scores go to "B134 Sleep functions" for mBERT (F1=0.906) and "B230 Hearing functions" for MedRoBERTa (F1=0.947). "D455 Exercise tolerance functions" performs worst for both (mBERT F1=0.074, MedRoBERTa F1=0.143). In the All Data setting, "B230 Hearing functions" scores best for mBERT (F1=1.000) and "B440 Respiration functions" for MedRoBERTa (F1=0.930).

"None", "D840-859 Work and employment", and "B152 Emotional functions" all score 0.000. The Conversational Data models achieve F1=1.000 on "B530 Weight maintenance functions", "B134 Sleep functions", and "B230 Hearing functions". Also, they score well on "B280 Sensations of pain" (F1=0.941). The LLM prompted models perform best on "B280 Sensations of pain" (zero-shot F1=0.956, few-shot F1=0.974) and "B1300 Energy level" (zero-shot F1=0.943, few-shot F1=0.963).

Figure 5.3 shows the confusion matrices for the LLM-prompted models under strict filtering and lenient evaluation. In the zero-shot setting (Figure 5.3a), "B1300 Energy level" (TP=25), "D450 Walking" (TP=59), and "B280 Sensations of pain" (TP=54) perform well. "B140 Attention functions" (TP=6), "B455 Exercise tolerance functions" (TP=3), and "D840-859 Work and employment" (TP=0) perform poorly. "None" is frequently confused with other categories. In the few-shot setting (Figure 5.3b), "B152 Emotional functions" (TP=50) and "B230 Hearing functions" (TP=19) also perform well. Here, categories are mistaken for "None" but not for one another.

5.2.3 Evaluation setting 3 (lenient filtering, strict evaluation)

The third evaluation setting combines lenient filtering with strict evaluation. Predictions are evaluated on a turn-by-turn basis, but within segments that have been extended with 10 turns before the first category annotation and 10 turns after the corresponding level annotation to provide additional context. This evaluation setting is expected to produce the weakest results of the four settings, as the larger segments

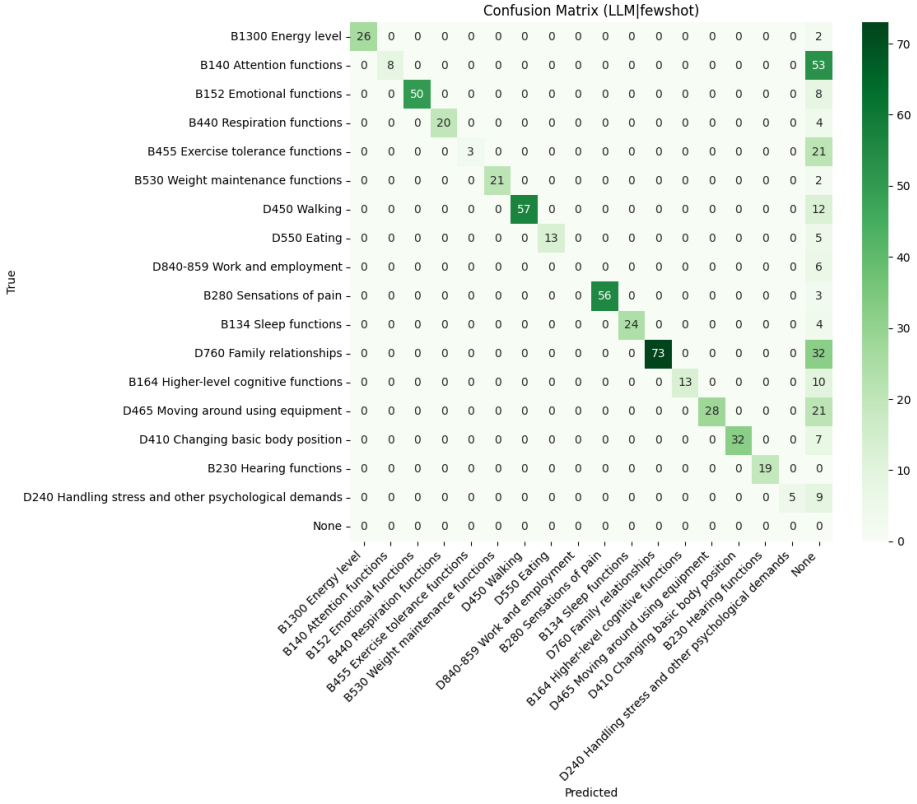
	Baseline						**All data**						**Support**
	mBERT			MedRoBERTa.nl			mBERT			MedRoBERTa.nl			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
B1300 Energy level	**1.000**	0.667	0.800	**1.000**	0.667	0.800	**1.000**	0.704	0.826	**1.000**	0.667	0.800	28
B140 Attention functions	**1.000**	0.117	0.209	**1.000**	0.117	0.209	**1.000**	0.133	0.235	**1.000**	0.117	0.209	61
B152 Emotional functions	**1.000**	0.607	0.756	**1.000**	0.696	0.821	**1.000**	0.839	0.913	**1.000**	0.804	0.891	58
B440 Respiration functions	**1.000**	0.696	0.821	**1.000**	0.696	0.821	**1.000**	**0.870**	**0.930**	**1.000**	0.870	0.930	24
D455 Exercise tolerance functions	**1.000**	0.038	0.074	**1.000**	0.077	0.143	**1.000**	0.077	0.143	**1.000**	0.077	0.143	24
B530 Weight maintenance functions	**1.000**	0.667	0.800	**1.000**	0.750	0.857	**1.000**	0.833	0.909	**1.000**	0.792	0.884	23
D450 Walking	**1.000**	0.771	0.871	**1.000**	0.743	0.852	**1.000**	0.843	0.915	**1.000**	0.814	0.898	69
B550 Eating	**1.000**	0.565	0.722	**1.000**	0.652	0.789	**1.000**	0.696	0.821	**1.000**	0.652	0.789	18
D840-859 Work and employment	**1.000**	0.500	0.667	**1.000**	0.500	0.667	0.000	0.000	0.000	0.000	0.000	0.000	6
B280 Sensations of pain	**1.000**	0.661	0.796	**1.000**	0.644	0.784	**1.000**	0.814	0.897	**1.000**	0.780	0.876	59
B134 Sleep functions	**1.000**	**0.828**	**0.906**	**1.000**	0.828	0.906	**1.000**	0.793	0.885	**1.000**	0.793	0.885	28
D760 Family relationships	**1.000**	0.717	0.835	**1.000**	0.768	0.869	**1.000**	0.646	0.785	**1.000**	0.616	0.762	105
B164 Higher-level cognitive functions	**1.000**	0.522	0.686	**1.000**	0.435	0.606	**1.000**	0.609	0.757	**1.000**	0.565	0.722	23
D465 Moving around using equipment	**1.000**	0.532	0.694	**1.000**	0.553	0.712	**1.000**	0.574	0.730	**1.000**	0.553	0.712	49
D410 Changing basic body position	**1.000**	0.500	0.667	**1.000**	0.425	0.596	**1.000**	0.550	0.710	**1.000**	0.575	0.730	39
B230 Hearing functions	**1.000**	0.800	0.889	**1.000**	**0.900**	**0.947**	**1.000**	**1.000**	**1.000**	**1.000**	0.800	0.889	19
D240 Handling stress and other psychological demands	**1.000**	0.400	0.571	**1.000**	0.333	0.500	**1.000**	0.333	0.500	**1.000**	0.333	0.500	14
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
micro avg	1.000	0.580	0.734	1.000	0.594	0.745	1.000	0.640	0.780	1.000	0.612	0.759	647
macro avg	0.944	0.533	0.653	0.944	0.544	0.660	0.889	0.573	0.664	0.889	0.545	0.646	647
weighted avg	1.000	0.580	0.704	1.000	0.594	0.713	0.991	0.640	0.744	0.991	0.612	0.725	647

	Conversational data						**LLM prompt generation**						
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
B1300 Energy level	**1.000**	0.667	0.800	**1.000**	0.667	0.800	**1.000**	0.893	0.943	**1.000**	0.929	0.963	28
B140 Attention functions	**1.000**	0.062	0.118	**1.000**	0.062	0.118	**1.000**	0.098	0.179	**1.000**	0.131	0.232	61
B152 Emotional functions	0.000	0.000	0.000	0.000	0.000	0.000	**1.000**	0.759	0.863	**1.000**	0.862	0.926	58
B440 Respiration functions	**1.000**	0.667	0.800	**1.000**	0.667	0.800	**1.000**	0.792	0.884	**1.000**	0.833	0.909	24
D455 Exercise tolerance functions	0.000	0.000	0.000	**1.000**	0.167	0.286	**1.000**	0.125	0.222	**1.000**	0.125	0.222	24
B530 Weight maintenance functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.696	0.821	**1.000**	0.913	0.955	23
D450 Walking	**1.000**	0.833	0.909	**1.000**	0.500	0.667	**1.000**	0.855	0.922	**1.000**	0.826	0.905	69
B550 Eating	**1.000**	0.571	0.727	**1.000**	0.571	0.727	**1.000**	0.889	0.941	**1.000**	0.722	0.839	18
D840-859 Work and employment	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	6
B280 Sensations of pain	**1.000**	0.889	0.941	**1.000**	0.889	0.941	**1.000**	**0.915**	**0.956**	**1.000**	0.949	0.974	59
B134 Sleep functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.786	0.880	**1.000**	0.857	0.923	28
D760 Family relationships	**1.000**	0.739	0.850	**1.000**	0.696	0.821	**1.000**	0.648	0.786	**1.000**	0.695	0.820	105
B164 Higher-level cognitive functions	**1.000**	0.500	0.667	**1.000**	0.500	0.667	**1.000**	0.391	0.562	**1.000**	0.565	0.722	23
D465 Moving around using equipment	**1.000**	0.750	0.857	**1.000**	0.750	0.857	**1.000**	0.490	0.658	**1.000**	0.571	0.727	49
D410 Changing basic body position	**1.000**	0.800	0.889	**1.000**	0.800	0.889	**1.000**	0.718	0.836	**1.000**	0.821	0.901	39
B230 Hearing functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.684	0.812	**1.000**	**1.000**	**1.000**	19
D240 Handling stress and other psychological demands	0.000	0.000	0.000	0.000	0.000	0.000	**1.000**	0.357	0.526	**1.000**	0.357	0.526	14
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
micro avg	1.000	0.624	0.768	1.000	0.607	0.755	1.000	0.635	0.777	1.000	0.692	0.818	647
macro avg	0.722	0.527	0.587	0.778	0.515	0.587	0.889	0.561	0.655	0.889	0.620	0.697	647
weighted avg	0.940	0.624	0.709	0.991	0.607	0.705	0.991	0.635	0.738	0.991	0.692	0.780	647

Table 5.3: Classification report of the category predictions with strict filtering and lenient evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing category per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.



(a) Zero-shot



(b) Few-shot

Figure 5.3: Confusion matrices for the categories prediction task of Prompt-Based Generation in strict filtering and lenient evaluation.

increase the likelihood of incorrect predictions under strict evaluation. The results for the third evaluation setting are presented in Table 5.4 and confirm this expectation.

Macro average F1 ranges from 0.214 to 0.246, with macro precision around 0.200 and macro recall above 0.350, which is noticeably lower than under strict filtering due to an increase of "None" turns in the gold data. For the encoder models, the balance between precision and recall switches compared to the previous two settings, while the LLM-prompted models follow the same pattern as before.

The All Data setting performs best (mBERT macro F1=0.244, MedRoBERTa macro F1=0.246). The Conversational Data model also outperforms the baseline (Conversations Only mBERT macro F1=0.243, MedRoBERTa macro F1=0.239; Baseline mBERT macro F1=0.230, MedRoBERTa macro F1=0.216). The LLM prompted models perform worst overall (zero-shot macro F1=0.216, few-shot macro F1=0.214). Notably, they achieve lower precision than the other experimental settings but higher recall. mBERT outperforms MedRoBERTa on the baseline and Conversational Data settings, while MedRoBERTa is slightly better on All Data. For the LLM settings, zero-shot outperforms few-shot.

Across all settings, "B134 Sleep functions" is consistently the best performing category (baseline: mBERT F1=0.409, MedRoBERTa F1=0.393; All Data: mBERT F1=0.457, MedRoBERTa F1=0.471; Conversational Data: mBERT F1=0.467, MedRoBERTa F1=0.475; LLM: zero-shot F1=0.407, few-shot F1=0.432).

"None" and "D840-859 Work and employment" score 0.000 across almost all settings. Consistently poor performers include "D455 Exercise tolerance functions" and "B140 Attention functions", both scoring below 0.100 across all models and settings.

Figure 5.4 shows the confusion matrices for the All Data fine-tuned model under lenient filtering and strict evaluation. The results are similar across both models. "B440 Respiration functions" (mBERT TP=49; MedRoBERTa TP=49) and "D450 Walking" (mBERT TP=137; MedRoBERTa TP=116) perform well. "B152 Emotional functions" and "D760 Family relationships" have a high number of true positives (mBERT: TP=128 and TP=179 respectively; MedRoBERTa: TP=178 and TP=174 respectively), but also a high number of false negatives where "None" is predicted instead (mBERT: FN=166 and FN=180; MedRoBERTa: FN=170 and FN=181).

Values "B140 Attention functions" is frequently confused with "B152 Emotional functions", "B455 Exercise tolerance functions" with "D450 Walking", and "D240 Handling stress and other psychological demands" with "B152 Emotional functions".

5.2.4 Evaluation setting 4 (lenient filtering, lenient evaluation)

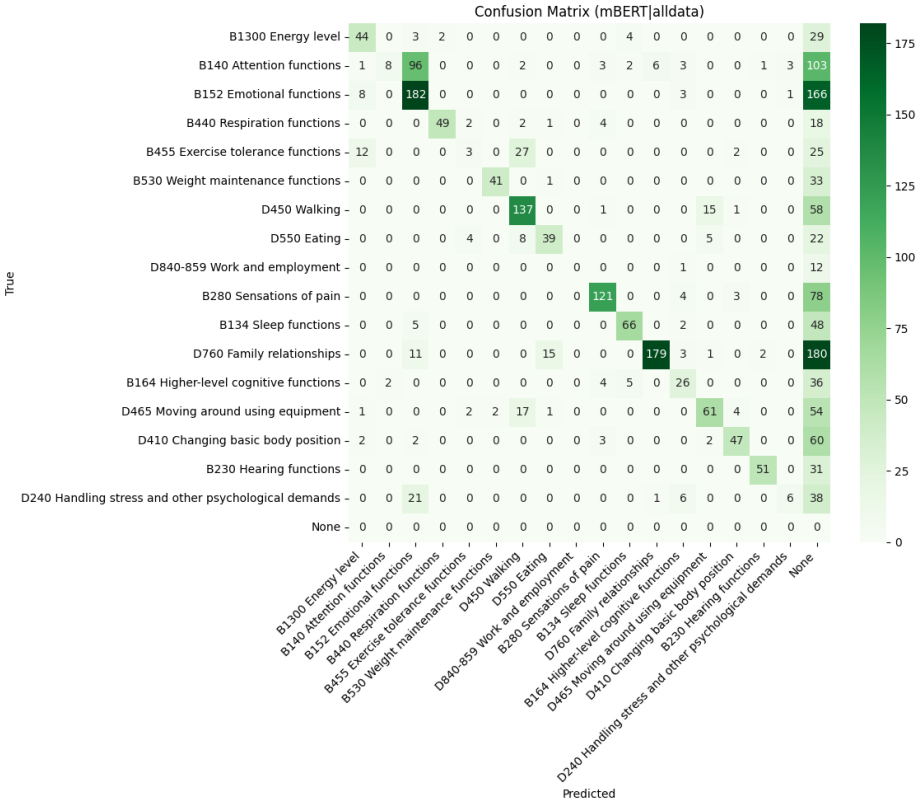
The fourth evaluation setting combines lenient filtering with lenient evaluation. The extended segments are used, but instead of evaluating predictions on a turn-by-turn basis, a prediction is considered correct if the predicted category appears anywhere within the corresponding segment. This evaluation setting is expected to produce the best results of the four settings, as lenient evaluation reduces the penalty for turn-level mismatches while the extended segments provide additional context. The results for the fourth evaluation setting are presented in Table 5.5 and confirm this expectation.

Macro F1 ranges from 0.663 to 0.809. Across all models, macro precision (average = 0.868) remains higher than macro recall (average = 0.694), consistent with the first two evaluation settings. Macro precision is similar to that observed under strict filtering

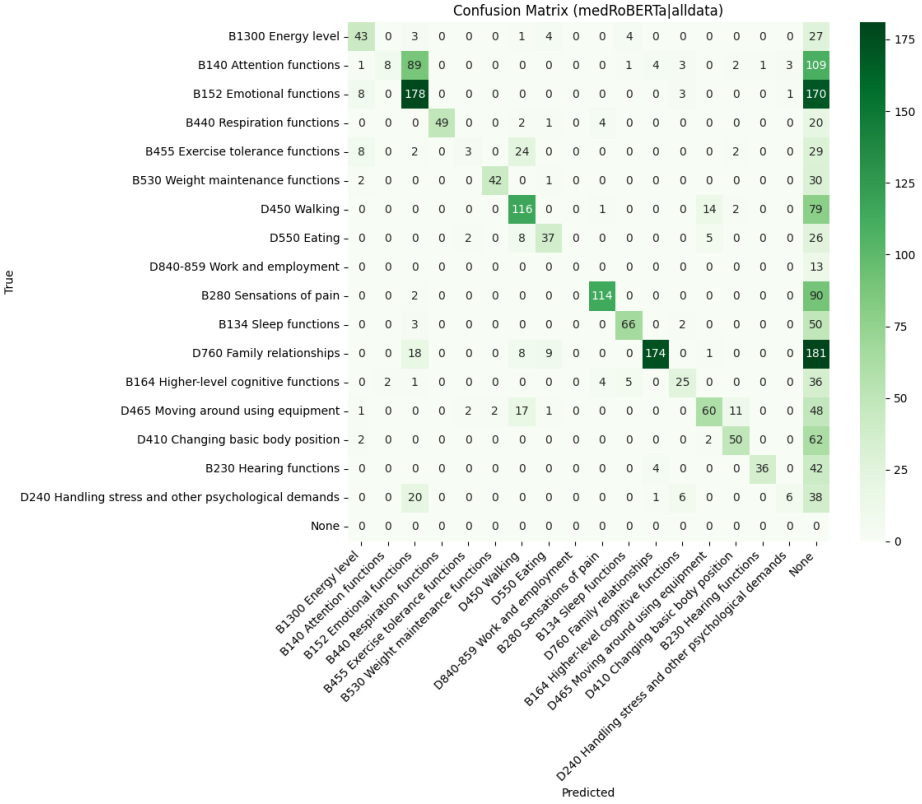
	Baseline						**All data**						**Support**
	mBERT			MedRoBERTa.nl			mBERT			MedRoBERTa.nl			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
B1300 Energy level	0.247	0.512	0.333	0.216	0.512	0.304	0.205	0.537	0.296	0.219	0.524	0.309	82
B140 Attention functions	0.348	0.037	0.066	0.286	0.037	0.065	0.216	0.037	0.062	0.308	0.037	0.065	228
B152 Emotional functions	0.227	0.383	0.285	0.226	0.408	0.291	0.236	0.506	0.322	0.248	0.494	0.330	360
B440 Respiration functions	0.248	0.421	0.312	0.128	0.421	0.196	0.225	**0.645**	0.333	0.227	**0.645**	0.336	76
D455 Exercise tolerance functions	0.038	0.029	0.033	0.086	0.044	0.058	0.083	0.044	0.058	0.120	0.044	0.065	69
B530 Weight maintenance functions	**0.375**	0.360	0.367	**0.307**	0.413	0.352	0.209	0.547	0.303	0.247	0.560	0.343	75
D450 Walking	0.257	0.505	0.341	0.244	0.462	0.319	0.207	0.646	0.314	0.184	0.547	0.276	212
B550 Eating	0.230	0.467	0.308	0.157	0.493	0.238	0.205	0.520	0.294	0.192	0.493	0.276	78
D840-859 Work and employment	0.068	0.308	0.111	0.068	0.385	0.115	0.000	0.000	0.000	0.000	0.000	0.000	13
B280 Sensations of pain	0.231	0.466	0.309	0.229	0.451	0.303	0.247	0.587	0.348	0.265	0.553	0.358	206
B134 Sleep functions	0.324	**0.554**	**0.409**	0.305	**0.554**	**0.393**	**0.393**	0.545	**0.457**	**0.415**	0.545	**0.471**	121
D760 Family relationships	0.316	0.471	0.378	0.304	0.512	0.381	0.312	0.458	0.371	0.339	0.445	0.385	391
B164 Higher-level cognitive functions	0.107	0.260	0.152	0.073	0.219	0.110	0.089	0.356	0.142	0.105	0.342	0.160	73
D465 Moving around using equipment	0.196	0.401	0.263	0.197	0.423	0.268	0.199	0.430	0.272	0.199	0.423	0.271	142
D410 Changing basic body position	0.148	0.371	0.211	0.164	0.328	0.218	0.188	0.405	0.257	0.197	0.431	0.270	116
B230 Hearing functions	0.125	0.463	0.196	0.127	0.573	0.208	0.331	0.622	0.432	0.340	0.439	0.383	82
D240 Handling stress and other psychological demands	0.046	0.155	0.071	0.064	0.085	0.073	0.231	0.085	0.124	0.316	0.085	0.133	72
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
micro avg	0.217	0.382	0.277	0.208	0.390	0.271	0.231	0.445	0.304	0.240	0.423	0.306	2396
macro avg	0.196	0.342	0.230	0.177	0.351	0.216	0.199	0.387	0.244	0.218	0.367	0.246	2396
weighted avg	0.242	0.382	0.273	0.224	0.390	0.264	0.238	0.445	0.290	0.260	0.423	0.293	2396

	Conversational data						**LLM prompt generation**					
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
B1300 Energy level	0.203	0.537	0.294	0.222	0.512	0.310	0.209	**0.829**	0.333	0.171	0.902	0.287
B140 Attention functions	0.267	0.037	0.064	0.267	0.037	0.064	0.145	0.050	0.075	0.133	0.078	0.098
B152 Emotional functions	0.223	0.494	0.307	0.237	0.528	0.327	0.200	0.581	0.298	0.163	0.625	0.259
B440 Respiration functions	0.223	**0.645**	0.331	0.216	**0.645**	0.323	0.199	0.658	0.306	0.189	0.750	0.302
D455 Exercise tolerance functions	0.051	0.029	0.037	0.073	0.044	0.055	0.055	0.074	0.063	0.038	0.074	0.051
B530 Weight maintenance functions	0.244	0.560	0.340	0.237	0.560	0.333	0.174	0.493	0.257	0.139	0.773	0.236
D450 Walking	0.186	0.594	0.284	0.179	0.533	0.268	0.175	0.627	0.274	0.170	0.632	0.267
B550 Eating	0.214	0.520	0.304	0.185	0.493	0.269	0.173	0.640	0.272	0.143	0.560	0.228
D840-859 Work and employment	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
B280 Sensations of pain	0.249	0.587	0.350	0.253	0.568	0.350	0.253	0.782	0.382	0.222	0.825	0.349
B134 Sleep functions	**0.404**	0.554	**0.467**	**0.420**	0.545	**0.475**	**0.311**	0.587	**0.407**	**0.308**	0.727	0.432
D760 Family relationships	0.297	0.496	0.371	0.296	0.435	0.352	0.260	0.573	0.358	0.229	0.578	0.328
B164 Higher-level cognitive functions	0.081	0.356	0.132	0.083	0.356	0.134	0.039	0.219	0.066	0.044	0.384	0.079
D465 Moving around using equipment	0.199	0.408	0.267	0.187	0.394	0.253	0.143	0.366	0.206	0.162	0.472	0.241
D410 Changing basic body position	0.167	0.431	0.241	0.167	0.405	0.237	0.138	0.603	0.224	0.134	0.733	0.227
B230 Hearing functions	0.388	0.573	0.463	0.354	0.561	0.434	0.202	0.500	0.288	0.254	**0.915**	**0.398**
D240 Handling stress and other psychological demands	0.222	0.085	0.122	0.231	0.085	0.124	0.065	0.099	0.078	0.058	0.099	0.073
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
micro avg	0.225	0.444	0.299	0.224	0.427	0.294	0.189	0.505	0.275	0.169	0.570	0.261
macro avg	0.201	0.384	0.243	0.200	0.372	0.239	0.152	0.427	0.216	0.142	0.507	0.214
weighted avg	0.238	0.444	0.286	0.239	0.427	0.282	0.191	0.505	0.266	0.174	0.570	0.257

Table 5.4: Clarification report of the category predictions with lenient filtering and strict evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing category per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.



(a) mBERT



(b) MedRoBERTa

Figure 5.4: Confusion matrices for the categories prediction task of Finetuning (All Data) in lenient filtering and strict evaluation.

with lenient evaluation, whereas macro recall is approximately 0.2 higher for most models.

In this setting, the baseline performs better than the fine-tuned models (Baseline mBERT macro F1=0.809, MedRoBERTa macro F1=0.784; All Data mBERT macro F1=0.754, MedRoBERTa macro F1=0.749; Conversations Only mBERT macro F1=0.663, MedRoBERTa macro F1=0.674). The LLM prompted models perform similarly to the All Data fine-tuned models, while outperforming the Conversational Data fine-tuned models (zero-shot macro F1=0.738, few-shot macro F1=0.759). mBERT outperforms MedRoBERTa on the Baseline and the All Data setting, while MedRoBERTa slightly outperforms mBERT on the Conversations Only setting. Few-shot outperforms zero-shot for the LLM settings.

Across most models, "None" and "D840-859 Work and employment" consistently score 0.000. "D240 Handling stress and other psychological demands" also scores 0.000 in the Conversational Data setting.

In the baseline, "D450 Walking", "B230 Hearing functions", "D760 Family relationships", and "B440 Respiration functions" all score above 0.970 for both models, while "B140 Attention functions" and "D455 Exercise tolerance functions" stay below 0.270. The All Data setting shows similarly strong scores for "D450 Walking", "B530 Weight maintenance functions", and "B230 Hearing functions", and the same weak categories.

In the Conversational Data setting, many categories reach F1=1.000, including "B1300 Energy level", "B530 Weight maintenance functions", "B280 Sensations of pain", "B134 Sleep functions", and "B230 Hearing functions", among others. The LLM prompted models score well on "B280 Sensations of pain" (zero-shot F1=0.991, few-shot F1=1.000) and "B230 Hearing functions" (few-shot F1=1.000), but poorly on "B140 Attention functions" (zero-shot F1=0.306, few-shot F1=0.373).

Figure 5.5 shows the confusion matrices for the baseline under lenient filtering and lenient evaluation. For mBERT (Figure 5.5a), "B152 Emotional functions" (TP=49), "D450 Walking" (TP=67), and "D760 Family relationships" (TP=95) perform well. "B140 Attention functions" (TP=8), "B455 Exercise tolerance functions" (TP=4), and "D840-859 Work and employment" (TP=4) perform poorly. All categories are at some point confused with "None", but not with one another. The same patterns hold for MedRoBERTa.

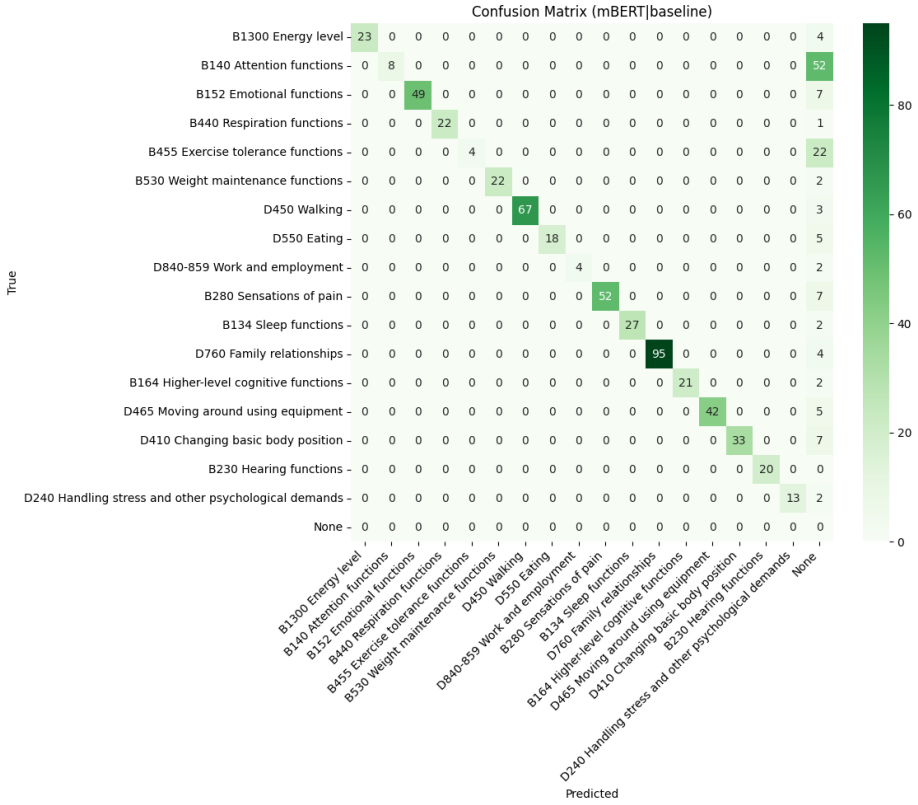
To summarise, mBERT outperformed MedRoBERTa in most evaluation settings, though MedRoBERTa often had an edge in the baseline. The best setting overall was lenient filtering with lenient evaluation, while lenient filtering with strict evaluation performed worst. Under strict filtering, the Prompt-Based Generation performed best. Under lenient filtering, the All Data fine-tuned model did best, with the Baseline also performing strongly under lenient evaluation. Overall, the fine-tuned encoders outperformed the Baseline, and in the lenient filtering settings also outperformed the Prompt-Based Generation.

Furthermore, precision under lenient evaluation was consistently high, indicating that predicted categories were usually correct. Under strict filtering with strict evaluation, precision was generally slightly higher than recall. In contrast, under strict filtering with lenient evaluation, recall was generally higher than precision. Since only one seed was used to train the models due to time constraints, results may vary if

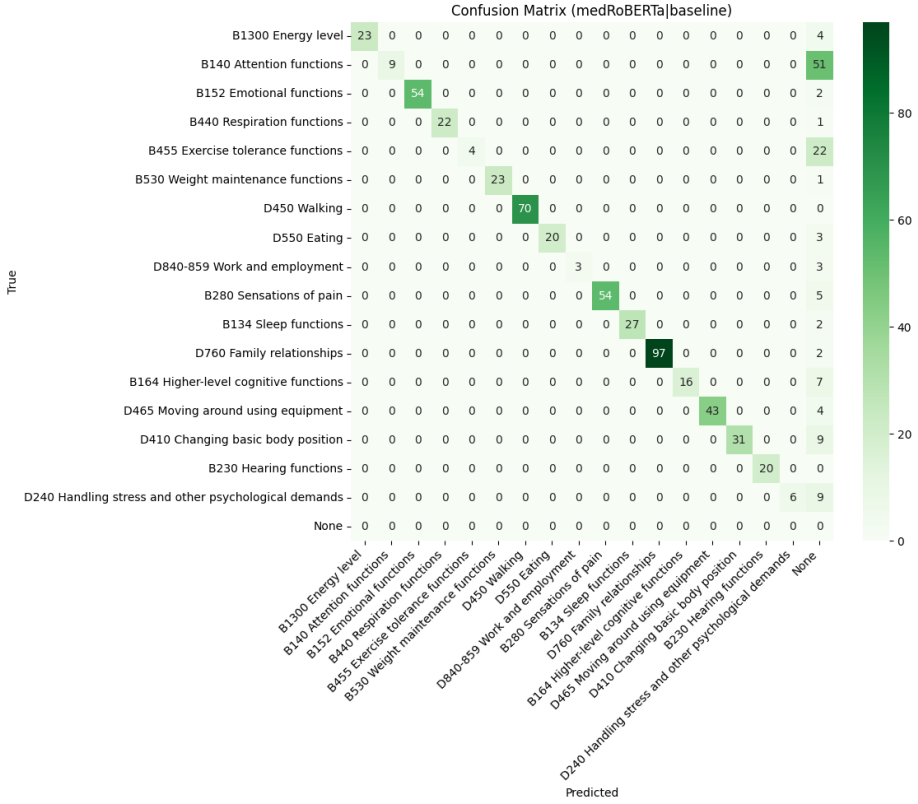
	Baseline						**All data**						**Support**
	mBERT			MedRoBERTa.nl			mBERT			MedRoBERTa.nl			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
B1300 Energy level	**1.000**	0.852	0.920	**1.000**	0.852	0.920	**1.000**	0.926	0.962	**1.000**	0.889	0.941	27
B140 Attention functions	**1.000**	0.133	0.235	**1.000**	0.150	0.261	**1.000**	0.150	0.261	**1.000**	0.133	0.235	60
B152 Emotional functions	**1.000**	0.875	0.933	**1.000**	0.964	0.982	**1.000**	0.946	0.972	**1.000**	0.964	0.982	56
B440 Respiration functions	**1.000**	0.957	0.978	**1.000**	0.957	0.978	**1.000**	0.957	0.978	**1.000**	0.957	0.978	23
D455 Exercise tolerance functions	**1.000**	0.154	0.267	**1.000**	0.154	0.267	**1.000**	0.154	0.267	**1.000**	0.154	0.267	26
B530 Weight maintenance functions	**1.000**	0.917	0.957	**1.000**	0.958	0.979	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	24
D450 Walking	**1.000**	0.957	0.978	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.971	0.986	70
B550 Eating	**1.000**	0.783	0.878	**1.000**	0.870	0.930	**1.000**	0.870	0.930	**1.000**	0.870	0.930	23
D840-859 Work and employment	**1.000**	0.667	0.800	**1.000**	0.500	0.667	0.000	0.000	0.000	0.000	0.000	0.000	6
B280 Sensations of pain	**1.000**	0.881	0.937	**1.000**	0.915	0.956	**1.000**	0.983	0.991	**1.000**	0.966	0.983	59
B134 Sleep functions	**1.000**	0.931	0.964	**1.000**	0.931	0.964	**1.000**	0.897	0.945	**1.000**	0.897	0.945	29
D760 Family relationships	**1.000**	0.960	0.979	**1.000**	0.980	0.990	**1.000**	0.960	0.979	**1.000**	0.929	0.963	99
B164 Higher-level cognitive functions	**1.000**	0.913	0.955	**1.000**	0.696	0.821	**1.000**	0.826	0.905	**1.000**	0.783	0.878	23
D465 Moving around using equipment	**1.000**	0.894	0.944	**1.000**	0.915	0.956	**1.000**	0.915	0.956	**1.000**	0.894	0.944	47
D410 Changing basic body position	**1.000**	0.825	0.904	**1.000**	0.775	0.873	**1.000**	0.850	0.919	**1.000**	0.900	0.947	40
B230 Hearing functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	20
D240 Handling stress and other psychological demands	**1.000**	0.867	0.929	**1.000**	0.400	0.571	**1.000**	0.333	0.500	**1.000**	0.333	0.500	15
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
micro avg	1.000	0.804	0.891	1.000	0.807	0.893	1.000	0.815	0.898	1.000	0.804	0.891	647
macro avg	0.944	0.754	0.809	0.944	0.723	0.784	0.889	0.709	0.754	0.889	0.702	0.749	647
weighted avg	1.000	0.804	0.856	1.000	0.807	0.855	0.991	0.815	0.856	0.991	0.804	0.849	647

	Conversational data						**LLM prompt generation**					
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
B1300 Energy level	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.929	0.963	**1.000**	0.929	0.963
B140 Attention functions	**1.000**	0.125	0.222	**1.000**	0.125	0.222	**1.000**	0.180	0.306	**1.000**	0.230	0.373
B152 Emotional functions	0.000	0.000	0.000	0.000	0.000	0.000	**1.000**	0.862	0.926	**1.000**	0.983	0.991
B440 Respiration functions	**1.000**	0.833	0.909	**1.000**	0.833	0.909	**1.000**	0.958	0.979	**1.000**	0.958	0.979
D455 Exercise tolerance functions	0.000	0.000	0.000	**1.000**	0.167	0.286	**1.000**	0.250	0.400	**1.000**	0.292	0.452
B530 Weight maintenance functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.913	0.955	**1.000**	**1.000**	**1.000**
D450 Walking	**1.000**	**1.000**	**1.000**	**1.000**	0.833	0.909	**1.000**	0.957	0.978	**1.000**	0.913	0.955
B550 Eating	**1.000**	0.714	0.833	**1.000**	0.714	0.833	**1.000**	0.889	0.941	**1.000**	0.778	0.875
D840-859 Work and employment	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
B280 Sensations of pain	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**0.983**	**0.991**	**1.000**	**1.000**	**1.000**
B134 Sleep functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.893	0.943	**1.000**	0.893	0.943
D760 Family relationships	**1.000**	0.957	0.978	**1.000**	0.957	0.978	**1.000**	0.724	0.840	**1.000**	0.867	0.929
B164 Higher-level cognitive functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.565	0.722	**1.000**	0.739	0.850
D465 Moving around using equipment	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.714	0.833	**1.000**	0.776	0.874
D410 Changing basic body position	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.923	0.960	**1.000**	0.923	0.960
B230 Hearing functions	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.895	0.944	**1.000**	**1.000**	**1.000**
D240 Handling stress and other psychological demands	0.000	0.000	0.000	0.000	0.000	0.000	**1.000**	0.429	0.600	**1.000**	0.357	0.526
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
micro avg	1.000	0.786	0.880	1.000	0.786	0.880	1.000	0.750	0.857	1.000	0.799	0.888
macro avg	0.722	0.646	0.663	0.778	0.646	0.674	0.889	0.670	0.738	0.889	0.702	0.759
weighted avg	0.940	0.786	0.815	0.991	0.786	0.825	0.991	0.750	0.824	0.991	0.799	0.858

Table 5.5: Clarification report of the category predictions with lenient filtering and lenient evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing category per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.



(a) mBERT



(b) MedRoBERTa

Figure 5.5: Confusion matrices for the categories prediction task of Baseline in lenient filtering and lenient evaluation.

generalised across multiple seeds.

The best performing setting (evaluation setting 4) exceeds the macro F1 of Chen and Vossen (2026), which plateaued at 0.67-0.68, compared to 0.67-0.81 in this work. At the category level, "D760 Family relationships" performs generally well here but scored low in Chen and Vossen (2026), while the Prompt-Based Generation results are more similar to theirs overall. Compared to Kim et al. (2022), results are similar depending on the model and setting. Their note-level results outperform sentence-level results, which aligns with the benefit of using lenient evaluation seen here. "D450 Walking" performs well in both works, while "B455 Exercise tolerance functions" and "D840-859 Work and employment" score poorly in both.

5.3 ICF levels

ICF level prediction is evaluated under the same four conditions as before. Results are reported for two input variants, namely *catpred*, where predicted ICF categories are used as context, and *catgold*, where gold categories are used. The metrics are MAE, MSE, and RMSE, where lower scores indicate better performance.

5.3.1 Evaluation setting 1 (strict filtering, strict evaluation)

Table 5.6 presents the results for the first evaluation setting, which combines strict filtering with strict evaluation. Similar to the category prediction task, turn-level evaluation is expected to be challenging for conversational data.

The baseline performs best overall. Using predicted categories as context, mBERT (MAE=1.110, MSE=1.543, RMSE=1.242) slightly outperforms MedRoBERTa (MAE=1.146, MSE=1.626, RMSE=1.275). Using gold categories, the pattern reverses: MedRoBERTa (MAE=1.154, MSE=1.628, RMSE=1.276) performs slightly better than mBERT (MAE=1.163, MSE=1.666, RMSE=1.291).

The Conversational Data model performs worse than the baseline for both context types. Using predicted categories, mBERT scores MAE=1.913 and MedRoBERTa MAE=1.505. Using gold categories, mBERT scores MAE=1.960 and MedRoBERTa MAE=1.726. This is also the only fine-tuned setting where MedRoBERTa consistently outperforms mBERT.

The LLM-prompted models perform worst overall. Few-shot prompting (MAE=2.031, MSE=6.463, RMSE=2.542) outperforms zero-shot (MAE=2.266, MSE=7.431, RMSE=2.726).

5.3.2 Evaluation setting 2 (strict filtering, lenient evaluation)

The second evaluation setting combines strict filtering with lenient evaluation. The results are presented in Table 5.7. This evaluation setting is expected to perform better than the previous setting, as predictions are no longer required to match the exact turn of the gold annotation. The results generally confirm this expectation, although the improvements are small in some cases. For example, MedRoBERTa fine-tuned on All Data using gold categories achieves an MAE of 1.910 under strict evaluation and 1.907 under lenient evaluation.

The Baseline again performs best, with both models scoring around 1.100 on MAE and 1.250 on RMSE. The All Data model scores higher, with mBERT around 1.600 (MAE) and 1.900 (RMSE), and MedRoBERTa around 1.900 (MAE) and 2.200 (RMSE).

	Baseline				**All data**			
	mBERT		MedRoBERTa		mBERT		MedRoBERTa	
	catpred	catgold	catpred	catgold	catpred	catgold	catpred	catgold
MAE	**1.110**	1.163	1.146	1.154	1.514	1.645	1.828	1.910
MSE	**1.543**	1.666	1.626	1.628	3.343	3.820	4.643	4.947
RMSE	**1.242**	1.291	1.275	1.276	1.828	1.955	2.155	2.224

	Conversational data				**LLM prompt generation**	
	mBERT		MedRoBERTa		Zero-shot	Few-shot
	catpred	catgold	catpred	catgold		
MAE	1.913	1.960	1.505	1.726	2.266	2.031
MSE	4.716	4.921	3.279	4.037	7.431	6.463
RMSE	2.172	2.218	1.811	2.009	2.726	2.542

Table 5.6: Results of level predictions for strict filtering and strict evaluation. catpred = predicted categories as context, catgold = gold categories as context, MAE = Mean Absolute Error, MSE = Mean Squared Error, RMSE = Root Mean Squared Error. The best performing scores are shown in **bold**. The best performing score per experimental setting and model is underlined.

The Conversational Data model performs slightly worse than All Data. mBERT scores around 1.875 (MAE) and 2.125 (RMSE), while MedRoBERTa performs slightly better at around 1.600 (MAE) and 1.850 (RMSE). The LLM-prompted models perform worst, with MAE=2.082 and RMSE=2.451 for both zero-shot and few-shot.

In the Baseline, mBERT outperforms MedRoBERTa using catpred, while MedRoBERTa is better using catgold. In the All Data setting, mBERT performs better in both cases. In the Conversational Data setting, MedRoBERTa performs better. Across all models and settings, catpred consistently outperforms catgold.

5.3.3 Evaluation setting 3 (lenient filtering, strict evaluation)

The third evaluation setting combines lenient filtering with strict evaluation. The results are presented in Table 5.8. As expected, most models perform worse than in the previous evaluation settings, as the larger segments increase the likelihood of turn-level mismatches. One exception is the Conversation Only setting, which performs slightly better than under strict filtering with strict evaluation.

The baseline again performs best overall. Using predicted categories, mBERT (MAE=1.147) outperforms MedRoBERTa (MAE=1.174). Using gold categories, the pattern reverses: MedRoBERTa (MAE=1.162) performs slightly better than mBERT (MAE=1.171). The LLM-prompted models perform worst, with zero-shot (MAE=2.324) behind few-shot (MAE=2.072).

In the All Data setting, mBERT outperforms MedRoBERTa for both context types (mBERT catpred MAE=1.797, catgold MAE=1.965; MedRoBERTa catpred MAE=2.045, catgold MAE=2.096). In the Conversational Data setting, MedRoBERTa performs better (mBERT catpred MAE=1.913, catgold MAE=1.944; MedRoBERTa catpred MAE=1.473, catgold MAE=1.486).

	Baseline				**All data**			
	mBERT		MedRoBERTa		mBERT		MedRoBERTa	
	catpred	catgold	catpred	catgold	catpred	catgold	catpred	catgold
MAE	**1.061**	1.151	1.091	1.142	1.579	1.629	1.874	1.907
MSE	**1.450**	1.644	1.530	1.607	3.522	3.677	4.740	4.878
RMSE	**1.204**	1.282	1.237	1.268	1.877	1.918	2.177	2.209

	Conversational data				**LLM prompt generation**	
	mBERT		MedRoBERTa		Zero-shot	Few-shot
	catpred	catgold	catpred	catgold		
MAE	1.822	1.939	1.462	1.712	2.082	2.082
MSE	4.289	4.836	3.059	3.991	6.007	6.007
RMSE	2.071	2.199	1.749	1.988	2.451	2.451

Table 5.7: Results of level predictions for strict filtering and lenient evaluation. catpred = predicted categories as context, catgold = gold categories as context, MAE = Mean Absolute Error, MSE = Mean Squared Error, RMSE = Root Mean Squared Error. The best performing scores are shown in **bold**. The best performing score per experimental setting and model is underlined.

	Baseline				**All data**			
	mBERT		MedRoBERTa		mBERT		MedRoBERTa	
	catpred	catgold	catpred	catgold	catpred	catgold	catpred	catgold
MAE	**1.147**	1.171	1.174	1.162	1.797	1.965	2.045	2.096
MSE	**1.614**	1.672	1.660	1.635	4.523	5.234	5.504	5.741
RMSE	**1.270**	1.293	1.288	1.278	2.127	2.288	2.346	2.396

	Conversational data				**LLM prompt generation**	
	mBERT		MedRoBERTa		Zero-shot	Few-shot
	catpred	catgold	catpred	catgold		
MAE	1.913	1.944	1.473	1.486	2.324	2.072
MSE	4.776	4.902	3.111	3.246	7.710	6.569
RMSE	2.185	2.214	1.764	1.802	2.777	2.563

Table 5.8: Results of level predictions for lenient filtering and strict evaluation. catpred = predicted categories as context, catgold = gold categories as context, MAE = Mean Absolute Error, MSE = Mean Squared Error, RMSE = Root Mean Squared Error. The best performing scores are shown in **bold**. The best performing score per experimental setting and model is underlined.

5.3.4 Evaluation setting 4 (lenient filtering, lenient evaluation)

The fourth evaluation setting combines lenient filtering with lenient evaluation. The results are presented in Table 5.9. This evaluation setting was expected to produce the best performance, but the results do not support this expectation. For most experimental settings, strict filtering with lenient evaluation outperforms lenient filtering with lenient evaluation. The only exception is MedRoBERTa fine-tuned on Conversation Only, which performs better in this evaluation setting than in the second evaluation setting.

The baseline again performs best. mBERT scores MAE=1.095 (catpred) and 1.178 (catgold), and MedRoBERTa scores MAE=1.117 (catpred) and 1.169 (catgold). The Conversational Data model outperforms the All Data model for both mBERT (MAE catpred=1.871 vs 1.932) and MedRoBERTa (MAE catpred=1.421 vs 2.055). The LLM-prompted models perform worst, with MAE=2.173 and RMSE=2.558 for both zero-shot and few-shot.

In the Conversational Data setting, MedRoBERTa outperforms mBERT. In the All Data setting, mBERT performs better on MAE, while MedRoBERTa is better on MSE and RMSE using catgold. In the baseline, mBERT performs better using catpred and MedRoBERTa using catgold. Catpred consistently outperforms catgold across all settings.

	Baseline				**All data**			
	mBERT		MedRoBERTa		mBERT		MedRoBERTa	
	catpred	catgold	catpred	catgold	catpred	catgold	catpred	catgold
MAE	**1.095**	1.178	1.117	1.169	1.932	2.192	2.055	2.203
MSE	**1.520**	1.703	1.581	1.666	5.045	6.332	5.446	6.169
RMSE	**1.233**	1.305	1.257	1.291	2.246	2.516	2.334	2.484

	Conversational data				**LLM prompt generation**	
	mBERT		MedRoBERTa		Zero-shot	Few-shot
	catpred	catgold	catpred	catgold		
MAE	1.871	1.945	1.421	1.472	2.173	2.173
MSE	4.434	4.899	2.909	3.081	6.544	6.544
RMSE	2.118	2.213	1.705	1.755	2.558	2.558

Table 5.9: Results of level predictions for lenient filtering and lenient evaluation. catpred = predicted categories as context, catgold = gold categories as context, MAE = Mean Absolute Error, MSE = Mean Squared Error, RMSE = Root Mean Squared Error. The best performing scores are shown in **bold**. The best performing score per experimental setting and model is underlined.

To sum up, strict filtering with lenient evaluation performed best for level prediction. The baseline outperformed all fine-tuned settings, suggesting that conversational training data did not improve level prediction. mBERT and MedRoBERTa performed similarly, with mBERT slightly ahead. Few-shot prompting outperformed zero-shot in the strict evaluation settings, while the two gave identical results under lenient evaluation. Both were outperformed by the encoder models. As previously mentioned, the

use of different seeds while training could have given different outcomes. These results could thus not be generalisable.

Compared to Jakubauskaitė (2025), the baseline of this work is outperformed by their sentence-level and note-level results. Fine-tuning on Conversational Data Only and All Data performed worse still. The same trend can be observed when comparing to Kim et al. (2022).

5.4 Relative time

Relative time prediction is evaluated under the same four conditions as before. No difference was found between using predicted or gold categories, or predicted or gold levels. Classification reports and confusion matrices were therefore identical across these variants, and only one generic version is reported per setting. For the best performing model in each setting, the confusion matrix is shown in the main text. All others are in Appendix E.

5.4.1 Evaluation setting 1 (strict filtering, strict evaluation)

The results for the first evaluation setting are presented in Table 5.10. This setting combines strict filtering with strict evaluation. As in the previous prediction tasks, moderate performance is expected under turn-level evaluation.

Macro average F1 scores are low across all models: mBERT scores 0.236, MedRoBERTa 0.218, zero-shot prompting 0.208, and few-shot prompting 0.211. For all models except mBERT, recall was slightly higher than precision, though the difference was small. The fine-tuned models outperform the LLM-prompted variants, and mBERT slightly outperforms MedRoBERTa. Among the prompted models, few-shot performs better than zero-shot.

"Present" is the best performing label across all models. The fine-tuned models score an F1-score between 0.642 and 0.652, and the prompted models between 0.573 and 0.597. Precision is higher than recall for all models, though the gap is smaller in the prompted models (mBERT: precision=0.873, recall=0.521; zero-shot: precision=0.680, recall=0.523).

Both "Future" and "None" receive F1=0.000. "None" has no support in the test set, but "Future" has a support of 26 instances.

The confusion matrices in Figure 5.6 show that "Present" is predicted most often (mBERT TP=609, MedRoBERTa TP=633), but is frequently confused with "None" (mBERT FN=463, MedRoBERTa FN=438) and "Past" (mBERT FN=84, MedRoBERTa FN=87). "Past" is predicted correctly some of the time but is often confused with "Present". "Future" is never predicted correctly.

5.4.2 Evaluation setting 2 (strict filtering, lenient evaluation)

The results for the second evaluation setting are presented in Table 5.11. This setting combines strict filtering with lenient evaluation. As expected, the results improve compared with the first evaluation setting, as evaluating predictions across an entire segment is better suited to conversational data than turn-level evaluation.

	Fine-tuning (Conversations Only)						**Prompt-based Generation**						**Support**
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
Past	0.225	0.420	0.293	0.157	0.435	0.231	0.148	**0.580**	0.235	0.171	**0.638**	0.269	83
Present	**0.873**	**0.521**	**0.652**	**0.791**	**0.540**	**0.642**	**0.680**	0.532	**0.597**	**0.664**	0.504	**0.573**	1168
Future	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	28
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
Micro avg	0.747	0.503	0.601	0.634	0.521	0.572	0.527	0.523	0.525	0.528	0.500	0.514	1279
Macro avg	0.275	0.235	0.236	0.237	0.244	0.218	0.207	0.278	0.208	0.209	0.285	0.211	1279
Weighted avg	0.815	0.503	0.616	0.735	0.521	0.603	0.632	0.523	0.562	0.620	0.500	0.542	1279

Table 5.10: Results of time predictions for strict filtering and strict evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing label per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.

Macro average F1 scores range from 0.297 to 0.324. Across all models, recall is again slightly higher than precision, with an average recall of 0.357 and an average precision of 0.303. This is higher than in under strict evaluation. The fine-tuned models again outperform the prompted models. mBERT (macro F1=0.324) slightly outperforms MedRoBERTa (macro F1=0.320), and few-shot (macro F1=0.314) outperforms zero-shot (macro F1=0.297).

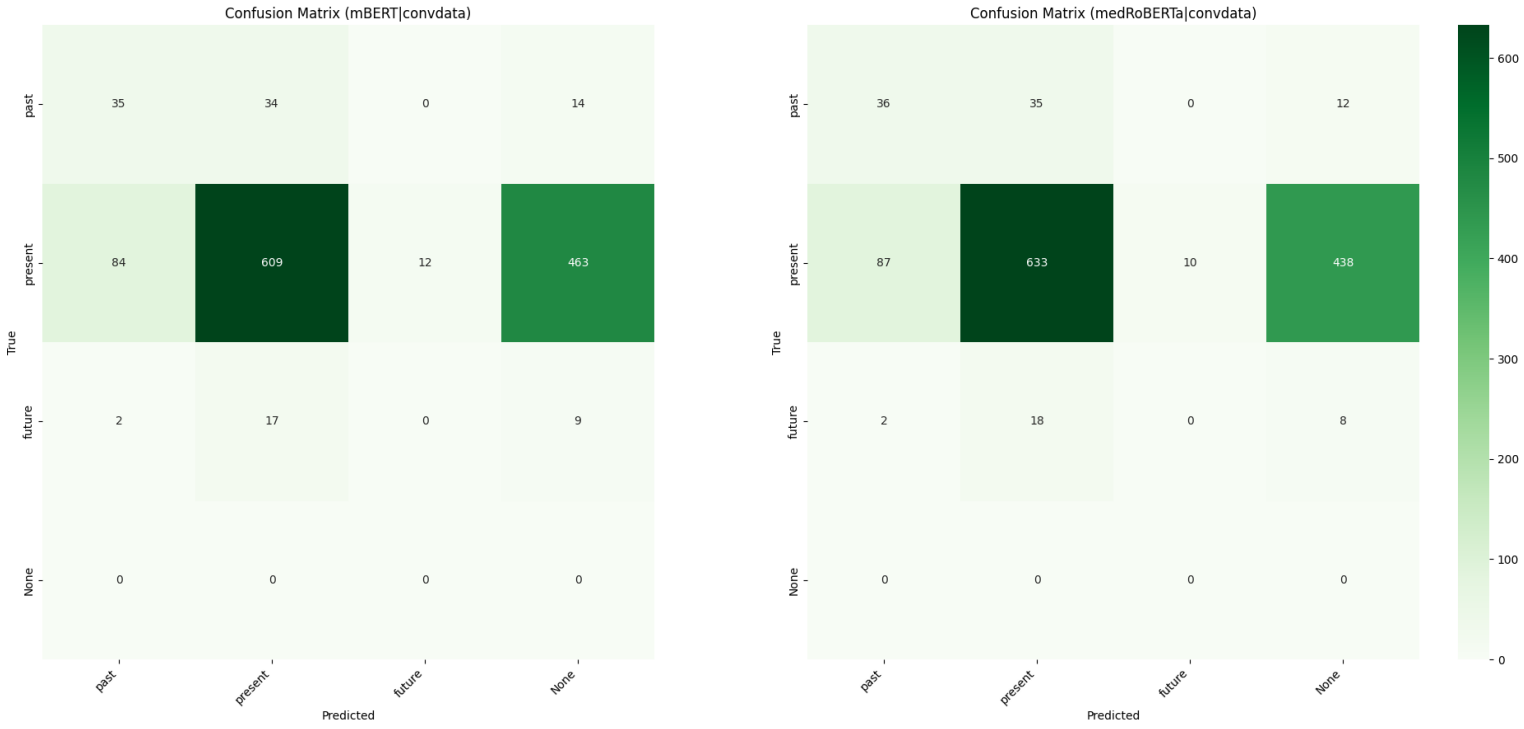
"Present" again performs best (mBERT F1=0.908, MedRoBERTa F1=0.911; zero-shot F1=0.822, few-shot F1=0.841). MedRoBERTa performs better than mBERT on "Present", but mBERT is stronger on the underrepresented "Past" label (mBERT F1=0.387, MedRoBERTa F1=0.367). Few-shot outperforms zero-shot on both labels. "Future" and "None" score 0.000 across all models.

	Fine-tuning (Conversations Only)						**Prompt-based Generation**						**Support**
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
Past	0.303	0.536	0.387	0.274	0.554	0.367	0.257	0.625	0.365	0.293	0.696	0.413	81
Present	**0.921**	**0.896**	**0.908**	**0.922**	**0.901**	**0.911**	**0.942**	**0.729**	**0.822**	**0.927**	**0.770**	**0.841**	657
Future	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	32
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
Micro avg	0.810	0.840	0.825	0.796	0.846	0.820	0.757	0.700	0.727	0.772	0.743	0.757	770
Macro avg	0.306	0.358	0.324	0.299	0.364	0.320	0.300	0.338	0.297	0.305	0.367	0.314	770
Weighted avg	0.842	0.840	0.838	0.840	0.846	0.839	0.856	0.700	0.759	0.846	0.743	0.781	770

Table 5.11: Results of time predictions for strict filtering and lenient evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing label per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.

The confusion matrices in Figure 5.7 show that both models perform best on "Present" (mBERT TP=515, MedRoBERTa TP=518), but frequently confuse it with "Past" and vice versa.

"None" is rarely predicted. "Future" is consistently misclassified as "Present" or



(a) mBERT (b) MedRoBERTa

Figure 5.6: Confusion matrices for the relative time prediction task of Finetuning (Conversation Only) in strict filtering and strict evaluation.

"Past".

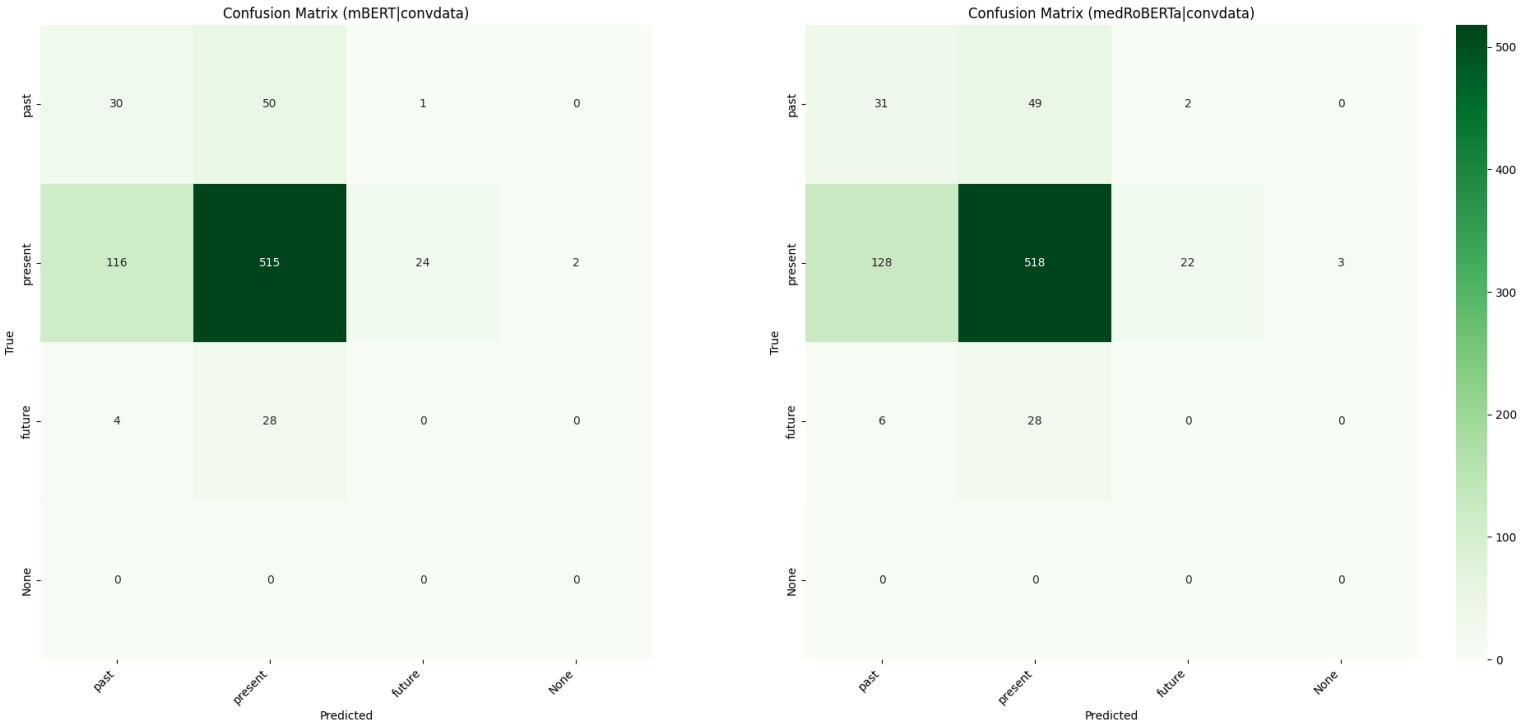
5.4.3 Evaluation setting 3 (lenient filtering, strict evaluation)

The results for the third evaluation setting are presented in Table 5.12. This setting combines lenient filtering with strict evaluation. As expected, the larger segments increase the likelihood of turn-level mismatches, resulting in lower performance than in the previous evaluation settings.

Macro average F1 scores are very low, ranging from 0.136 (both prompted models) to 0.167 (mBERT). The balance between recall and precision follows the same pattern as the other strict evaluation setting. For all models except mBERT (macro precision=0.199, macro recall=0.191), recall (average=0.276) overtakes precision (average=0.113). However, the results are much lower than the previous settings. Overall, mBERT performs best, followed by MedRoBERTa (F1=0.146), with both prompted models behind.

"Present" is again the strongest label. mBERT scores F1=0.494, MedRoBERTa F1=0.460, zero-shot F1=0.423, and few-shot F1=0.391. For "Present", precision exceeds recall in the fine-tuned models, but recall beats precision in the prompted models.

For "Past", recall is higher than precision across all models. mBERT scores F1=0.174



(a) mBERT (b) MedRoBERTa

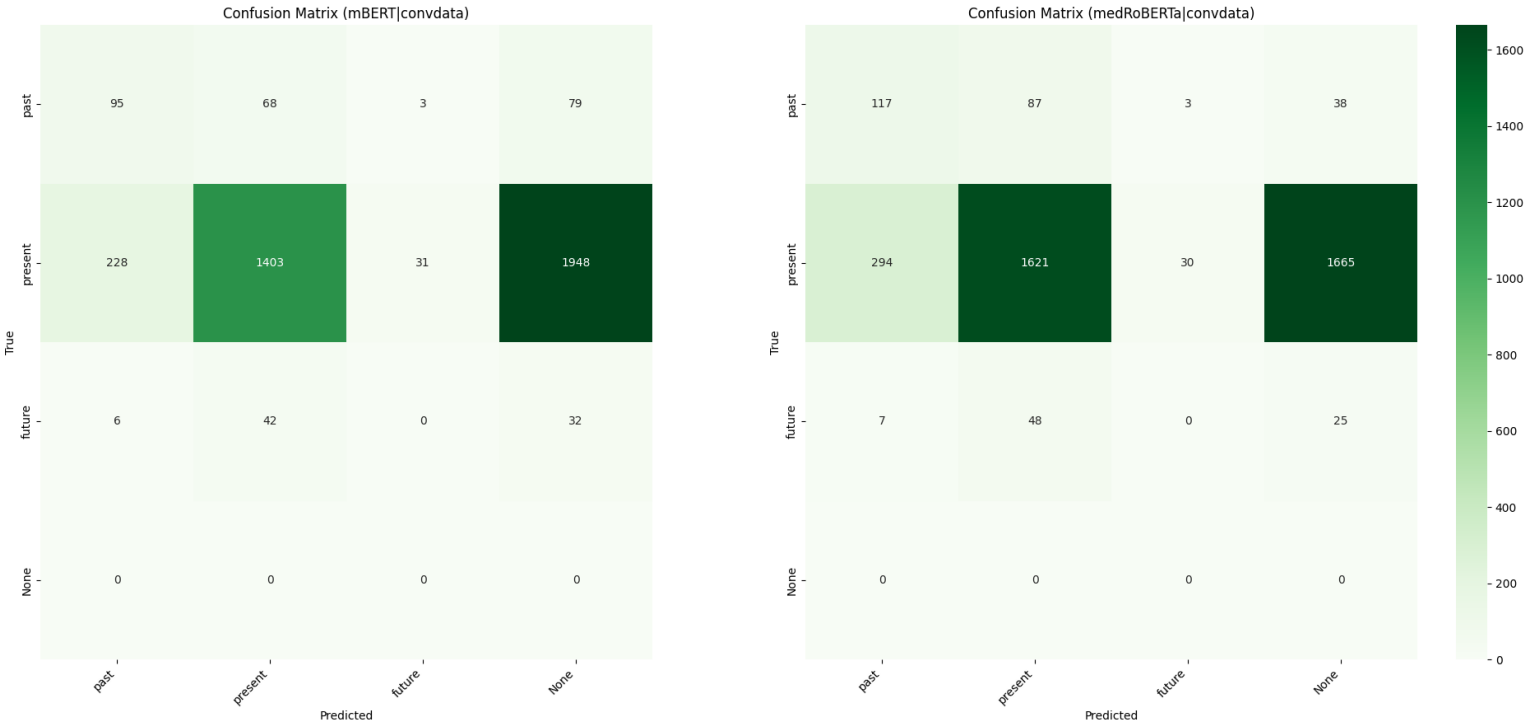
Figure 5.7: Confusion matrices for the relative time prediction task of Finetuning (Conversation Only) in strict filtering and lenient evaluation.

and MedRoBERTa F1=0.122, while few-shot (F1=0.152) slightly outperforms zero-shot (F1=0.121). "Future" and "None" score 0.000 across all models.

	Fine-tuning (Conversations Only)						**Prompt-based Generation**						**Support**
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
Past	0.113	0.376	0.174	0.070	**0.476**	0.122	0.067	**0.629**	0.121	0.085	**0.690**	0.152	245
Present	**0.681**	**0.387**	**0.494**	**0.477**	0.444	**0.460**	**0.340**	0.560	**0.423**	**0.317**	0.512	**0.391**	3610
Future	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	80
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
Micro avg	0.506	0.378	0.433	0.318	0.437	0.368	0.253	0.552	0.347	0.246	0.511	0.332	3935
Macro avg	0.199	0.191	0.167	0.137	0.230	0.146	0.102	0.297	0.136	0.101	0.300	0.136	3935
Weighted avg	0.634	0.378	0.465	0.444	0.437	0.431	0.317	0.552	0.397	0.297	0.511	0.369	3935

Table 5.12: Results of time predictions for lenient filtering and strict evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing label per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.

The confusion matrices in Figure 5.8 show that "Present" performs best (mBERT



(a) mBERT

(b) MedRoBERTa

Figure 5.8: Confusion matrices for the relative time prediction task of Finetuning (Conversation Only) in lenient filtering and strict evaluation.

TP=1403; MedRoBERTa TP=1621) but still misses many instances, which are predicted as "None" (mBERT FN=1948; MedRoBERTa FN=1665) or "Past" (mBERT FN=228; MedRoBERTa FN=294). "Past" is predicted correctly on many occasions (mBERT TP=95; MedRoBERTa TP=117) but is still often confused with "Present" (mBERT FN=68; MedRoBERTa FN=87) and "None" (mBERT FN=79; MedRoBERTa FN=38). "Future" is never predicted correctly and is most often confused with "Present" (mBERT FN=42; MedRoBERTa FN=48).

5.4.4 Evaluation setting 4 (lenient filtering, lenient evaluation)

The results for the fourth evaluation setting are presented in Table 5.13. This setting combines lenient filtering with lenient evaluation. It is expected to achieve the best performance, as predictions are matched across larger conversational segments.

Macro average F1 scores range from 0.354 to 0.365. Recall is again higher than precision for all models, averaging 0.539 and 0.317 respectively. Both measures are higher than in the previous evaluation settings. The fine-tuned models and prompted models perform similarly here. MedRoBERTa performs best, followed closely by few-shot prompting, while zero-shot performs worst.

"Present" again performs best, with F1 scores ranging from 0.963 to 0.979. Recall

and precision are close for this label, with recall slightly higher in most cases. "Past" is the second strongest label: mBERT scores F1=0.417, few-shot F1=0.416, zero-shot F1=0.410, and MedRoBERTa F1=0.396. Recall is much higher than precision for all models on this label.

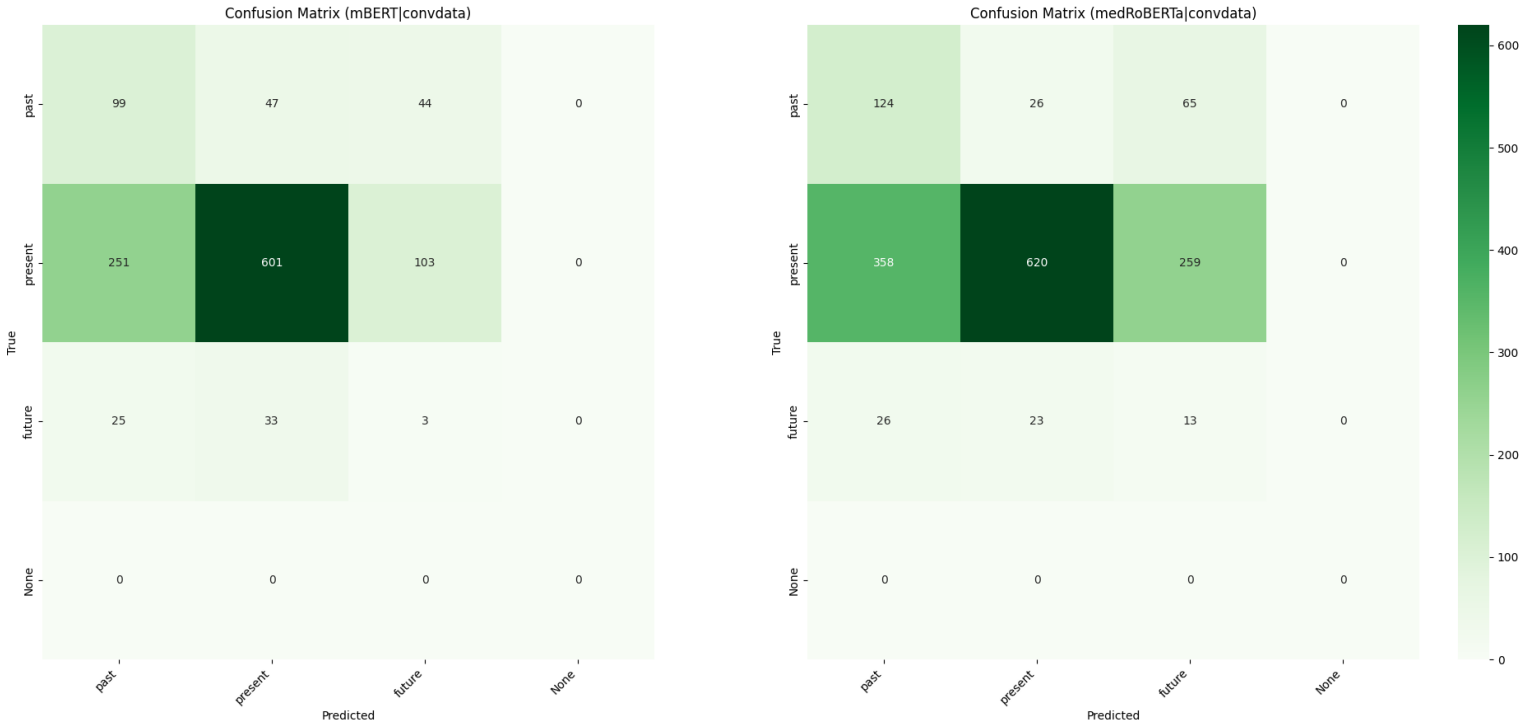
"None" scores 0.000. "Future" performs slightly better, with F1 scores ranging from 0.041 to 0.086; recall again greatly exceeds precision, and MedRoBERTa performs best in this class.

	Fine-tuning (Conversations Only)						**Prompt-based Generation**						**Support**
	mBERT			MedRoBERTa.nl			Zero-shot			Few-shot			
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
Past	0.292	0.728	0.417	0.253	0.912	0.396	0.264	0.912	0.410	0.267	0.941	0.416	215
Present	**0.965**	**0.968**	**0.966**	**0.960**	**0.998**	**0.979**	**0.975**	**0.952**	**0.963**	**0.960**	**0.998**	**0.979**	955
Future	0.029	0.125	0.047	0.047	0.542	0.086	0.023	0.208	0.041	0.033	0.333	0.059	61
None	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0
Micro avg	0.659	0.900	0.761	0.535	0.969	0.689	0.556	0.922	0.694	0.551	0.968	0.703	1231
Macro avg	0.321	0.455	0.357	0.315	0.613	0.365	0.316	0.518	0.354	0.315	0.568	0.364	1231
Weighted avg	0.819	0.900	0.842	0.809	0.969	0.850	0.822	0.922	0.839	0.811	0.968	0.852	1231

Table 5.13: Results of time predictions for lenient filtering and lenient evaluation. P = Precision, R = Recall, F1 = F1-score. The best performing label per model and experimental setting is shown in **bold**. The best performing model and experimental setting for each category is underlined.

The confusion matrices in Figure 5.9 show that "Present" performs well (mBERT TP=601; MedRoBERTa TP=620) but is frequently confused with "Past" (mBERT FN=251; MedRoBERTa FN=358) and "Future" (mBERT FN=103; MedRoBERTa FN=259). "Past" is predicted correctly on many occasions (mBERT TP=99; MedRoBERTa TP=124) but is also confused with "Present" (mBERT FN=47; MedRoBERTa FN=26) and "Future" (mBERT FN=44; MedRoBERTa FN=65). "Future" rarely receives the correct prediction (mBERT TP=3; MedRoBERTa TP=13) and is most often confused with "Past" (mBERT FN=25; MedRoBERTa FN=26) and "Present" (mBERT FN=33; MedRoBERTa FN=23).

To summarise, lenient filtering with lenient evaluation produced the best results for relative time prediction. The Conversation Only fine-tuned models generally outperformed the prompt-based approach. mBERT outperformed MedRoBERTa across all evaluation settings except in evaluation setting 4 (lenient filtering, lenient evaluation). Zero-shot and few-shot prompting achieved comparable performance, although few-shot performed slightly better in the best-performing evaluation setting. As with the previous prediction tasks, only a single random seed was used for training, limiting the generalisability of the results. Across most evaluation settings, recall was slightly higher than precision. Under lenient filtering with lenient evaluation, the difference between recall and precision was larger than in the other evaluation settings, with recall exceeding precision.



(a) mBERT

(b) MedRoBERTa

Figure 5.9: Confusion matrices for the relative time prediction task of Finetuning (Conversation Only) in lenient filtering and lenient evaluation.

5.5 Use-case: event-centred knowledge graph

Table 5.14 presents the summary statistics of the predicted ICF levels for each time point, category, and speaker. The predictions were generated by the MedRoBERTa Conversational Data model using lenient filtering, as this model achieved the best overall performance for relative time prediction. Table 5.15 presents the corresponding summary statistics for the gold labels.

The summary statistics are reported as the mean and sample standard deviation (mean ± standard deviation). The number of level mentions is shown between brackets. The gold labels contain substantially more mentions than the predicted labels and cover more categories at each time point.

Time point t_0 contains the highest number of level mentions for both the predicted labels (33 mentions) and the gold labels (76 mentions). This time point includes the T0 conversation together with past references from T1. Time point t_3 contains only one predicted mention and three gold mentions, all referring to future events from T2. Across all time points, the patient contributes more observations than the healthcare professional, with 42 and 24 predicted mentions, respectively, and 117 and 43 gold mentions.

Several categories appear across multiple time points in the predicted labels, including "D760 Family relationships", "D450 Walking", and "B280 Sensations of pain". Other categories occur at only one time point. For example, "B152 Emotional functions" and "B530 Weight maintenance functions" appear only at t_1, while "B230 Hearing functions" appears only at t_0. Some categories are not predicted at all, including "D240 Handling stress and other psychological demands", "D840-859 Work and employment", and "B140 Attention functions".

The gold labels show a similar pattern for categories that occur frequently across time points. However, "B152 Emotional functions" and "B530 Weight maintenance functions" are observed at multiple time points in the gold labels. "D240 Handling stress and other psychological demands" and "D840-859 Work and employment" also occur, although only rarely. Every category is represented at least once in the gold labels.

For the predicted labels, the highest mean levels are observed for "D465 Moving around using equipment" (approximately 3.5) and "D450 Walking" (approximately 2.6). The lowest mean levels are observed for "B134 Sleep functions" (approximately 0.95) and "B550 Eating" (approximately 0.98). Most categories remain relatively stable over time. The largest changes are observed for "D450 Walking", where the healthcare professional's mean decreases by almost 1.5 levels between t_0 and t_1, and "D760 Family relationships", where the patient's mean increases by approximately 0.5 levels from t_0 to t_1 before decreasing by almost one level at t_2.

The gold labels show relatively high mean levels for "D240 Handling stress and other psychological demands" and "B530 Weight maintenance functions". However, these categories contain only one mention per time point. "D760 Family relationships" also has a relatively high mean across multiple observations. "D465 Moving around using equipment" and "D450 Walking" again show relatively high mean levels. In contrast to the predicted labels, "B134 Sleep functions" has a mean level above 3.0 in the gold labels. The largest changes over time are observed for "B1300 Energy level", where the patient's mean increases from level 1 during the first time points to level 4 at t_2. "D450 Walking" also shows noticeable changes over time. For the healthcare professional, the mean level increases by approximately 1.5 levels between t_{-1} and t_0 before decreasing by about one level between t_0 and t_1.

Several categories show considerable variation within individual time points. For the predicted labels, "D410 Changing basic body position" has a standard deviation of 1.130. High variation is also observed for "B280 Sensations of pain" for the patient at t_1 ($\sigma = 1.355$), "D450 Walking" for the healthcare professional at t_0 ($\sigma = 0.720$), and "D760 Family relationships" for the healthcare professional at t_2 ($\sigma = 0.779$). Categories with a single observation per time point have no variation by definition. Examples include "B550 Eating" and "B164 Higher-level cognitive functions". Categories with the largest numbers of observations include "B280 Sensations of pain" and "D760 Family relationships", each with seven observations at t_1.

For the gold labels, the highest variation is observed for "B134 Sleep functions" for the patient at t_2 ($\sigma = 1.528$) and "B280 Sensations of pain" for the patient at t_0 ($\sigma = 1.323$). As with the predicted labels, many categories contain only a single observation per speaker and time point, including "D240 Handling stress and other psychological demands" and "B530 Weight maintenance functions". "D760 Family re-

lationships" and "B280 Sensations of pain" each contain a large number of observations.

For the predicted labels, the patient's utterances generally have higher mean levels than those of the healthcare professional. Some categories are discussed only by the patient, including "B440 Respiration functions", "B550 Eating", "D410 Changing basic body position", and "B230 Hearing functions" at t_0. Third-party speakers occur only at t_{-1} and t_0, with one and two observations, respectively.

For the gold labels, the differences between the patient and the healthcare professional are less consistent, with higher mean levels alternating between speakers across categories and time points. Third-party speakers are present only at t_0 and t_1.

	t0			t1			t2		t3		t4	
Category	**Patient**	**HP**	**3rd party**	**Patient**	**HP**	**3rd party**	**Patient**	**HP**	**Patient**	**HP**	**Patient**	**HP**
B1300 Energy level	–	–	–	1.040 ± 0.058 (4)	0.994 ± 0.035 (3)	–	–	1.204 (1)	0.963 (1)	–	–	–
B140 Attention functions	–	–	–	–	–	–	–	–	–	–	–	–
B152 Emotional functions	–	–	–	–	–	–	1.035 ± 0.015 (2)	1.047 (1)	–	–	–	–
B440 Respiration functions	–	–	–	1.074 ± 0.083 (2)	–	–	–	–	1.015 ± 0.025 (2)	–	–	–
D455 Exercise tolerance functions	–	–	–	–	–	–	–	–	–	–	–	–
B530 Weight maintenance functions	–	–	–	–	–	–	–	1.091 (1)	–	–	–	–
D450 Walking	–	–	3.214 (1)	–	3.266 ± 0.720 (2)	2.457 (1)	3.297 ± 0.189 (3)	0.991 (1)	–	–	–	–
B550 Eating	–	–	–	0.829 (1)	–	–	1.130 (1)	–	1.435 (1)	–	–	–
D840–859 Work and employment	–	–	–	–	–	–	–	–	–	–	–	–
B280 Sensations of pain	–	–	–	1.166 ± 0.184 (5)	1.162 (1)	1.084 (1)	1.709 ± 1.355 (4)	1.192 ± 0.051 (2)	–	1.103 (1)	–	–
B134 Sleep functions	–	–	–	–	0.950 (1)	–	–	–	–	–	–	–
D760 Family relationships	1.083 (1)	1.253 (1)	–	1.539 ± 0.331 (6)	1.212 (1)	–	2.085 (1)	–	1.109 ± 0.032 (2)	1.473 ± 0.779 (3)	–	1.012 (1)
B164 Higher-level cognitive functions	–	–	–	–	1.355 (1)	–	–	–	–	1.791 (1)	–	–
D465 Moving around using equipment	–	–	–	3.618 (1)	–	–	–	3.794 ± 0.007 (2)	3.092 ± 0.980 (2)	–	–	–
D410 Changing basic body position	–	–	–	2.678 ± 1.130 (2)	–	–	–	–	–	–	–	–
B230 Hearing functions	–	–	–	1.039 (1)	–	–	–	–	–	–	–	–
D240 Handling stress and other psychological demands	–	–	–	–	–	–	–	–	–	–	–	–

Table 5.14: Mean predicted level, standard deviation, and number of observations per predicted category, speaker, and time point. Values are reported as mean ± standard deviation, with the number of observations in parentheses. The standard deviation is excluded when only one observation is available. HP = healthcare professional; – = no observations.

	t0		t1			t2			t3		t4	
Category	**Patient**	**HP**	**Patient**	**HP**	**3rd party**	**Patient**	**HP**	**3rd party**	**Patient**	**HP**	**Patient**	**HP**
B1300 Energy level	1.000 (1)	2.000 (1)	2.000 ± 0.000 (3)	–	–	1.000 (1)	2.000 (1)	–	4.000 (1)	1.000 (1)	–	–
B140 Attention functions	–	–	3.333 ± 1.155 (3)	–	–	1.000 (1)	–	–	1.000 (1)	–	–	–
B152 Emotional functions	–	–	3.692 ± 0.751 (13)	4.000 ± 0.000 (2)	–	3.750 ± 0.500 (4)	4.000 ± 0.000 (3)	–	4.000 ± 0.000 (2)	–	–	–
B440 Respiration functions	–	–	4.000 ± 0.000 (2)	–	–	2.500 ± 0.707 (2)	–	–	1.333 ± 0.577 (3)	–	–	–
D455 Exercise tolerance functions	–	2.000 (1)	–	–	–	2.000 (1)	2.000 ± 0.000 (2)	–	–	–	–	–
B530 Weight maintenance functions	–	–	4.000 (1)	4.000 (1)	–	4.000 (1)	4.000 (1)	–	4.000 (1)	–	–	–
D450 Walking	–	2.000 (1)	2.500 ± 0.577 (4)	3.500 ± 0707 (2)	4.000 ± 0.000 (2)	2.800 ± 0.837 (5)	2.500 ± 0.837 (6)	–	3.000 ± 0.000 (2)	–	–	–
B550 Eating	–	–	4.000 (1)	–	–	–	–	–	–	–	–	–
D840–859 Work and employment	–	–	4.000 (1)	–	–	–	–	–	–	–	–	–
B280 Sensations of pain	2.000 (1)	–	3.000 ± 1.323 (9)	2.667 ± 0.577 (3)	–	2.444 ± 1.014 (9)	2.000 ± 0.000 (3)	–	4.000 ± 0.000 (2)	–	1.000 (1)	–
B134 Sleep functions	–	–	3.000 ± 1.000 (3)	3.000 ± 1.414 (2)	–	3.667 ± 0.577 (3)	–	–	2.667 ± 1.528 (3)	–	–	–
D760 Family relationships	4.000 (1)	–	4.000 ± 0.000 (10)	4.000 (1)	–	4.000 ± 0.000 (2)	4.000 ± 0.000 (2)	–	4.000 ± 0.000 (3)	4.000 ± 0.000 (2)	–	–
B164 Higher-level cognitive functions	–	–	0.000 (1)	1.000 (1)	–	–	1.000 (1)	2.000 (1)	0.000 (1)	–	–	–
D465 Moving around using equipment	–	–	4.000 ± 0.000 (2)	–	–	3.000 (1)	–	–	3.000 (1)	–	–	–
D410 Changing basic body position	–	–	3.000 ± 1.414 (2)	4.000 ± 0.000 (2)	–	3.000 ± 1.414 (2)	–	–	–	4.000 (1)	1.000 (1)	0.000 (1)
B230 Hearing functions	–	–	3.000 ± 1.414 (2)	3.500 ± 0.707 (2)	–	–	–	–	4.000 (1)	–	–	–
D240 Handling stress and other psychological demands	–	–	4.000 (1)	–	–	–	–	–	4.000 (1)	–	–	–

Table 5.15: Mean gold level, standard deviation, and number of observations per gold category, speaker, and time point. Values are reported as mean ± standard deviation, with the number of observations in parentheses. The standard deviation is excluded when only one observation is available. HP = healthcare professional; – = no observations.

Chapter 6

Discussion

This chapter discusses the results of the thesis. It begins with a general interpretation of the overall findings, before examining each task in more detail: ICF category prediction, level regression, and relative time classification. Furthermore, the results of the knowledge graph are explained. Next, the quality of the annotated data is discussed, covering inter-annotator agreement, training data quality, and transcription quality. The research question and subquestions are then answered, and the hypotheses are evaluated against the results. The chapter closes with a discussion of the limitations of this work.

6.1 Interpretation of results

Overall, model performance was moderate to poor across all three tasks. This is partly expected: spoken conversations are unstructured and difficult to model, and the transcriptions were imperfect. The lenient evaluation conditions were introduced to account for this, and the gap between strict and lenient results confirms that the models often predict the right label but in the wrong turn. This was also observed for disagreement of human annotators. The following paragraphs discuss the most important trends, after which the findings per task are examined in more detail.

Overall ranking of model settings: The All Data fine-tuning setting performed best for ICF category classification. This corresponds to the hypothesis that a model benefits from exposure to both medical content from EHR data and conversational structure from the conversation data. The baseline performed surprisingly well, particularly on the level regression task. One likely explanation is that the quality of the synthetic conversational training annotations was not high enough to give the fine-tuned models a consistent advantage. The Conversational Data model performed steadily across tasks and did particularly well on relative time classification, but lacked the medical grounding that likely limited its performance on the other two tasks. The LLM-prompted models performed worst overall, though an exception is discussed in Section 6.1.1. Further refinement of the prompts could potentially improve their performance. The few-shot prompted model was the same model used to generate the synthetic training annotations. Its poor performance thus raises concerns about the quality of those annotations: if the model struggled to correctly label the test data, it likely produced inconsistent training labels as well, which could have affected all fine-tuned models downstream.

mBERT vs MedRoBERTa: mBERT outperformed MedRoBERTa in most conditions. For ICF category classification, mBERT did better in the fine-tuned settings, while MedRoBERTa had an edge in the baseline. This is broadly in line with my expectations: MedRoBERTa's medical pre-training is most useful when trained only on EHR data, but mBERT's multilingual Wikipedia knowledge becomes more useful once conversational training data is added. For ICF levels and relative time, mBERT generally scored higher overall, though the pattern was less consistent for levels. These differences are discussed further in Sections 6.1.2 and 6.1.3

Few-shot vs zero-shot: Finally, few-shot prompting outperformed zero-shot in most settings. The few-shot examples showed the model what correct annotations look like, including common mistakes, which likely gave it a clear advantage over zero-shot prompting.

6.1.1 ICF categories

Lenient evaluation produced the best results across all models for both precision and recall, particularly when combined with lenient filtering. The combination of lenient filtering with strict evaluation produced the worst results.

Lenient evaluation performing best makes sense intuitively. In a conversation, an ICF category is rarely discussed in a single turn, rather, it often spans several turns. Whether a human or a model places the label on turn n or turn $n+1$ is partly a matter of interpretation. A mismatch of even one turn can heavily penalise strict evaluation, even when the prediction is essentially correct. Lenient evaluation accounts for this by checking whether the category appears anywhere in the surrounding segment.

This also explains the almost perfect precision under lenient evaluation. If the model predicts a category that does appear somewhere in the segment, it is counted as correct, so false positives are rare. The models appear to correctly identify which segment is about which category, even if the exact turn is slightly off. The higher recall under lenient evaluation can be explained by the same principle. Predictions that would be counted as false negatives under strict evaluation are considered correct if they occur somewhere else within the same segment.

Strict filtering performs slightly worse than lenient filtering under lenient evaluation. This is likely because lenient filtering keeps the turns leading up to the annotated segment, providing useful contextual information that helps the model predict more accurately.

The poor results under lenient filtering with strict evaluation can be explained by the increased number of turns the model must predict on. More turns means more opportunity for false positives, where the model incorrectly predicts a label for the turn, and false negatives, where the model fails to label the turn. The number of mismatches severely lowers precision and recall. This also explains why this setting performs worse than strict filtering with strict evaluation, where the smaller number of turns reduces the chance of false predictions.

Across all four evaluation settings, several patterns are consistent. "None" always scores 0.000 because its support in the filtered test set is 0, meaning there can be no true positives. "D840-859 Work and employment" scores 0.000 in all fine-tuned models and the LLM. This is unsurprising given that the patients are elderly and no longer working. When the label does occur, it is likely because the patient is referring to past

work or to a family member's employment. The baseline predicts this label correctly in some cases, which suggests that the EHR training data provides a clearer pattern for this category. The conversational training data appears to cloud this, causing the fine-tuned models to stop predicting it altogether.

"B140 Attention functions" and "D455 Exercise tolerance functions" also score low throughout. Both are rare in the training data, so the models had limited opportunity to learn patterns for them. "B140 Attention functions" is particularly problematic because it appears frequently in the test set, meaning the model is asked to predict it often despite having seen little of it during training.

Among the stronger performing labels, "B280 Sensations of pain" stands out, particularly for the LLM-prompted models. Its definition is clear and it is unlikely to be confused with other categories, as can also be seen in the confusion matrices. "B230 Hearing functions" is predicted well by the fine-tuned models despite not being highly frequent in the training data. Like "B280", it has a very distinct definition and no closely related categories it could be confused with. "B134 Sleep functions" performs well especially in evaluation setting 3. This is likely because sleep-related vocabulary is relatively distinctive compared to other categories, and the discussions about sleep often span several turns.

Category confusion patterns are further analysed in Section 7.4.

The best results of this thesis (evaluation setting 4: lenient filtering, lenient evaluation) exceed those of Chen and Vossen (2026), which reflect the richer contextual information that conversations provide compared to EHR sentences. However, the other three evaluation settings (evaluation settings 1, 2, and 3) perform worse, which highlights how complex conversational dialogue is. Strict turn-level evaluation penalises the models heavily in a way that does not indicate their actual understanding.

6.1.2 ICF levels

For level regression, strict filtering with lenient evaluation performed best in most cases. This is likely because lenient evaluation allows the correct level to appear anywhere in the segment, reducing the penalty for small mismatches. At the same time, strict filtering keeps the segment short and focused, preventing new mentions of the same category from introducing additional mismatches. The annotation density analysis in Section 3.8.3 supports this: on average, more than half of the turns in a strict segment contain a level annotation, and the average segment is nearly three turns long. This suggests that multiple level assignments often occur within a single segment, which is plausible since speakers can revise their description of a condition across turns.

Lenient filtering produced fewer level annotations per segment, which could explain why the evaluation settings using lenient filtering were outperformed by those using strict filtering. The additional context turns introduced by lenient filtering created more opportunities for mismatches without proportionally adding more level annotations.

The Baseline performed best across all evaluation settings, which suggests that the conversational training data did not help with level prediction. A likely explanation is that the EHR level annotations are more consistent and unambiguous than the synthetic conversational ones. Adding noisy conversational annotations may therefore have hurt performance rather than improved it. For most settings, the All Data model performed

better than Conversational Data Only for mBERT, while MedRoBERTa showed the opposite pattern. This could be because mBERT benefits from having more training data overall, while MedRoBERTa, which already has a strong medical foundation, is able to make better use of the conversational data alone. In evaluation setting 4, the Conversational Data model outperformed both All Data models for both mBERT and MedRoBERTa. The LLM-prompted models performed worst across all settings. This is particularly notable given that the same model was used to generate the synthetic training annotations.

In all settings, RMSE exceeded MAE, indicating that some larger errors were present and being penalised by the squaring in RMSE. This gap was smaller for the baseline than for the other models, suggesting that the baseline produced more consistent predictions with fewer extreme outliers. On a scale of 0 to 4, an MAE of around 1.1 for the baseline means the model is off by roughly one level on average. This difference suggests there is meaningful room for improvement, particularly for the fine-tuned and LLM-prompted models where errors are even larger.

Level 0 was excluded from the training data because it was substantially overrepresented in the synthetic annotations relative to the test data. The Conversational Data models therefore had no exposure to level 0 at all, while the baseline and All Data models did contain some knowledge of it from the EHR data. Despite this, the Conversational Data model performed comparably to All Data, which suggests that level 0 was rare enough in the test data that not knowing it did not significantly damage performance. The LLM, however, was still able to predict level 0 during inference, since it did not have the same training restrictions. Given that it also over-assigned this level during annotation, it likely continued to do so during inference, which may have contributed to its poor regression scores. A more detailed analysis of this is presented in Section 7.3.

Regarding context, using predicted categories consistently led to better results than using gold categories, with only one exception. This suggests that the models predict levels more reliably when the surrounding context contains consistent category annotations. In contrast, if the gold categories do not align with the predicted levels, this can negatively impact the outcome. The relationship between category prediction quality and level prediction is discussed further in Section 7.

Compared to Jakubauskaitė (2025) and Kim et al. (2022), adding conversational training data did not improve results enough to match their scores. This is likely a combination of two factors: the complexity of spoken dialogue, which makes level prediction harder than in EHR text, and the quality of the synthetic annotations, which introduced noise rather than useful signal. Level annotation is inherently more ambiguous than category annotation even in written data, so the impact of noisy training labels is likely greater here than for the other tasks.

6.1.3 Relative time

For relative time prediction, lenient filtering with lenient evaluation performed best, followed by strict filtering with lenient evaluation, then strict filtering with strict evaluation. Lenient filtering with strict evaluation again performed worst. This pattern is seen in the category classification task as well, and can be explained by the same mechanism: more turns to predict on leads to more false positives and negatives at the

turn level.

Lenient filtering with lenient evaluation likely performed best because it provided the most context around each segment. The correct time point was sometimes specified in a turn preceding the explicit level annotation, and the additional context made this accessible to the model. The recall in this setting was higher than in the previous evaluation settings, suggesting that the models missed fewer relevant cases. The larger gap between precision and recall suggests that the additional context improved recall more than precision. This allowed the models to identify more relevant cases, while the improvement in precision was smaller.

"None" and "Future" consistently scored 0.000 or close to it. "None" had no support in the filtered test set, so no true positives were possible. "Future" was rare in the training data, which explains why fine-tuned models struggled to predict it. The LLM-prompted models also failed to predict it reliably. The slightly higher scores for "Future" under lenient filtering with lenient evaluation suggest that the models do occasionally predict it, but with many mismatches.

"Present" performed well across all settings. It is the majority label in both the training data (apart from "None") and the filtered test set, which explains why the fine-tuned models adapted to it well. For the LLM-prompted models, the prompt instructed the model to default to "Present" when a label needed to be assigned but no other time point could be determined, which likely boosted its score on this label.

Unlike the level task, the relative time predictions showed no difference between using gold or predicted categories and levels as context. This suggests that the fine-tuned models did not rely meaningfully on the previous labels when assigning a time point, but instead based their predictions primarily on the text of the current turn itself.

6.2 Use case: knowledge graph

The results presented in Section 5.5 demonstrate the potential of an event-centred knowledge graph for structuring patient information over time. The following discussion compares the predicted and gold knowledge graphs. It focuses on the categories they contain, how these change over time, and the differences between speakers.

Categories such as "D760 Family relationships", "D450 Walking", and "B280 Sensations of pain" appear across multiple time points in both the predicted and gold labels. This is expected, as these topics are discussed frequently in interRAI interviews. Although the predicted labels cover fewer categories and contain fewer observations than the gold labels, they capture many of the same frequently occurring ones. This suggests that the pipeline successfully identifies the main topics discussed, even if less frequent categories are sometimes missed.

For the predicted labels, "D450 Walking" and "D465 Moving around using equipment" have the highest mean levels, suggesting that mobility is among the more impaired domains for this patient. The gold labels show a similar pattern for these categories. However, there are also differences between the predicted and gold labels. "B134 Sleep functions" has one of the lowest predicted mean levels but one of the highest in the gold labels, indicating that level estimation is not yet accurate for all

categories.

Time point t_0 contains the most observations in both the predicted and gold labels, combining the present of T0 with past references from T1. Time point t_1 contains fewer observations despite combining information from three conversations, which may reflect that functional levels were discussed less explicitly at that point.

For the predicted labels, the patient's utterances generally receive higher mean levels than those of the healthcare professional in the predicted labels. This may reflect different perspectives on functioning. Some categories appear only in the patient's speech, suggesting the healthcare professional followed the questionnaire more closely while the patient introduced additional topics. In the gold labels, the pattern alternates more between speakers, suggesting that the differences between speakers are less consistent in the manually annotated data than in the predicted labels. Third-party observations are rare in both cases.

Several categories change over time. "D450 Walking" fluctuates in both the predicted and gold labels, suggesting variation in reported mobility. "D760 Family relationships" also varies in the predicted labels. In the gold labels, "B1300 Energy level" increases from level 1 at earlier time points to level 4 at t_2, though this is based on very few observations and should be interpreted cautiously. Overall, the predicted labels capture several of the same temporal patterns as the gold labels, though the magnitude of changes is not always the same

These observations should be interpreted with care. Many categories have only one or two observations per time point, limiting the reliability of the summary statistics. Some categories are absent at certain time points, making longitudinal comparison impossible. The relative time labels are also coarse: past, present, and future map to six-month intervals, whereas the actual references may span days or years. This is sufficient for a proof of concept but would need refinement for clinical use.

Despite these limitations, this kind of summary could be useful in practice. It gives healthcare professionals a quick overview of which ICF domains were discussed, how they were rated, and how they changed over time. Representing this information in a knowledge graph makes it open to queries. For example, a professional could ask how often a patient discussed a given category, or compare how the patient and the healthcare professional rated the same domain differently across conversations.

6.3 Quality of annotated data

6.3.1 Inter-annotator agreement

The IAA scores reported in Section 5 indicate limited agreement overall. Category agreement was fair but low, relative time agreement was slightly higher, and level agreement was slightly higher than that. However, the relatively low scores suggest that the annotations, as well as the test set itself, should be interpreted with care.

One reason for the low scores is that Cohen's (weighted) κ measures agreement on a turn-by-turn basis. In a conversation, however, annotators may agree on the content of a segment while disagreeing on exactly which turn carries the label. This makes (weighted) κ a strict and potentially unfair measure for conversational data. A more detailed analysis of agreement within segments is presented in Section 7.1.

The annotation distances show a similar pattern. For most files, the distance between annotators was relatively large. In some cases, this was because one annotator assigned an annotation that the other did not. As a result, the distance was calculated relative to the end of the conversation, which increased the reported distances.

Overall, the IAA results suggest that strict turn-by-turn evaluation is a challenging evaluation method. Even the human annotators did not consistently agree at this level of detail.

6.3.2 Training data quality

The quality of the synthetic training annotations was another source of noise. The annotation distributions did not always match those of the gold test set. For ICF levels in particular, the training data contained far more 0-ratings than the test data, which could have introduced a bias towards lower level predictions and was therefore taken out. This is investigated further in Section 7.3.

In addition, all three tasks suffered from severe class imbalance. The majority of turns in both training and test data carry no ICF label at all. Rare categories and levels were seen too little during training for the models to learn reliable patterns, which is reflected in their near-zero scores on labels such as "D840-859 Work and employment" and "B140 Attention functions".

6.3.3 Transcription quality

The quality of the transcriptions also influenced results. Annotators reported that many parts of the conversations were difficult or impossible to annotate due to incomprehensible audio transcription. This was caused by a combination of unclear recordings, patients mumbling, and imperfect transcription model output which resulting in repetitions and unrecognised words. The transcription temperature was kept low to limit hallucination, but a slightly higher setting might allow the model to produce more coherent output. Multiple transcription approaches were tested, but none resolved these issues entirely.

6.4 Research Question

The main research question asked whether language models can automatically assign ICF labels from interRAI interviews. The following paragraphs address each subquestion in turn before answering the main question.

Subquestion (a): How well can fine-tuned language models extract ICF labels from conversational healthcare data? The results give a mixed answer. For ICF category classification, fine-tuning on both EHR and conversational data gave the best performance, confirming the hypothesis that exposure to both domains is beneficial. Domain adaptation was thus possible for this task. For ICF level regression, however, fine-tuning on conversational data did not help. The baseline trained only on EHR data performed best. The hypothesis was therefore only partially confirmed. Domain adaptation worked for categories but not for levels. For relative time classification, the fine-tuned model outperformed the prompt-based approach, which is a positive result given that this task was only trained on conversational data.

Subquestion (b): How well do language models with different pretraining domains generalise to ICF classification in conversational medical settings? The results broadly confirmed the hypothesis. mBERT outperformed MedRoBERTa in most fine-tuned settings, while MedRoBERTa had an advantage in the baseline. This is consistent with the expectation that mBERT's broader linguistic flexibility is more useful when conversational data is added, while MedRoBERTa's medical pretraining is most valuable in a purely medical setting. Few-shot prompting outperformed zero-shot in most settings, also as expected. The one clear exception to the hypothesis was the results of the LLM. Contrary to expectations, the prompt-based models typically performed worst, suggesting that the unstructured and noisy nature of the conversational data made it particularly difficult for the decoder to identify reliable patterns.

Main research question. Language models can assign ICF labels and levels from interRAI interviews to a generally reliable degree, but performance varied considerably across tasks and evaluation conditions. Category prediction showed the most promise, particularly under lenient evaluation. Level prediction proved to be considerably harder, and relative time prediction mostly relied on majority labels. A consistent finding across all tasks was that several labels, particularly rare ones such as "D840-859 Work and employment" and "Future", were almost never predicted correctly. These pull the overall scores down and indicate a pressing need for further development.

The next chapter presents an error analysis to better understand the types of mistakes made by the models that contribute to these results.

Apart from the classification tasks, this thesis also demonstrated that evaluation method matters substantially. Strict turn-by-turn evaluation severely penalises models even when their predictions are essentially correct. This suggests that standard NLP evaluation metrics are not well suited for conversational annotation tasks. Defining suitable evaluation methods for this type of data is an important open problem.

Additionally, the predictions of the best-performing model were used to populate an event-centred knowledge graph, providing a structured and queryable representation of patient functioning over time. This shows that the output of the classification pipeline can already be used for a practical application, despite its current performance.

6.5 Limitations

Several limitations of this thesis are worth noting.

The test set was annotated by a single baseline annotator, whose reliability was assessed through comparison with additional annotators on a subset of files. Ideally, all annotators would have annotated all files and disagreements would have been resolved through discussion. The current setup means that any systematic bias in the baseline annotator's judgement is carried through to the evaluation.

Related to this, the annotation guidelines were not always straightforward to apply. Deciding which category belongs to a given turn, and whether a level should be assigned, was sometimes ambiguous in conversational language. Even within the annotator team, there was discussion on how to interpret certain cases. There was not always one objectively correct annotation.

The size of the test set was also limited by the time available for manual annotation. A larger and more diverse test set would give a more reliable picture of model

performance.

The evaluation segments were defined automatically based on the annotation process. This means that some informative parts of the conversation may not have made it into the filtered test set, and the same prediction can be counted as correct under lenient evaluation but incorrect under strict evaluation. The reported performance thus depends partly on how the evaluation segments are defined.

Also, the segmentation strategy was applied only to the test data and not to the training data. As a result, the training data may also have contained large sections of conversation that were not relevant to the annotation task. Excluding these sections could have improved the quality of the training data and, consequently, the fine-tuning of the models.

Finally, the synthetic annotations used for training depended on the chosen prompts. Different prompting strategies could produce different training data, and the reported performance is not necessarily the maximum achievable with LLM-generated annotations.

Chapter 7

Error Analysis

In this chapter, an error analysis is performed on common errors which may have influenced the results of this thesis. This chapter investigates four sources of error. First, inter-annotator agreement is examined at the segment level. Second, the effect of disjointness in the gold labels, where a category is annotated but no level is specified, is investigated. Third, the over-assignment of level 0 in the synthetic annotations is analysed. Finally, categories that are frequently confused with one another are examined in more detail.

The data in this thesis is medical data. Due to privacy reasons, no actual examples are provided in this analysis. Upon request, access to the evaluated chunks is possible.

7.1 Inter-annotator agreement

The turn-by-turn IAA analysis in Chapter 5 used a strict approach that may not be appropriate for conversational data, where annotators may agree on the content of a segment but place their labels on slightly different turns. This section takes a segment-level perspective, comparing annotations within strict and lenient segments rather than turn-by-turn. It also analyses specific disagreements to identify areas where the annotation guidelines could be improved.

7.1.1 Category agreement

For each file, the baseline annotator's segments were identified and the segment category was noted. For each segment, it was then checked whether the additional annotator had assigned the same category anywhere within that segment. The results are shown in Figure 7.1.

Agreement at the segment level is still low. In some files, the proportion of agreement does not exceed 0.30 for either filtering condition. Overall, strict segments show about 0.33 proportional agreement and lenient segments about 0.49. The higher agreement under lenient filtering is expected, since longer segments give the additional annotator more opportunities to assign the same category. Even so, the overall agreement is low, suggesting genuine uncertainty about category boundaries.

The most common source of disagreement, in both filtering conditions, was whether to assign a category at all. The baseline annotator frequently marked a category that the additional annotator did not, which was especially penalised under strict filtering.

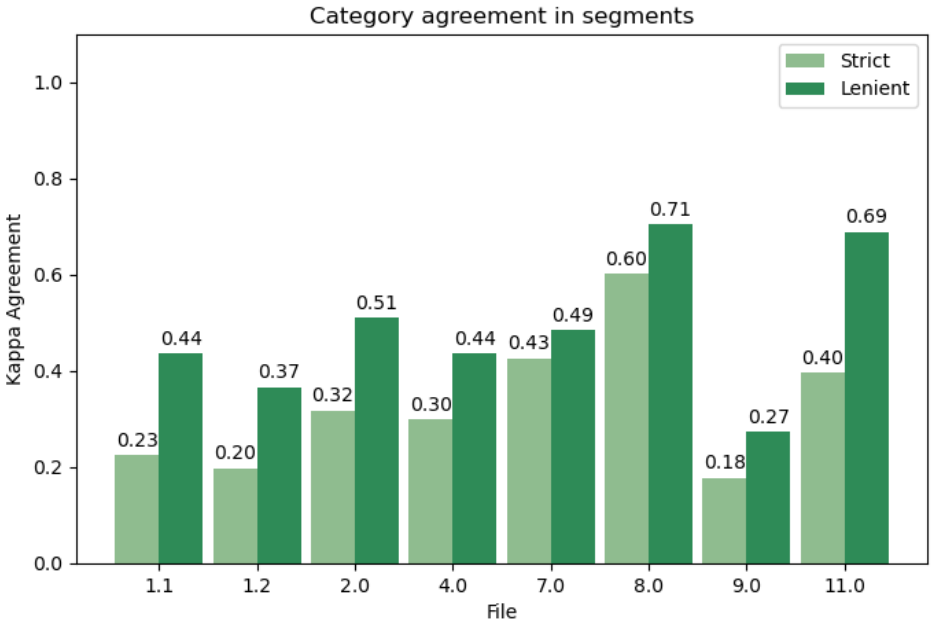


Figure 7.1: Proportion of category κ agreement on segments between the baseline annotator and the additional annotator.

Several specific categories were often problematic in the samples. "D760 Family relationships" caused disagreement when patients mentioned a family member without describing an interaction. For example, mentioning a daughter-in-law without saying she visited. The boundary of what counts as a family relationship event needs to be more clearly defined. "D450 Walking" and "D455 Exercise tolerance functions" were also confused with each other, as well as "B1300 Energy level" and "D455 Exercise tolerance functions". The latter two are easy to distinguish when a turn focuses on an activity rather than the energy required to perform it. In the lenient filtering, "B140 Attention functions" and "B152 Emotional functions" were frequently confused, which mirrors the confusion seen in the model predictions (see Section 5.2). Similarly, "B164 Higher-level cognitive functions" and "B134 Sleep functions" were sometimes used interchangeably. Clearer boundary descriptions for these category pairs in the annotation guidelines would likely improve agreement.

One additional pattern was that annotators sometimes became too focused on the dominant category in a segment, missing other categories that were briefly discussed within it. Annotators should remain alert to category switches even within a densely annotated segment. There was also some disagreement about how many consecutive turns should be labelled with a category, which suggests that clearer instructions on annotation span are needed.

7.1.2 Level agreement

For level agreement, the level assigned to the segment category was compared between the baseline and additional annotator within each segment. Segments where the additional annotator had not assigned a level were excluded. The mean difference in level

per file is shown in Figure 7.2.

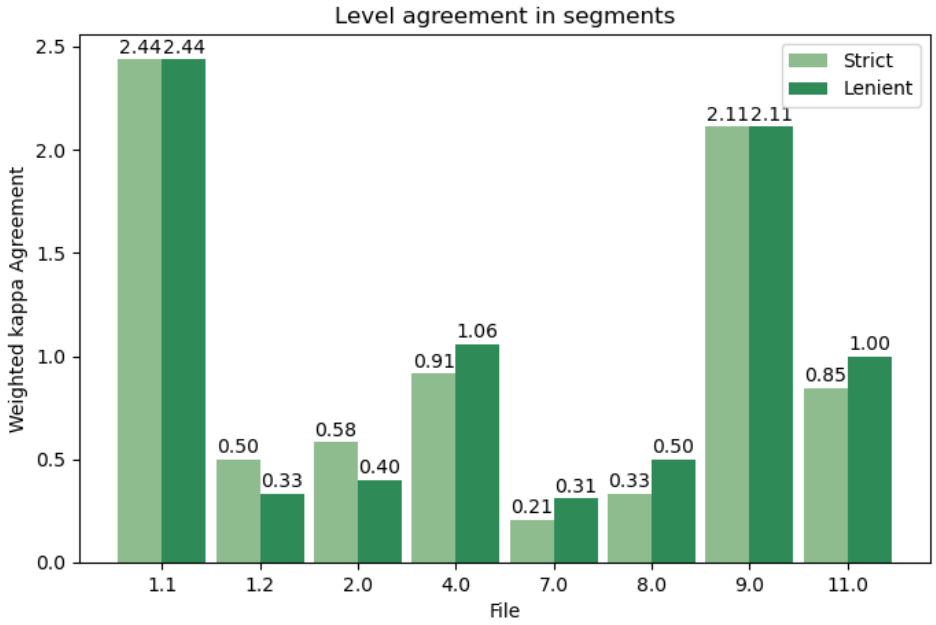


Figure 7.2: Mean difference of level weighted κ agreement on segments between baseline annotator and additional annotator.

For most files, the mean difference is below 1 level. Files 1.1 and 9 are exceptions, with differences above 2. These files show identical scores for strict and lenient filtering, likely because no segments were excluded due to missing level annotations, meaning all mismatches were kept. For files 4 and 11, annotators disagreed by around 1 level. For the remaining files, disagreement was between 0.3 and 0.5 levels.

The most notable finding was that some annotators reversed the level scale, treating 0 as full impairment and 4 as no impairment. This occurred for categories such as "B280 Sensations of pain", where it is easy to interpret 0 pain as a positive outcome and therefore assign 0. This issue was flagged by one annotator during the process, but the analysis shows it occurred more broadly and sometimes within a single segment. A clear discussion of the scale before annotation begins is essential to prevent this.

Besides scale reversal, annotators sometimes disagreed by one level, particularly for "B530 Weight maintenance functions", "B1300 Energy level", and "B152 Emotional functions". In the lenient filtering, annotators occasionally changed their level assignment within a segment, causing a mismatch with the annotation of the baseline. This is in line with the lower scores for lenient filtering in Figure 7.2. A possible solution would be to compare the average level per segment instead of comparing each individual level annotation within the segment. Some disagreements were also caused by annotators placing the level on a slightly different turn, which would have been penalised in the turn-by-turn IAA evaluation in Chapter 5.

7.1.3 Relative time agreement

For relative time, segments were compared in the same way. The results are shown in Figure 7.3. Segments without a time annotation and files with no time annotations at all were excluded.

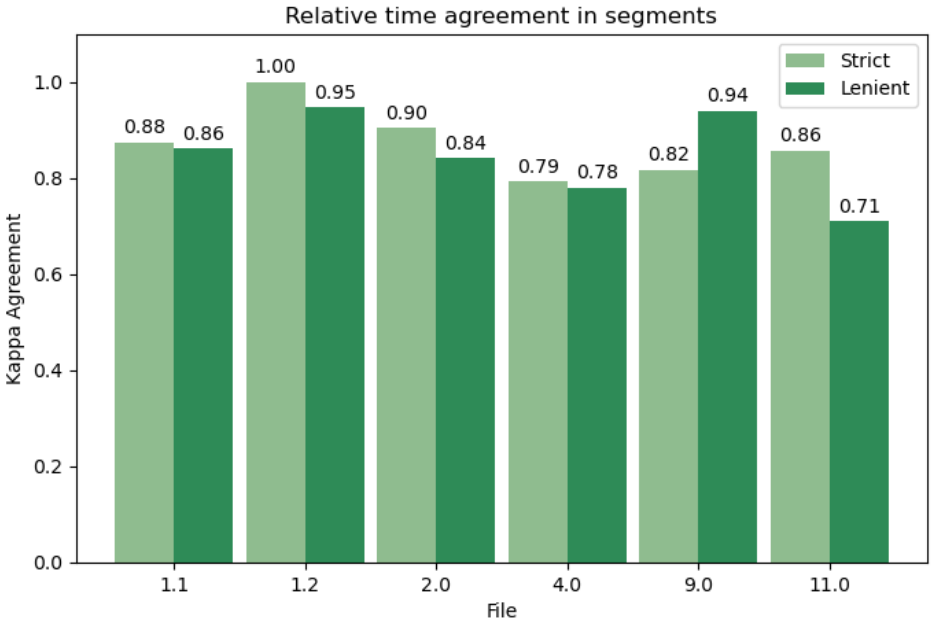


Figure 7.3: Proportion of relative time κ agreement on segments between baseline annotator and additional annotator.

Relative time agreement is considerably higher than category or level agreement, with proportional agreement of around 0.88 under strict filtering and 0.85 under lenient filtering. The slightly lower score under lenient filtering may reflect annotators revising their time judgement as the segment extends beyond the strict boundary. The higher agreement overall is likely partly due to the smaller number of possible labels.

The main source of confusion was the distinction between "past" and "present". This mirrors the model predictions, where these two labels were most frequently confused with each other. The definitions may need to be sharpened: "past" should refer to something that was the case but is no longer, while "present" covers anything currently happening, including ongoing routines. Annotators did not always apply this distinction consistently, possibly because relative time was a new annotation type for all of them.

Another issue was that annotators were not always consistent about which turns to annotate for relative time. The instruction was to only annotate time on turns where a level was also given, but annotators sometimes annotated time without a level, or a level without a time. In the strict filtering, there were also disagreements about where time annotations should start and end within a segment. In the lenient filtering, some additional annotators left turns entirely unannotated, causing more mismatches.

In conclusion, the segment-level IAA analysis shows that agreement is higher than the turn-by-turn scores in Chapter 5 suggest, but still far from perfect. The main issues are unclear category boundaries, inconsistent application of the level scale, and ambiguous instructions on when to start and stop annotating. These findings suggest that this test set's annotations should be treated as a starting point rather than definitive ground truth, and that the annotation guidelines still have room for refinement.

7.2 Disjointness

This section investigates the effect of disjointness on model predictions. A disjointed turn is one where an ICF category is annotated in the gold labels but no corresponding level is given. The question is whether the models correctly resist predicting a level in these cases, and whether this behaviour differs between strict and lenient filtering. The analysis uses predicted categories as context for level prediction, since these produced the best results overall in Chapter 5.

For each model, all disjointed turns were identified. The number of correctly predicted categories was counted, along with the number of false positive category predictions and the number of turns where the model predicted a level despite none being present in the gold annotations. The average predicted level was also calculated to check for systematic bias.

Table 7.1 shows the results. There were 407 disjointed turns under strict filtering (19.68% of turns) and 861 under lenient filtering (5.37% of turns). Strict filtering resulted in a higher proportion of correct category predictions for segments containing disjointed turns. Lenient filtering produced slightly more false positive predictions, where a category was predicted that did not appear in the gold annotations.

For all encoder models, the number of correctly predicted categories matched the number of level mismatches exactly. This means the models predicted a level for every turn where they also predicted a category, without exception. The models appear unable to not do a level prediction once a category has been assigned, even when the context suggests the level has not yet been specified. This is a systematic limitation, since the models have learned to treat category and level as inseparable, which does not hold in conversational data where a category can be mentioned across several turns before a level is given.

The Prompt-Based Generation showed slightly different behaviour. In some cases, it correctly refrained from assigning a level even when a category was present, particularly under lenient filtering. This suggests that the decoder has some capacity to interpret the context and recognise that a level has not yet been specified. However, the few-shot prompted model in particular produced more false positive category predictions.

The average predicted level was low across most models, with the exception of the baseline, which predicted levels closer to the middle of the 0–4 scale. The lower predictions of the fine-tuned models suggest a systematic bias towards lower levels. This is consistent with the label distribution of the training set shown in Section 3.9.3. Although level 0 was excluded from the training data because of over-annotation in the synthetic labels, level 1 still accounted for a large proportion of the training labels. In contrast, levels 3 and 4 occurred much less frequently. This imbalance likely made the models more likely to predict lower levels.

	Baseline				**Fine-tuning (All)**			
	Strict		Lenient		Strict		Lenient	
	mBERT	MedRoBERTa	mBERT	MedRoBERTa	mBERT	MedRoBERTa	mBERT	MedRoBERTa
Total disjoints	407	407	861	861	407	407	861	861
Correct categories	252	268	514	547	273	263	566	547
FP categories	82	73	193	179	75	69	166	161
Level mismatch	252	268	514	547	273	263	566	547
Mean predicted level	2.00	2.00	2.00	2.00	0.71	0.37	0.33	0.10
	Fine-tuning (Conversations)				**Prompt-based**			
	Strict		Lenient		Strict		Lenient	
	mBERT	MedRoBERTa	mBERT	MedRoBERTa	Zero-shot	Few-shot	Zero-shot	Few-shot
Total disjoints	407	407	861	861	407	407	861	861
Correct categories	275	272	571	565	200	252	434	532
FP categories	74	68	166	149	65	94	142	214
Level mismatch	275	272	571	565	170	70	359	132
Mean predicted level	1.00	1.03	1.00	1.02	1.48	1.44	1.57	1.63

Table 7.1: Category check for model predictions in disjoint turns where the category exists but no level is annotated.

A closer look at specific disjointed turns revealed a recurring pattern. The models frequently assigned levels to turns containing questions or leading statements, where the patient has not yet confirmed or specified any impairment. For example, a question asking whether someone walks with or without a cane, or a statement that someone does not use a hearing aid, was often treated by the models as if the level had already been given. The models should have used the surrounding context to recognise that the level would only be specified in a subsequent turn. This error occurred in both strict and lenient filtering conditions.

These findings highlight a key challenge in applying sentence-level classification to conversational data. The models do not track what information has and has not yet been established in the dialogue sufficiently. Addressing this would likely require either a more context-aware model or more targeted training examples that include disjointed category-level sequences.

7.3 Level 0 in the synthetic annotations

A recurring error in the synthetic training annotations was the over-assignment of ICF level 0. The LLM assigned 0 so frequently that it became the majority label in the training data, even though it did not occur proportionally as often in the test data. This section investigates the patterns behind this behaviour, and analyses the impact of prompt and example modification.

To identify patterns, the ratio of level 0 annotations was calculated across all files. The ten files with the highest ratio were selected, and for each file a sample of 12 turns with a high concentration of 0 annotations was extracted and inspected manually.

After identifying the patterns, each sample was re-annotated by the LLM with 20 turns of context before and after, using the same format as the original annotations. Several prompt variants and example sets were tested. The analysis focused on whether the annotations for the 12 sampled turns improved.

7.3.1 Repetition

An unmistakable source of the high level of 0 counts was repeated sentences in the transcription. Three of the ten examined files contained samples with repeating turns, and in each case the LLM assigned level 0 to every repeated instance of the same sentence. This drastically raised the proportion of level 0 in those files.

The repeated turns were not always incorrectly annotated. In one case, the repeated sentence described a patient still being in pain, for which level 0 could be appropriate. In another case, however, the repeated sentence stated that the patient had no pain at all. This case will be discussed further in Section 7.3.2. The issue is not only whether the level is correct, but whether assigning a level to every repeated instance is appropriate at all.

7.3.2 Negation

Many errors involved negations. In several samples, the LLM assigned level 0 to turns where the patient explicitly stated the absence of a problem. For example, a patient stating they do not feel pain was labelled with a low level, suggesting the model locked onto the word "pain" without correctly processing the negation.

A similar pattern occurred with turns about routine behaviour. A statement such as "she never misses a visit" was annotated as level 0, likely because the model focussed on "misses a visit" rather than on the negation that reverses its meaning. A human annotator would interpret this as the person always attending, not as a sign of impairment.

7.3.3 Common categories

Two categories appeared frequently in the sampled turns with incorrectly assigned level 0, namely "D760 Family relationships" and "B152 Emotional functions".

"D760 Family relationships" was a category that caused discussion among human annotators as well. Its boundaries are genuinely ambiguous, and it is not always clear what counts as a family relationship event and what the appropriate level is. This ambiguity likely affected the LLM in the same way.

For "B152 Emotional functions", there were cases where the errors involved statements about the absence of psychiatric illness. The LLM appears to have focused on the psychiatric condition itself rather than on the fact that it was not present, again reflecting a failure to handle negation as discussed in Section 7.3.2.

7.3.4 Transcription quality

Transcription errors also contributed to incorrect level assignments. In several samples, sentences spanned multiple turns due to transcription mistakes, making it unclear which turn should carry the level annotation. In other cases, sentences were cut off mid-way, leaving incomplete utterances that the model appeared to misinterpret by defaulting to level 0. Also, there were various scenarios in which individual words were transcribed incorrectly, constructing a whole sentence that is nonsensical. The LLM appeared to assign level 0 in these cases out of uncertainty.

7.3.5 False positives

A final pattern was that the LLM sometimes assigned level 0 to turns that should not have received any level at all. In these cases, the surrounding context contained turns with level 0, and the model appeared to carry this over to the current turn even when no level was needed. This suggests that the LLM used the recent annotation history as a standard, rather than evaluating each turn independently.

7.3.6 Prompt modification

Based on these patterns, several changes were made to the prompt and few-shot examples. The prompt was shortened, since an overly long prompt can cause the model to lose track of key instructions. Also, the instruction not to use 0 as a default label was moved to the end of the prompt, in the hope that the model would weight it more heavily during generation. Furthermore, an instruction was added to avoid assigning a level to every instance of a repeated sentence. Additionally, new few-shot examples were added covering negation cases and the problematic categories identified above. The altered prompts and examples can be found in Appendix C and D.

None of these changes produced a meaningful improvement. Turns that were annotated as 0 before remained 0 after the modifications. The instruction on repeated sentences occasionally caused the first two instances to be annotated differently from the rest, but all other instances were still assigned level 0. The negation examples had almost no effect. A minor improvement was observed for "D760 Family relationships" in some cases, but the same errors persisted in others.

The failure of prompt modifications to resolve this issue points to two possible problems. First, the LLM probably had trouble understanding the transcription due to the quality. Incomplete or nonsensical sentences cannot be reliably annotated regardless of the prompt. Second, ICF level annotation in conversational data is substantially difficult, and even human annotators disagreed on many cases. The over-assignment of level 0 in the training data is therefore both a modelling limitation and a reflection of the task's inherent ambiguity. This likely contributed to the overall weaker level prediction results reported in Section 5 as well.

7.4 Category confusion

This section examines cases where the models confused one category with another. Under lenient evaluation, confusion was mostly between categories and "None". Under strict evaluation, confusion also occurred between specific category pairs. The three most common pairs were "B140 Attention functions" and "B152 Emotional functions", "D455 Exercise tolerance functions" and "D450 Walking", and "D240 Handling stress and other psychological demands" and "B152 Emotional functions". These confusions partly align with Chen and Vossen (2026), who also reported confusion between "D240" and "B152". The other pairs do not exactly match their findings, but involve similar category overlaps.

Confusion with "None" is not analysed in depth here. It is most likely a consequence of the high proportion of "None" in the training data for each task (categories, levels, and relative time), which biases the models towards predicting it as a default. Contrarily, the models also occasionally predict an annotation when the correct label is "None", resulting in false positives.

To investigate the three confusion pairs, all turns where the model predicted a different label than the gold label were gathered for each pair. A sample was drawn across models, datasets, and filtering conditions, and inspected for patterns.

A general finding was that the models tended to predict the more frequent category over the less frequent one. Since the models saw more training examples for common categories, it is unsurprising that they default to those when a turn is ambiguous.

7.4.1 B140 Attention functions vs B152 Emotional functions

Turns with gold label "B140 Attention functions" but predicted label "B152 Emotional functions" were mostly about emotional reactions, namely patients getting upset or angry. Because the turn contained an emotional response, the models assigned the emotional functions category. The distinction is subtle however: attention functions relate to the ability to focus, while emotional functions relate to the experience and regulation of emotions. Turns that described an emotional reaction as a consequence of difficulty concentrating were particularly hard to classify correctly.

7.4.2 D455 Exercise tolerance functions vs D450 Walking

Confusion between these two categories was mostly caused by turns about walking. However, rather than describing the basic ability to walk, the turns focused on endurance. For example, whether a patient could walk longer distances. This is more characteristic of exercise tolerance than of walking, which is why the human annotator assigned "D455". The models focused on the walking itself and missed the endurance component.

7.4.3 D240 Handling stress and other psychological demands vs B152 Emotional functions

Turns in this confusion pair described stressful situations alongside the patient's emotional reaction to them. Human annotators assigned "D240" because the turn was about coping with a stressful situation, but the models focused on the emotional response, which included remarks about being scared, angry, or worried, and predicted "B152" instead. The models appear to weigh the surface-level emotional content more heavily than the underlying context.

In all these sampled cases, the predicted category was not entirely wrong: the turn did contain a hint of it. The distinction between the correct and predicted label was subtle, but the model consistently leaned towards the more noticeable surface feature. Adding explicit descriptions to the annotation guidelines, for example by explaining how each category differs from its most commonly confused neighbour, could improve both the training data quality and the model predictions.

Chapter 8

Conclusion

To conclude, this thesis investigated whether language models can automatically assign ICF labels from verbal interRAI interviews. The results show that this is possible to a meaningful degree, though performance varies across tasks, models, and evaluation conditions.

For ICF category prediction, fine-tuning on both EHR and conversational data proved most effective. For level regression, the baseline trained only on EHR data performed best, suggesting that the conversational training data did not provide additional useful information for this task. Relative time prediction was handled best by the models trained on conversational data only. The LLM-prompted models perform generally worst across all tasks. Among the encoder models, mBERT adapted better to the fine-tuned settings, while MedRoBERTa performed best in the baseline. Few-shot prompting consistently outperformed zero-shot.

The error analysis revealed several recurring challenges. Rare categories were poorly predicted across all models. Transcription quality had a grave impact on the synthetic training annotations. Annotator segment agreement was low, which puts the model results in perspective and confirms that the task is genuinely difficult even for humans.

This thesis also introduced a way of segmenting conversations into meaningful units for evaluation, combining strict and lenient filtering with strict and lenient evaluation. This revealed that the evaluation method has a large effect on the reported performance, and that standard turn-level evaluation is often too strict for conversational data. Further work is needed to refine these segmentation strategies.

The predictions of the best-performing model were additionally used to populate an event-centred knowledge graph, demonstrating a concrete downstream application of the classification pipeline.

Together, these contributions show that adapting language models to conversational clinical data is feasible, but requires careful attention to data quality, evaluation design, and task-specific annotation guidelines.

8.1 Future work

Building on this, several directions could expand on the findings of this thesis. These are grouped into four themes: data quality, modelling approaches, evaluation methods, and knowledge graphs as described below.

Data quality: The most direct improvement would come from better transcriptions. Improved audio quality and/or a better transcription model would benefit both the synthetic and gold annotations, and in turn the model performance. Future work could also explore mixing EHR and conversational data during training rather than fine-tuning sequentially, and could invest in a fully human-annotated training set to reduce noise from synthetic labels.

Modelling: Other model architectures could be explored, such as XLM-RoBERTa as an additional multilingual baseline, or ensemble methods that combine the strengths of both models used here. An end-to-end approach, which predicts category, level, and relative time together rather than as three separate tasks, could also be investigated, as it might allow the model to better understand the full context of a turn and the relationship between the three labels.

Evaluation: As mentioned, the segmentation strategy used to define strict and lenient evaluation boundaries could be refined in future work. Boundary detection by annotators based on topical shifts in the conversation, for example, might produce more meaningful segments than the automatic annotation-based approach used here. Also, refining the annotation guidelines to explicitly describe what distinguishes each category from others would improve inter-annotator agreement and the reliability of the gold labels. Furthermore, having more annotators label the same files would also improve the reliability of the gold labels. Lastly, evaluating on other clinical domains would test whether the approach generalises beyond the interRAI HC questionnaire context.

Knowledge graphs: Finally, future work cou8ld include more conversations and more patients to build a more reliable knowledge graph. A more detailed temporal representation than the current three time classes could also improve longitudinal tracking. This would further demonstrate the potential of the proposed pipeline to support automatic knowledge graph construction.

Appendix A

ICF-17 set

Category code	Category name	Description
B1300	Energy level	Mental functions that produce vigour and stamina
B140	Attention functions	Specific mental functions of focusing on an external stimulus or internal experience for the required period of time
B152	Emotional functions	Specific mental functions related to the feeling and affective components of the processes of the mind
B440	Respiration functions	Functions of inhaling air into the lungs, the exchange of gases between air and blood, and exhaling air
B455	Exercise tolerance functions	Functions related to respiratory and cardiovascular capacity as required for enduring physical exertion
B530	Weight maintenance functions	Functions of maintaining appropriate body weight, including weight gain during the development period
D450	Walking	Moving along a surface on foot, step by step, so that one foot is always on the ground, such as when strolling, sauntering, walking forwards, backwards, or sideways.
D550	Eating	Carrying out the coordinated tasks and actions of eating food that has been served, bringing it to the mouth and consuming it in culturally acceptable ways, cutting or breaking food into pieces, opening bottles and cans, using eating implements, having meals, feasting or dining.
D840-D859	Work and employment	Apprenticeship (work preparation); acquiring, keeping and terminating a job; remunerative employment; non-remunerative employment
B280	Sensations of pain	Sensation of unpleasant feeling indicating potential or actual damage to some body structure
B134	Sleep functions	General mental functions of periodic, reversible and selective physical and mental disengagement from one's immediate environment accompanied by characteristic physiological changes
D760	Family relationships	Creating and maintaining kinship relationships, such as with members of the nuclear family, extended family, foster and adopted family and step-relationships, more distant relationships such as second cousins, or legal guardians
B164	Higher-level cognitive functions	Specific mental functions especially dependent on the frontal lobes of the brain, including complex goal-directed behaviours such as decision-making, abstract thinking, planning and carrying out plans, mental flexibility, and deciding which behaviours are appropriate under what circumstances; often called executive functions
D465	Moving around using equipment	Moving the whole body from place to place, on any surface or space, by using specific devices designed to facilitate moving or create other ways of moving around, such as with skates, skis, scuba equipment, swim fins, or moving down the street in a wheelchair or a walker
D410	Changing basic body position	Getting into and out of a body position and moving from one location to another, such as rolling from one side to the other, sitting, standing, getting up out of a chair to lie down on a bed, and getting into and out of positions of kneeling or squatting
B230	Hearing functions	Sensory functions relating to sensing the presence of sounds and discriminating the location, pitch, loudness and quality of sounds
D240	Handling stress and other psychological demands	Carrying out simple or complex and coordinated actions to manage and control the psychological demands required to carry out tasks demanding significant responsibilities and involving stress, distraction, or crises, such as taking exams, driving a vehicle during heavy traffic, putting on clothes when hurried by parents, finishing a task within a time-limit or taking care of a large group of children
None		Does not belong to any of the ICF categories in the list

Table A.1: ICF-17 set

Appendix B

Annotation Guidelines

B.1 Old Categories (9)

B1300 Energy level Mental functions that produce vigour and stamina.

B140 Attention functions Specific mental functions of focusing on an external stimulus or internal experience for the required period of time.

Inclusions: functions of sustaining attention, shifting attention, dividing attention, sharing attention; concentration; distractibility

Exclusions: consciousness functions; energy and drive functions; sleep functions; memory functions; psychomotor functions; perceptual functions.

B152 Emotional functions Specific mental functions related to the feeling and affective components of the processes of the mind.

Inclusions: functions of appropriateness of emotion, regulation and range of emotion; affect; sadness, happiness, love, fear, anger, hate, tension, anxiety, joy, sorrow; lability of emotion; flattening of affect.

Exclusions: temperament and personality functions; energy and drive functions.

B440 Respiration functions Functions of inhaling air into the lungs, the exchange of gases between air and blood, and exhaling air.

Inclusions: functions of respiration rate, rhythm and depth; impairments such as apnoea, hyperventilation, irregular respiration, paradoxical respiration, and brochial spasm, and as in pulmonary emphysema; upper pulmonary obstruction, reduction in airflow through upper and lower airways.

Exclusions: respiratory muscle functions; additional respiratory functions; exercise tolerance functions.

B455 Exercise tolerance functions Functions related to respiratory and cardiovascular capacity as required for enduring physical exertion.

Inclusions: functions of physical endurance, aerobic capacity, stamina and fatiguability.

Exclusions: functions of the cardiovascular system; haematological system functions; respiration functions; respiratory muscle functions; additional respiratory functions.

B530 Weight maintenance functions Functions of maintaining appropriate body weight, including weight gain during the developmental period.

Inclusions: functions of maintenance of acceptable Body Mass Index (BMI); and impairments such as underweight, cachexia, wasting, overweight, emaciation and such as in primary and secondary obesity.

Exclusions: assimilation functions; general metabolic functions; endocrine gland functions.

D450 Walking Moving along a surface on foot, step by step, so that one foot is always on the ground, such as when strolling, sauntering, walking forwards, backwards, or sideways.

Inclusions: walking short or long distances; walking on different surfaces; walking around obstacles.

Exclusions: transferring oneself; moving around.

D550 Eating Indicating need for, and carrying out the coordinated tasks and actions of eating food that has been served, bringing it to the mouth and consuming it in culturally acceptable ways, cutting or breaking food into pieces, opening bottles and cans, using eating implements, having meals, feasting or dining.

Exclusion: drinking.

D840-D859 Work and employment

Apprenticeship (work preparation)

Engaging in programmes related to preparation for employment, such as performing the tasks required of an apprenticeship, internship, articling and in service training.

Exclusion: vocational training.

Acquiring, keeping and terminating a job

Seeking, finding and choosing employment, being hired and accepting employment, maintaining and advancing through a job, trade, occupation or profession, and leaving

a job in an appropriate manner.

Inclusions: seeking employment; preparing a resume or curriculum vitae; contacting employers and preparing interviews; maintaining a job; monitoring one's own work performance; giving notice; and terminating a job.

Remunerative employment
Engaging in all aspects of work, as an occupation, trade, profession or other form of employment, for payment, as an employee, full or part time, or self-employed, such as seeking employment and getting a job, doing the required tasks of the job, attending work on time as required, supervising other workers or being supervised, and performing required tasks alone or in groups.

Inclusions: self-employment, part-time and full-time employment.

Non-remunerative employment
Engaging in all aspects of work in which pay is not provided, full-time or part-time, including organized work activities, doing the required tasks of the job, attending work on time as required, supervising other workers or being supervised, and performing required tasks alone or in groups, such as volunteer work, charity work, working for a community or religious group without remuneration, working around the home without remuneration.

Exclusion: Domestic Life.

Work and employment, other specified and unspecified

B.2 New Categories (8)

B280 Sensations of pain Sensation of unpleasant feeling indicating potential or actual damage to some body structure.

Inclusions: sensations of generalized or localized pain, in one or more body part, pain in a dermatome, stabbing pain, burning pain, dull pain, aching pain; impairments such as myalgia, analgesia and hyperalgesia.

B134 Sleep functions General mental functions of periodic, reversible and selective physical and mental disengagement from one's immediate environment accompanied by characteristic physiological changes.

Inclusions: functions of amount of sleeping, and onset, maintenance and quality of sleep; functions involving the sleep cycle, such as in insomnia, hypersomnia and narcolepsy.

Exclusions: consciousness functions; energy and drive functions; attention functions; psychomotor functions.

D760 Family relationships Creating and maintaining kinship relationships, such as with members of the nuclear family, extended family, foster and adopted family and step relationships, more distant relationships such as second cousins, or legal guardians.

Inclusions: parent-child and child-parent relationships, sibling and extended family relationships.

B164 Higher-level cognitive functions Specific mental functions especially dependent on the frontal lobes of the brain, including complex goal-directed behaviours such as decision-making, abstract thinking, planning and carrying out plans, mental flexibility, and deciding which behaviours are appropriate under what circumstances; often called executive functions.

Inclusions: functions of abstraction and organization of ideas; time management, insight and judgement; concept formation, categorization and cognitive flexibility.

Exclusions: memory functions; thought functions; mental functions of language; calculation functions.

D465 Moving around using equipment Moving the whole body from place to place, on any surface or space, by using specific devices designed to facilitate moving or create other ways of moving around, such as with skates, skis, scuba equipment, swim fins, or moving down the street in a wheelchair or a walker.

Exclusions: transferring oneself; walking; moving around; using transportation; driving.

D410 Changing basic body position Getting into and out of a body position and moving from one location to another, such as rolling from one side to the other, sitting, standing, getting up out of a chair to lie down on a bed, and getting into and out of positions of kneeling or squatting.

Inclusion: changing body position from lying down, from squatting or kneeling, from sitting or standing, bending and shifting the body's centre of gravity.

Exclusion: transferring oneself.

B230 Hearing functions Sensory functions relating to sensing the presence of sounds and discriminating the location, pitch, loudness and quality of sounds.

Inclusions: functions of hearing, auditory discrimination, localization of sound source, lateralization of sound, speech discrimination; impairments such as deafness, hearing impairment and hearing loss.

Exclusions: perceptual functions and mental functions of language.

D240 Handling stress and other psychological demands Carrying out simple or complex and coordinated actions to manage and control the psychological demands required to carry out tasks demanding significant responsibilities and involving stress, distraction, or crises, such as taking exams, driving a vehicle during heavy traffic, putting on clothes when hurried by parents, finishing a task within a time-limit or taking care of a large group of children.

Inclusions: handling responsibilities; handling stress and crisis.

Appendix C

Prompts

Prompt clean up transcription

The cleaning prompt went through several iterations. Initially, the model assigned speaker labels based solely on the diarisation output from `Pyannote`, which did not always produce correct results. The model was therefore instructed to determine speaker labels independently from only the conversational content. Confidence scores were also added (one for the speaker assignment and one for the text coherence) along with explicit criteria defining what a low score means and what conditions must be met for a high score, making the scores more reliable. The final prompt was:

```
   You are a helpful assistant correcting the speaker labels in a diarized
interview transcript in Dutch.
The transcript is a chunk of a JSON object where each turn has a speaker
label, such as SPEAKER_00 or SPEAKER_01.
First, determine which speaker label corresponds to which person by looking
at the conversation as a whole.
{HP} is the person who asks the questions most of the time, and patient is
the person who answers most of the time.
A proxy may be present and should be labelled as 'Proxy_patient'.
Any other speakers should be labelled as 'Third Party'.
Once you have determined which speaker label corresponds to which person,
make a new field to each turn entiteld "speaker_llm" with the speaker label
that you think it is.
If you can't identify the speaker_llm, but only if you cannot identify them,
assign it the value None.
Do not change anything else.  Keep the structure and text exactly as they
are.
Additionally, add three extra fields to each turn:
- 'confidence_score_speaker':  indicates how certain you are about the assigned
speaker label.  This ranges from 0 to 1.  A low score indicates that there
might be something wrong with the speaker label, or that you are not confident
that it is the correct label.  A high score indicates that the speaker label
is likely correct.
- 'confidence_score_text':  indicates how certain you are that the text is
a correct grammatical, coherent sentence which follows logically based on
```

```
the previous 10 utterances.  A high score indicates a coherent sentence in
Dutch.
- 'notes', you can write any notes about the transcription, such as the presence
of noise or the presence of a proxy or third party.  You can also comment
on why you assigned certain confidence scores here.
Always fill in the 'confidence_score_speaker' and 'confidence_score_text'
field, but only fill in the 'notes' field if there is something worth noting.
If a word or phrase is continuously repeated in one turn or across multiple
turns, it could mean that the Whisper model got stuck.  This would result
in a low confidence score for the text.
Only return the corrected JSON object and nothing else.
```

Prompt assignment ICF-category

Later iterations of the prompt required explicit instructions on the output format. Without these, the model frequently returned inconsistent or malformed output. The model was therefore instructed to return a JSON object. Additionally, the model had to be restricted to the previously defined category list, since it would otherwise generate its own thematic labels. The final prompt is shown below.

```
   You are an annotation assistant.
You will receive turns from a conversation in Dutch between a healthcare
professional and an elderly person.
The turns are given in a JSON format.
Your task is to assign an ICF category to each turn based on the categories
in the detailed definitions.
You can fill in more than one category to each turn if applicable.
If you fill in None, you can NOT assign another category.
Only fill in the category if the turn fits into the description of that category
as listed in the detailed definitions.
ONLY assign a category if it is clearly and explicitly discussed in that
turn.
If it is unclear from the turn what category is discussed, do NOT fill in
a category.
If you are unsure, assign None.  It is better to assign None than t guess.
Do not assign a category to turns where only an introduction is given, unless
the turn clearly specifies a category.
ONLY use categories from the list above.  Do NOT invent or paraphrase categories.
Return the same number of turns as the input, with an ICF_category field added
to each turn.
The ICF_category field should contain a list of the applicable ICF categories,
and NO more.
There should NOT be more or less turns in the output than in the input.  Do
NOT add or remove turns.
Return only a valid JSON object with double quotes around all keys and string
values.  Do NOT use single quotes.  Use colons as delimiter.
Do not include explanations or markdown.
```

Prompt assignment ICF-level

This prompt was separated from the category annotation step because combining the two tasks caused the model to assign category labels instead of levels, or to return non-numerical output. The model was therefore explicitly instructed to only assign a number between 0 and 4, and only when a level was clearly specified in that turn. It was also instructed to return the level without any explanation. The final prompt is shown below.

```
   You are an annotation assistant.
You will receive turns from a conversation in Dutch between a healthcare
professional and an elderly person.
The turns are given in a JSON format.  Each turn may have no (None), one,
or more ICF categories assigned.
Your task is to find the impairment level for each ICF category that appears
in the conversation.
Go through the turns one by one.  When you encounter a turn with an ICF category,
look at the surrounding turns to find where the level of that category is
discussed.
Only assign a level if the turn is clearly discussing the level of function
for that specific category, not just any level or number mentioned in the
conversation.
If a turn mentions a level but it is about a different topic or category,
do not assign it.
The impairment level is a number from 0 to 4, where 0 means total impairment
and functioning is limited,and 4 means no impairment at all.
Only assign a level if it is explicitly and clearly stated in the conversation.
If no level is discussed, assign None.  It is better to assign None than
to guess.
Do NOT assign 0 by default.  0 is a valid level only if total impairment
is explicitly discussed.
If a turn discusses a good level of a previously discussed category, but
has not had that ICF category assigned, also update the ICF_category field
with the corresponding category.
The ICF_level field must contain a list with the same number of elements as
the ICF_category field.
Each level value should correspond to the category at the same position in
the ICF_category list.
If a ICF category is assigned, but no level is discussed, mark that element
in the list of the ICF_level with None.
Do not assign a level to turns where only a question is asked, unless the
answer is in the same turn.
Do not assign a level to turns where only an introduction is given, unless
the turn clearly specifies a category with a level.
Only use a number (0, 1, 2, 3, or 4) or None for ICF_level.  Do not add any
text or explanation.
Return the same number of turns as the input, with ICF_level updated for each
turn.
```

```
There should NOT be more or less turns in the output than in the input.  Do
NOT add or remove turns.
Return only a valid JSON object with double quotes around all keys and string
values.  Do NOT use single quotes.  Use colons as delimiter.
Do not include explanations or markdown.
```

Prompt assignment relative time

As with the previous prompts, the model had to be explicitly instructed to return a JSON object. It was also restricted to only the predefined relative time labels or "None", and instructed to only assign a time point if a level was specified in that turn. The final prompt is shown below.

```
   You are an annotation assistant.
You will receive turns from a conversation in Dutch between a healthcare
professional and an elderly person.
The turns are given in a JSON format.  Each turn may have ICF categories
and ICF levels assigned.
Your task is to determine the relative time for each turn where an ICF level
has been assigned.
Go through the turns one by one.
For each level in ICF_level assign a relative time.
Only assign a relative time if the turn has an ICF level that is not None.
If NO ICF level has been filled in at all, do NOT fill in the relative time
but mark it with [None].
The relative_time field must contain a list with the same number of elements
as the ICF_level field.
Each relative time value should correspond to the level at the same position
in the ICF_level list.
If the level has been filled out for one category, but not for another, fill
in the relative time for the level that has been specified.  Fill in None
for elements that have not been specified.
Choose only from the following options:
- None:  when no ICF level was determined for that turn.
- past:  something that happened in the past and is no longer happening.
- present:  something that is currently happening, including routines and
ongoing situations.
- future:  something that has not happened yet but will happen.
Every turn must have a relative_time field with a list.  This field should
be equally long as the ICF_levels list.
Do not add or remove turns.  There should NOT be more or less turns in the
output than in the input.
Return only a valid JSON object with double quotes around all keys and string
values.  Do NOT use single quotes.  Use colons as delimiter.
Do not include explanations or markdown.
```

Modified prompt - Error Analysis: length

For the error analysis, the level annotation prompt was shortened to test whether a more concise prompt improved the annotations. The revised prompt is shown below.

```
   You are an annotation assistant.
You will receive turns from a conversation in Dutch between a healthcare
professional and an elderly person.
The turns are given in a JSON format.  Each turn may have no (None), one,
or more ICF categories assigned.
Your task is to find the impairment level for each ICF category that appears
in the conversation.
The impairment level is a number from 0 to 4, where 0 means total impairment
and functioning is limited, and 4 means no impairment at all.
If no level is discussed, assign None.  It is better to assign None than
to guess.
Only use a number (0, 1, 2, 3, or 4) or None for ICF_level.  Do not add any
text or explanation.
Only assign a level if the turn is clearly discussing the level of function
for that specific category, not just any level or number mentioned in the
conversation.
The ICF_level field must contain a list with the same number of elements as
the ICF_category field.
Each level value should correspond to the category at the same position in
the ICF_category list.
Return the same number of turns as the input, with ICF_level updated for each
turn.
There should NOT be more or less turns in the output than in the input.  Do
NOT add or remove turns.
Return only a valid JSON object with double quotes around all keys and string
values.  Do NOT use single quotes.  Use colons as delimiter.
Do not include explanations or markdown.
Do NOT assign 0 by default.  0 is a valid level only if total impairment
is explicitly discussed.
```

Modified prompt - Error analysis: repetition

This prompt was developed for the error analysis to address the over-annotation of repeated sentences. It builds on the shortened prompt above.

```
   You are an annotation assistant.
You will receive turns from a conversation in Dutch between a healthcare
professional and an elderly person.
The turns are given in a JSON format.  Each turn may have no (None), one,
or more ICF categories assigned.
Your task is to find the impairment level for each ICF category that appears
in the conversation.
```

```
The impairment level is a number from 0 to 4, where 0 means total impairment
and functioning is limited, and 4 means no impairment at all.
If no level is discussed, assign None.  It is better to assign None than
to guess.
Only use a number (0, 1, 2, 3, or 4) or None for ICF_level.  Do not add any
text or explanation.
Only assign a level if the turn is clearly discussing the level of function
for that specific category, not just any level or number mentioned in the
conversation.
If the text in a turn or across turns is constantly repeating itself, only
assign the level to the first two repetitions and assign None to the remaining
turns that contain the same repetition.
The ICF_level field must contain a list with the same number of elements as
the ICF_category field.
Each level value should correspond to the category at the same position in
the ICF_category list.
Return the same number of turns as the input, with ICF_level updated for each
turn.
There should NOT be more or less turns in the output than in the input.  Do
NOT add or remove turns.
Return only a valid JSON object with double quotes around all keys and string
values.  Do NOT use single quotes.  Use colons as delimiter.
Do not include explanations or markdown.
Do NOT assign 0 by default.  0 is a valid level only if total impairment
is explicitly discussed.
```

Appendix D

Few-shot examples

Few-shot examples ICF-category

```
{
    "sentence": "Ik wandel niet meer heel vaak alleen, omdat ik
       bang ben om te vallen. Meestal ga ik dan op pad met mijn
       zus.",
    "ICF_category": ["D450 Walking", "D760 Family relationships"]
},
{
    "sentence": "Kunt u nog goed horen als iemand u een vraag
       stelt?",
    "ICF_category": ["B230 Hearing functions"]
},
{
    "sentence": "Ik bereid mijn eigen maaltijden voor.",
    "ICF_category": ["None"]
}
```

Few-shot examples ICF-level

```
{
    "sentence": "Ik wandel niet meer heel vaak alleen, omdat ik
       bang ben om te vallen. Meestal ga ik dan op pad met mijn
       zus.",
    "ICF_category": ["D450 Walking", "D760 Family relationships"]
       ,
    "ICF_level": ["1", "4"]
},
{
    "sentence": "Kunt u nog goed horen als iemand u een vraag
       stelt?",
    "ICF_category": ["B230 Hearing functions"],
    "ICF_level": ["None"]
},
{
    "sentence": "Ik slaap nooit meer goed. ",
    "ICF_category": ["B134 Sleep functions"],
```

```
    "ICF_level": ["0"]
},
{
    "sentence": "Ik heb nooit problemen met horen.",
    "ICF_category": ["B230 Hearing functions"],
    "ICF_level": ["4"]
},
{
    "sentence": "Ik bereid mijn eigen maaltijden voor.",
    "ICF_category": ["None"],
    "ICF_level": ["None"]
},
{
    "sentence": "Ik loop niet meer heel goed de trap op, maar ik
       slaap boven.",
    "ICF_category": ["D450 Walking", "B134 Sleep functions"],
    "ICF_level": ["3", "None"]
},
{
    "sentence": "U ervaart dus heel veel pijn, helemaal als u van
        uw buik naar uw rug rolt.",
    "ICF_category": ["B280 Sensations of pain", "D410 Changing
       basic body position"],
    "ICF_level": ["0", "3"]
},
{
    "sentence": "Het is vandaag 7 maart en ik ben bij
       correspondentnummer 999000.",
    "ICF_category": ["None"],
    "ICF_level": ["None"]
}
```

Few-shot examples relative time

```
{
    "sentence": "Voordat ik was gevallen liep ik altijd zelf naar
        de supermarkt. Dat doet nog steeds heel veel pijn.",
    "relative_time": ["past", "present"],
    "ICF_category": ["D450 Walking", "B280 Sensations of pain"],
    "ICF_level": ["4", "0"]
},
{
    "sentence": "Dus u eet elke dag nog een kommetje soep
       helemaal zelf. Gaat u daarna een blokje om?",
    "relative_time": ["present", "None"],
    "ICF_category": ["D550 Eating", "D450 Walking"],
    "ICF_level": ["4", "None"]
},
{
    "sentence": "Morgen komt mijn dochter langs en ik heb haar al
        heel lang niet meer gezien.",
    "relative_time": ["future"],
    "ICF_category": ["D760 Family relationships"],
```

```
    "ICF_level": ["2"]
},
{
    "sentence": "Heeft u wel eens moeite met ademhalen?",
    "relative_time": ["None"],
    "ICF_category": ["B440 Respiration functions"],
    "ICF_level": ["None"]
},
{
    "sentence": "Dan gaan we nu beginnen.",
    "relative_time": ["None"],
    "ICF_category": ["None"],
    "ICF_level": ["None"]
}
```

Added examples - Error analysis: negations

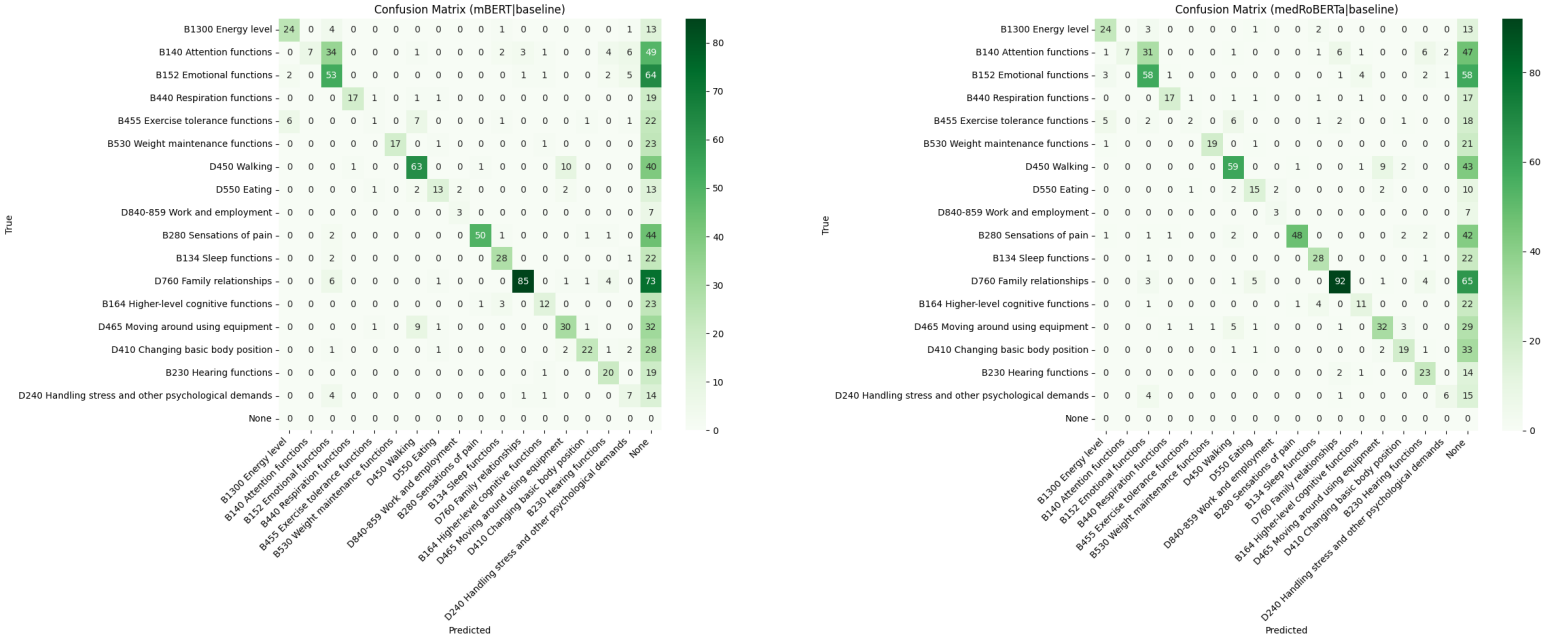
```
{
        "sentence": "Ik heb geen pijn meer",
        "ICF_category": ["B280 Sensations of pain"],
        "ICF_level": ["4"]
},
{
        "sentence": "En er zijn geen psychiatrische ziektes
           aanwezig.",
        "ICF_category": ["B152 Emotional functions"],
        "ICF_level": ["4"]
}
```

Added examples - Error analysis: common categories

```
{
    "sentence": "Mijn dochter komt elke week op de thee en dan
       kletsen we over de kleinkinderen",
    "ICF_category": ["D760 Family relationships"],
    "ICF_level": ["4"]
},
{
    "sentence": "En er zijn geen psychiatrische ziektes aanwezig
       .",
    "ICF_category": ["B152 Emotional functions"],
    "ICF_level": ["4"]
}
```

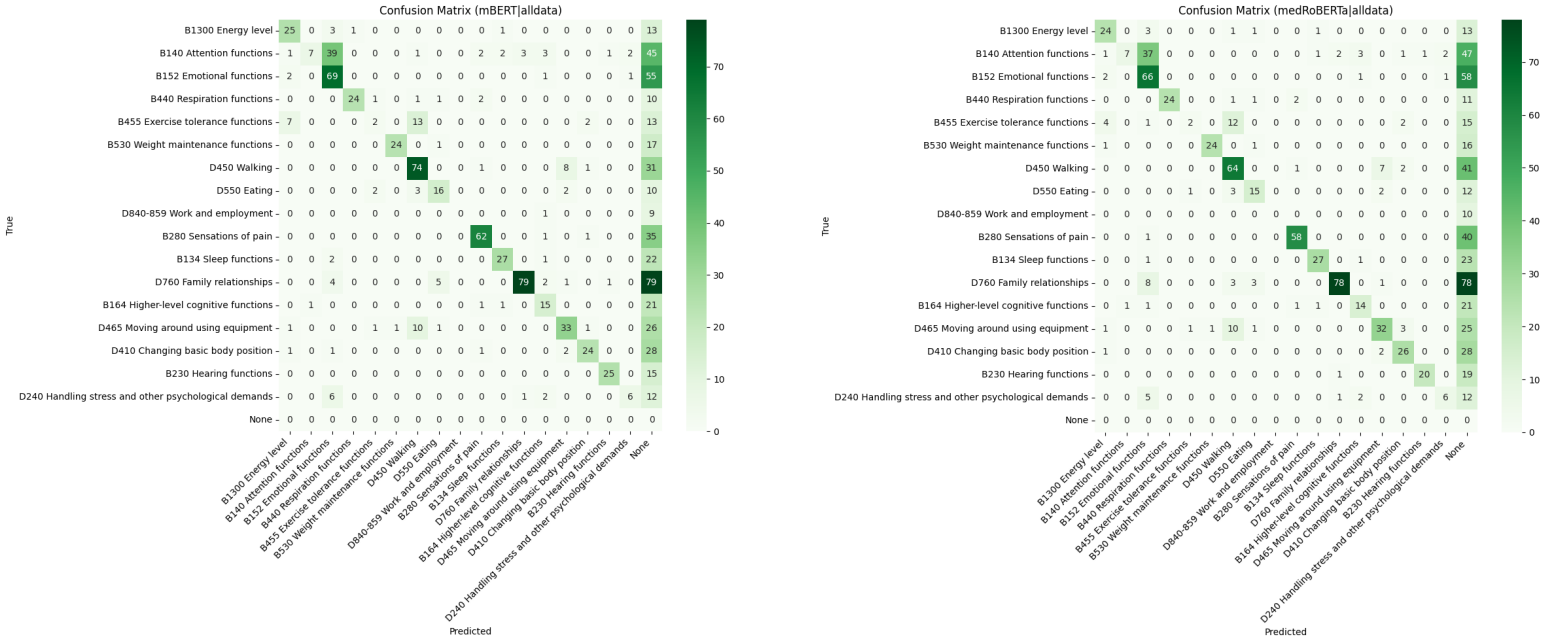
Appendix E

Additional Results



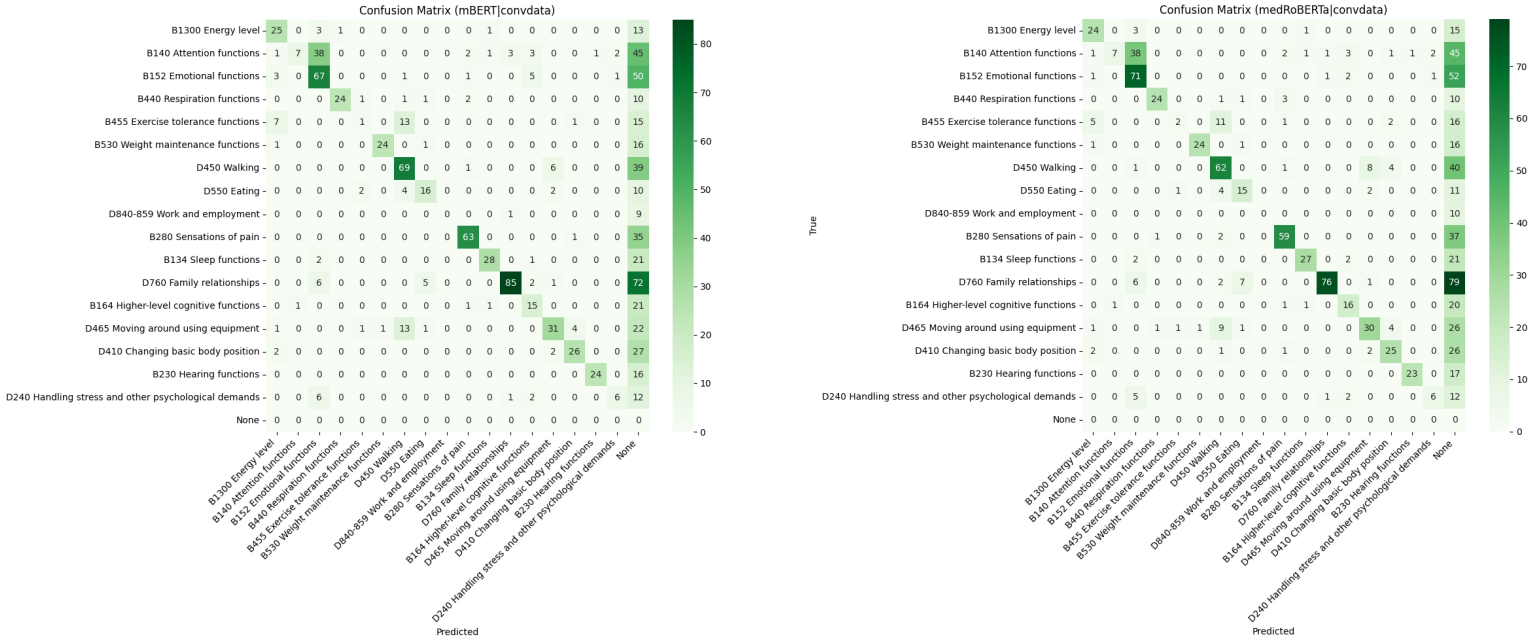
(a) mBERT (b) medRoBERTa

Figure E.1: Confusion matrices for the category prediction task of Baseline in strict filtering and strict evaluation.



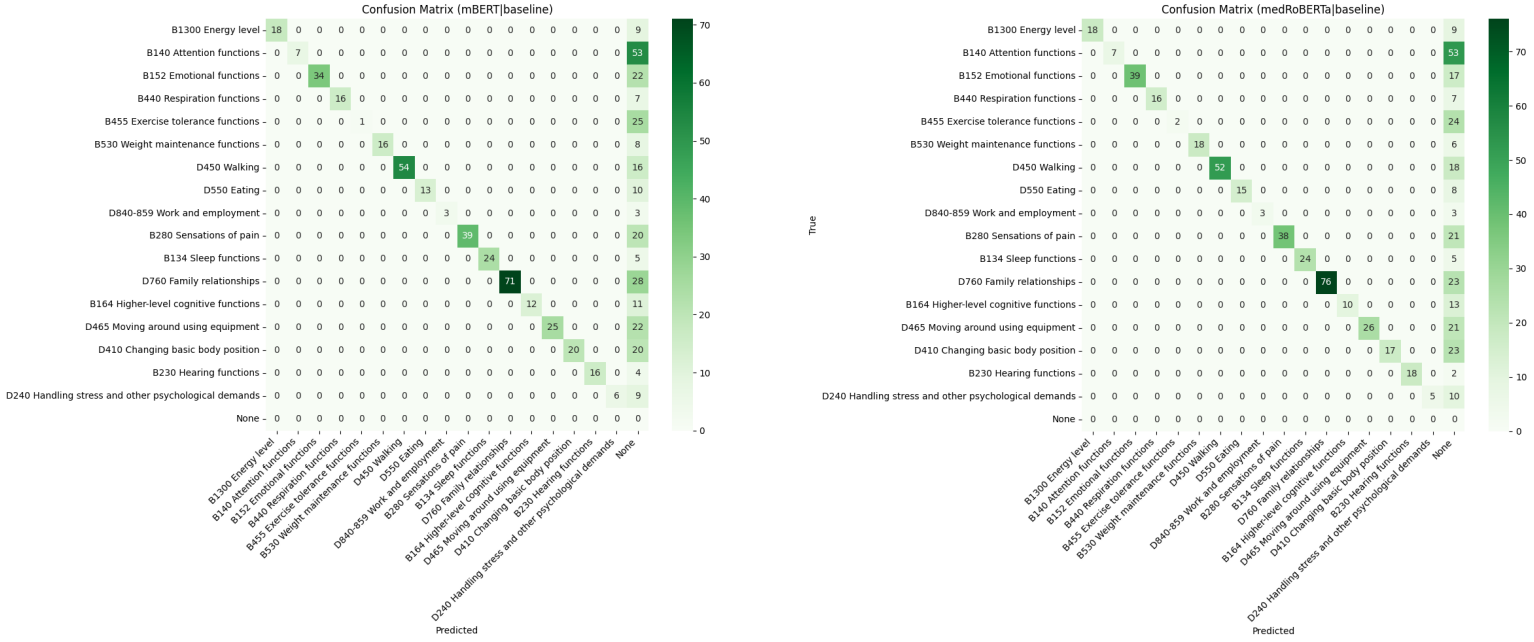
(a) mBERT (b) medRoBERTa

Figure E.2: Confusion matrices for the category prediction task of Finetuning (All Data) in strict filtering and strict evaluation.



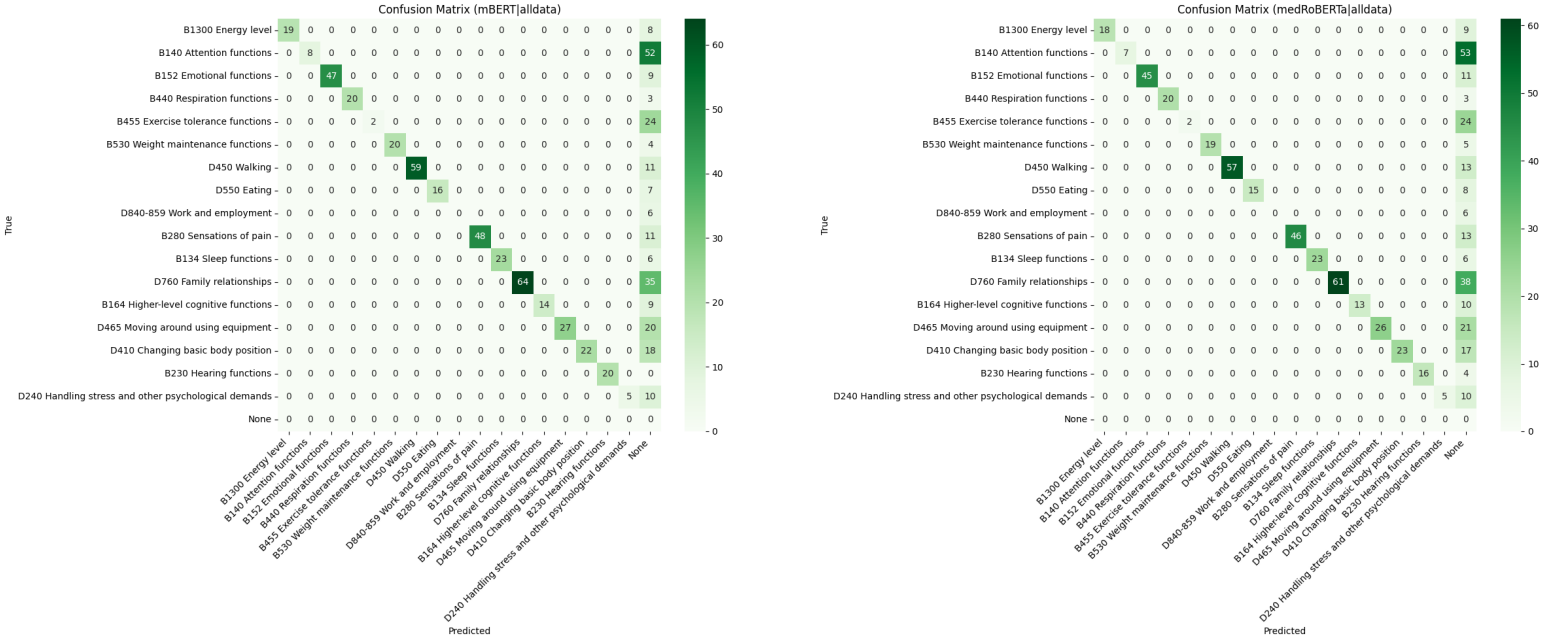
(a) mBERT (b) medRoBERTa

Figure E.3: Confusion matrices for the category prediction task of Finetuning (Conversations Only) in strict filtering and strict evaluation.



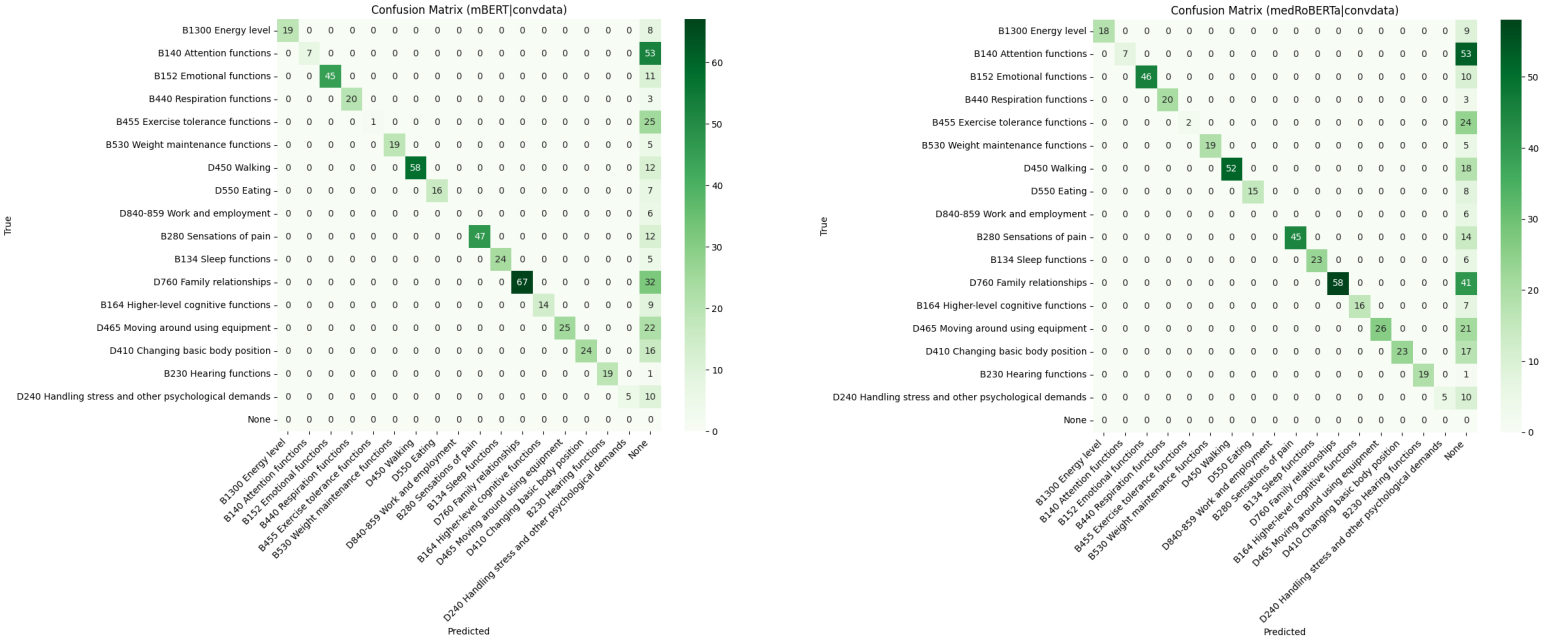
(a) mBERT (b) medRoBERTa

Figure E.4: Confusion matrices for the category prediction task of Baseline in strict filtering and lenient evaluation.



(a) mBERT (b) medRoBERTa

Figure E.5: Confusion matrices for the category prediction task of Finetuning (All Data) in strict filtering and lenient evaluation.

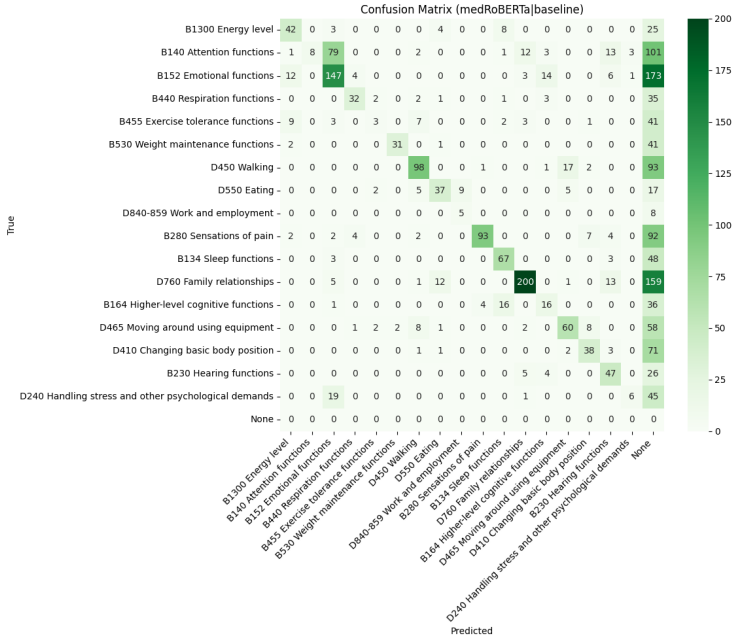


(a) mBERT (b) medRoBERTa

Figure E.6: Confusion matrices for the category prediction task of Finetuning (Conversations Only) in strict filtering and lenient evaluation.

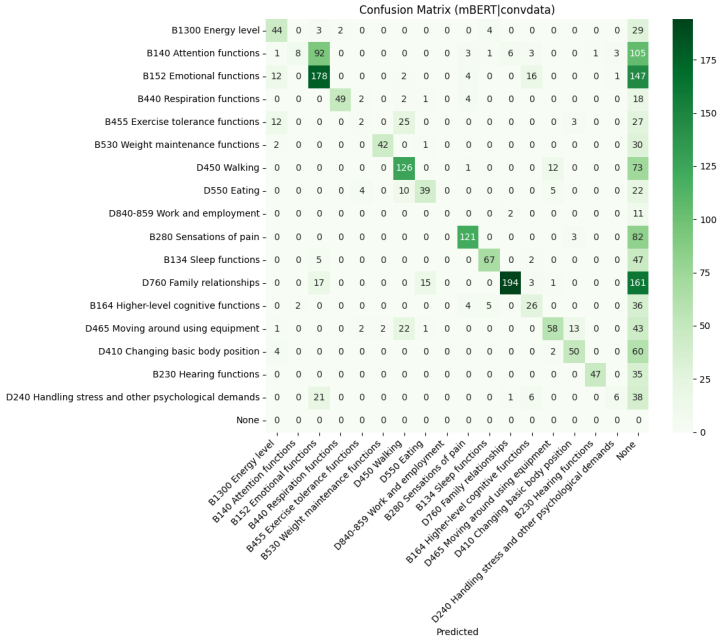


(a) mBERT

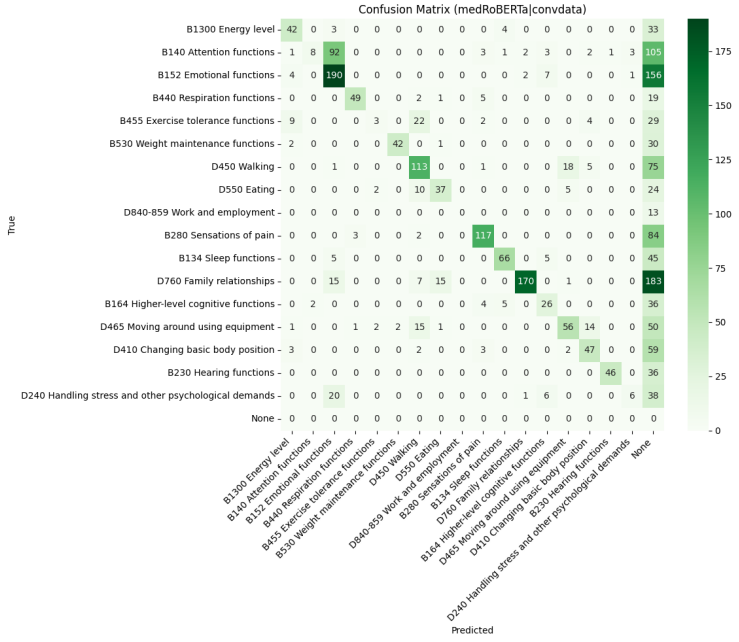


(b) medRoBERTa

Figure E.7: Confusion matrices for the category prediction task of Baseline in lenient filtering and strict evaluation.

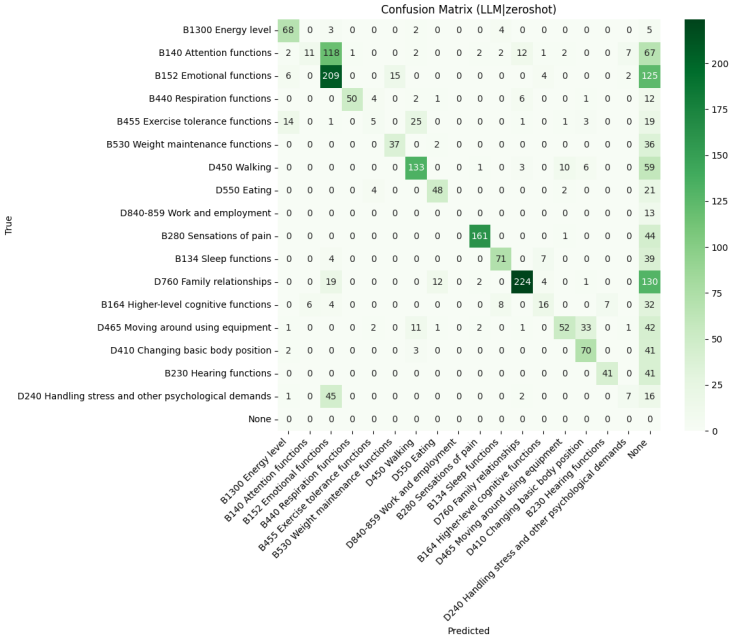


(a) mBERT

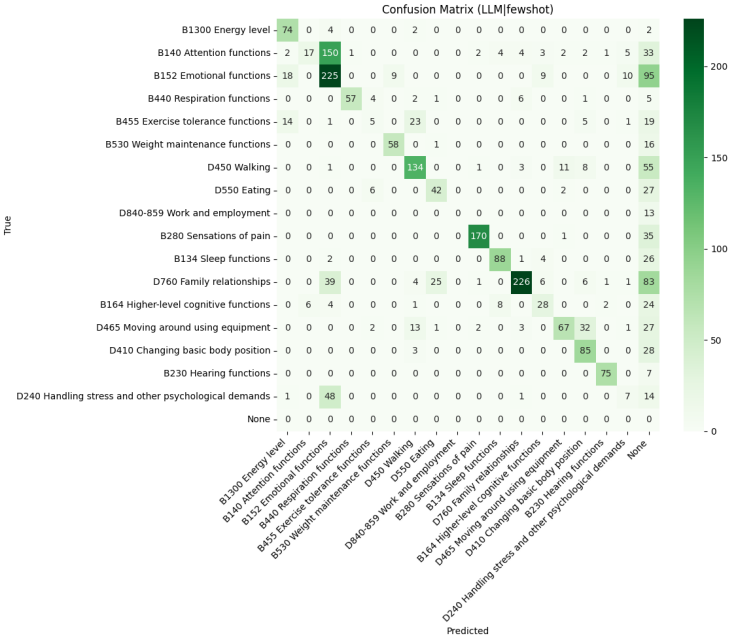


(b) medRoBERTa

Figure E.8: Confusion matrices for the category prediction task of Finetuning (Conversations Only) in lenient filtering and strict evaluation.

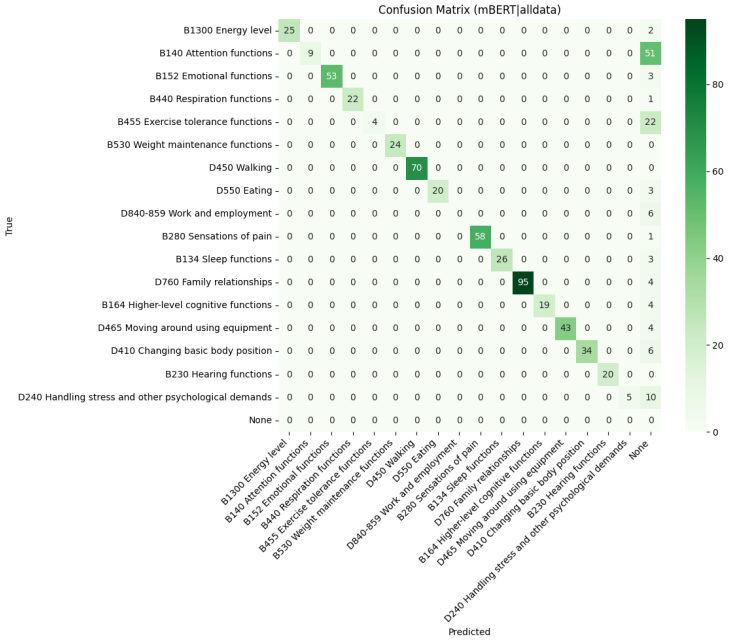


(a) Zero-shot

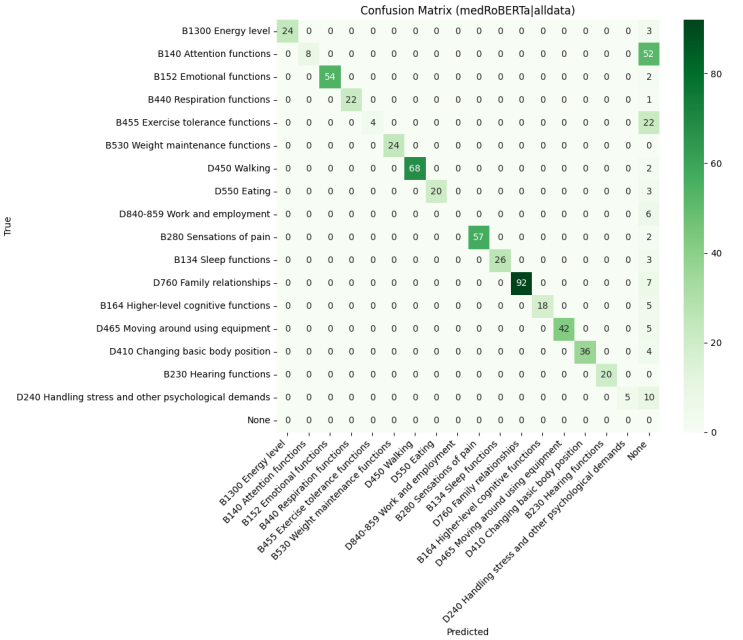


(b) Few-shot

Figure E.9: Confusion matrices for the category prediction task of Prompt-Based Generation in lenient filtering and strict evaluation.

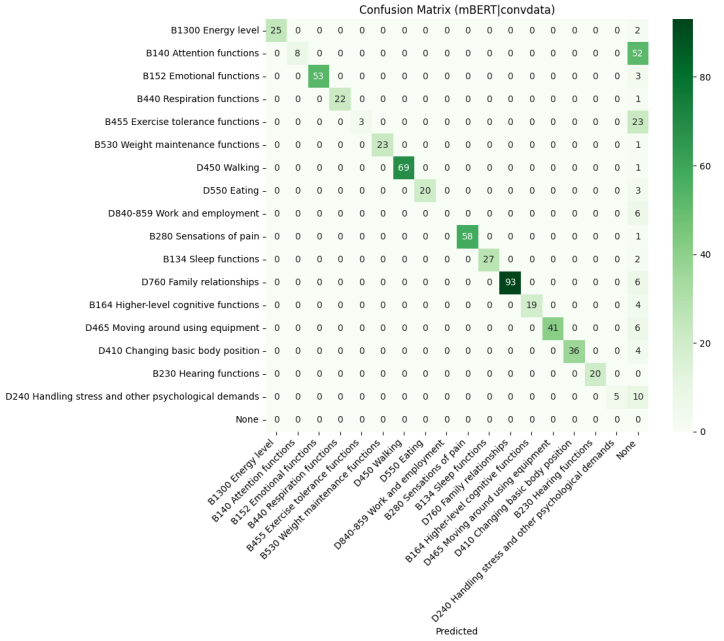


(a) mBERT

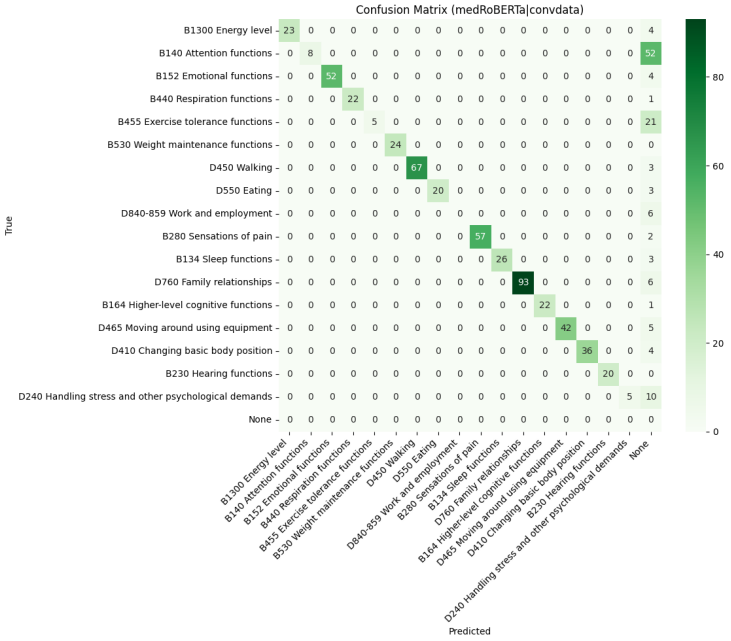


(b) medRoBERTa

Figure E.10: Confusion matrices for the category prediction task of Finetuning (All Data) in lenient filtering and lenient evaluation.

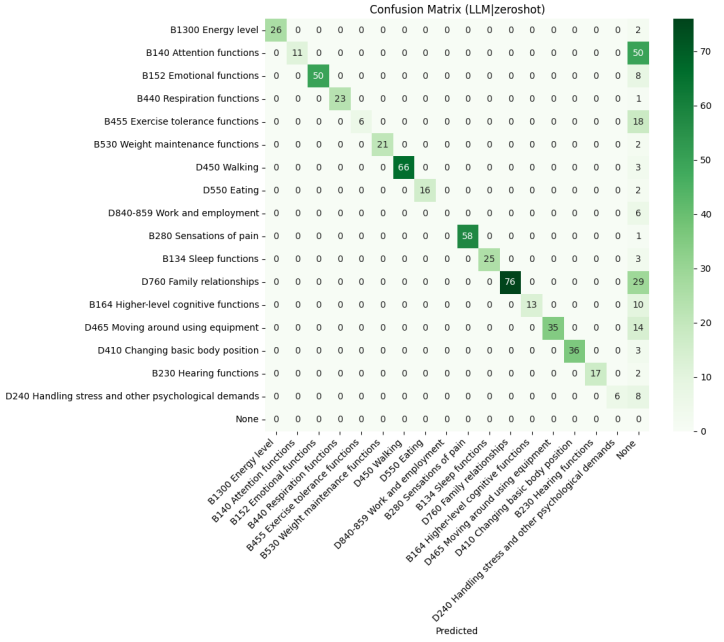


(a) mBERT

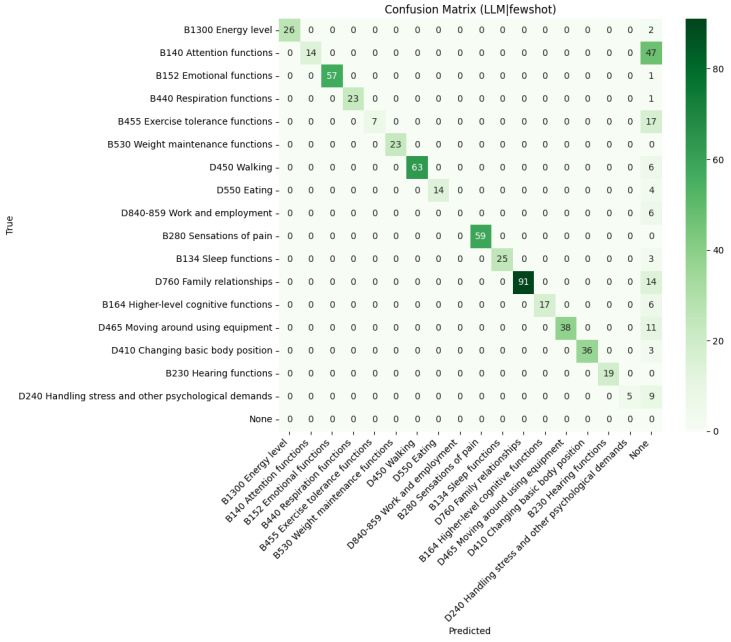


(b) medRoBERTa

Figure E.11: Confusion matrices for the category prediction task of Fineturning (Conversations Only) in lenient filtering and lenient evaluation.

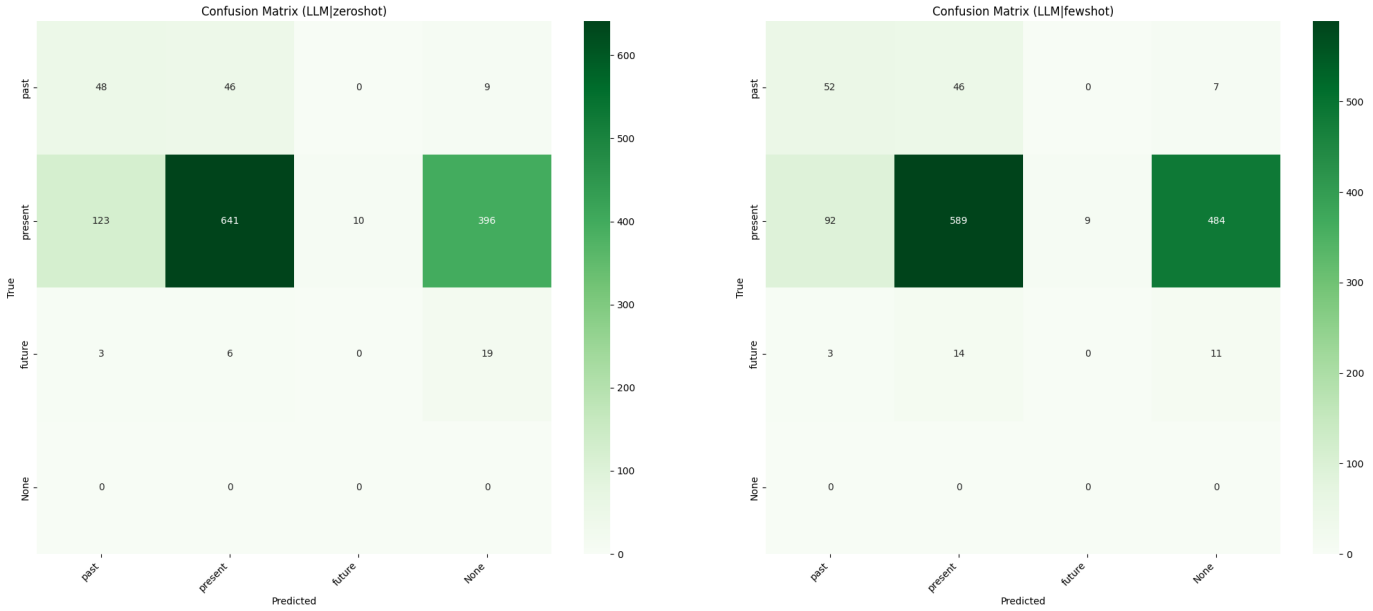


(a) Zero-shot



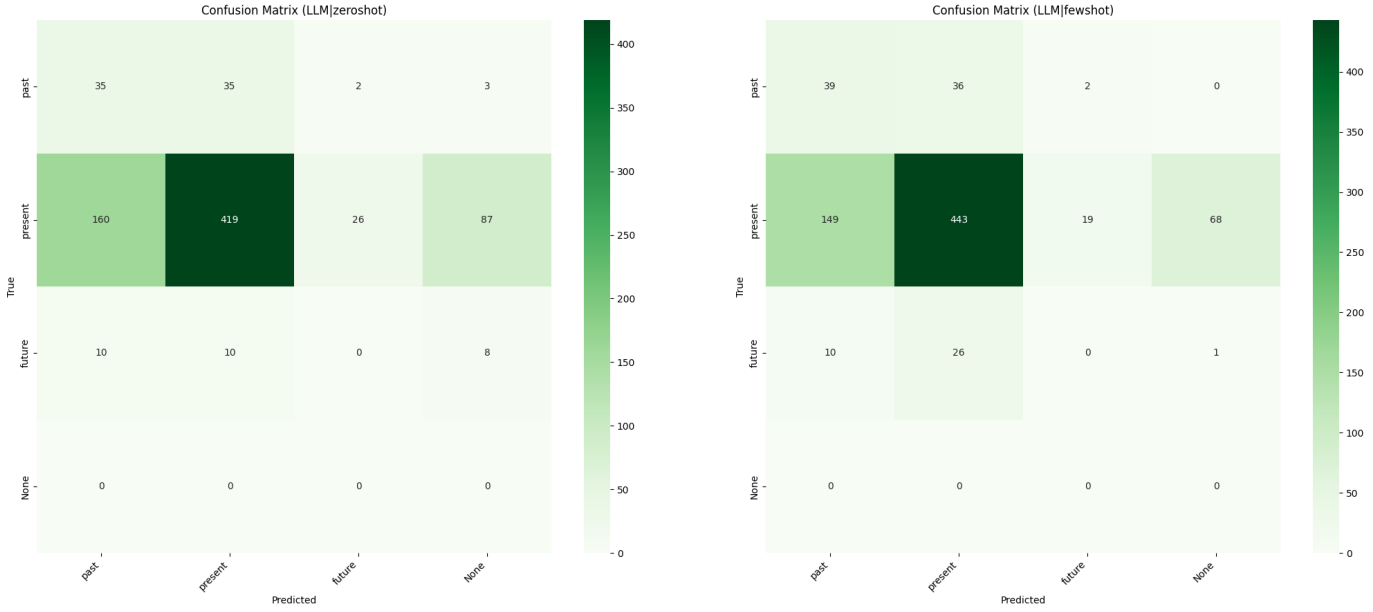
(b) Few-shot

Figure E.12: Confusion matrices for the category prediction task of Prompt-Based Generation in lenient filtering and lenient evaluation.



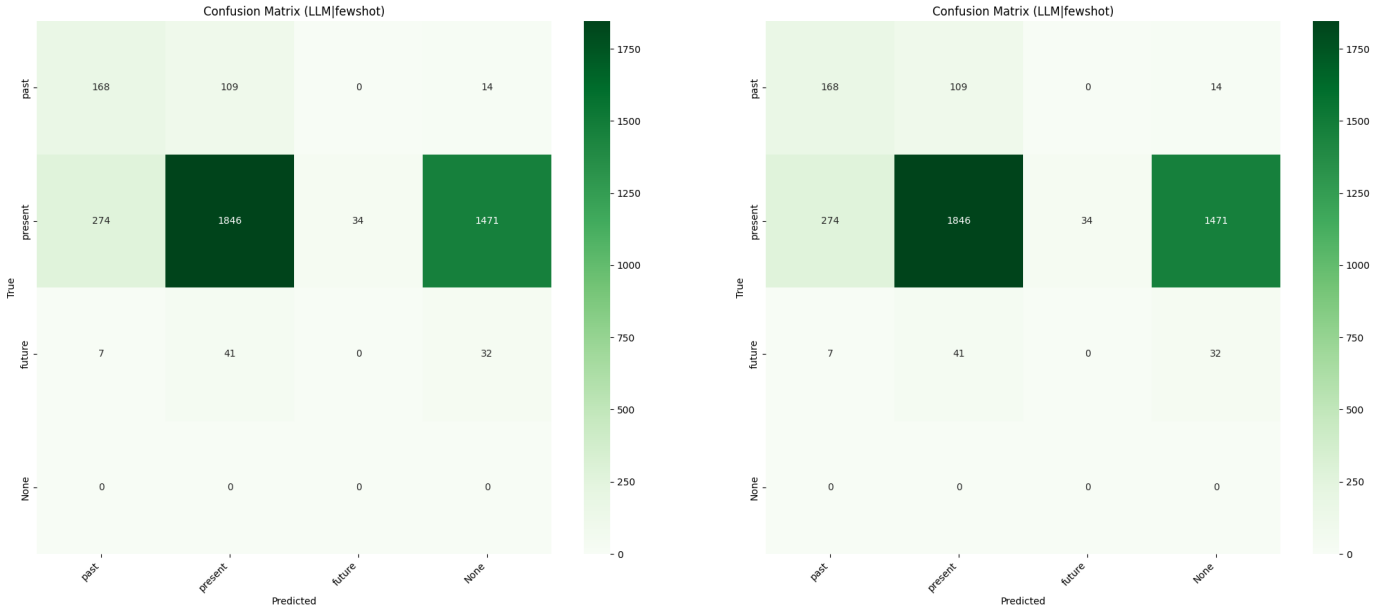
(a) Zero-shot (b) Few-shot

Figure E.13: Confusion matrices for the category prediction task of Prompt-Based Generation in strict filtering and strict evaluation.



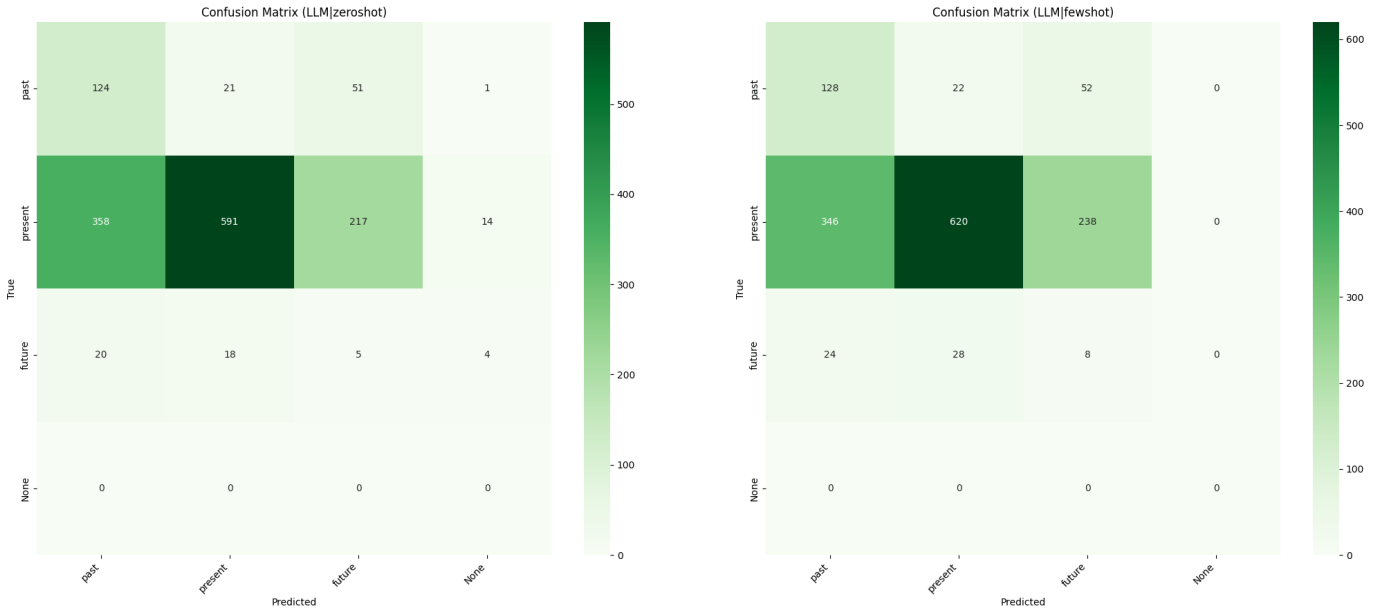
(a) Zero-shot (b) Few-shot

Figure E.14: Confusion matrices for the category prediction task of Prompt-Based Generation in strict filtering and lenient evaluation.



(a) Zero-shot (b) Few-shot

Figure E.15: Confusion matrices for the category prediction task of Prompt-Based Generation in lenient filtering and strict evaluation.



(a) Zero-shot (b) Few-shot

Figure E.16: Confusion matrices for the category prediction task of Prompt-Based Generation in lenient filtering and lenient evaluation.

References

A-PROOF Consortium. A-proof. `https://aproof.ai/`, 2025. Accessed: 2026-06-19.

S. Abdi, A. Spann, J. Borilovic, L. de Witte, and M. Hawley. Understanding the care and support needs of older people: a scoping review and categorisation using the who international classification of functioning, disability and health framework (icf). *BMC geriatrics*, 19(1):195, 2019.

W. Ali, M. Saleem, B. Yao, A. Hogan, and A.-C. N. Ngomo. A survey of rdf stores & sparql engines for querying knowledge graphs. *The VLDB Journal*, 31(3):1–26, 2022.

X. Amatriain, A. Sankar, J. Bing, P. K. Bodigutla, T. J. Hazen, and M. Kazi. Transformer models: an introduction and catalog. *arXiv preprint arXiv:2302.07730*, 2023.

S. I. Arnoult, L. Petram, and P. Vossen. Batavia asked for advice. pretrained language models for named entity recognition in historical texts. In *Proceedings of the 5th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature*, pages 21–30, 2021.

S. Bayat and G. Isik. Assessing the efficacy of lstm, transformer, and rnn architectures in text summarization. In *Proceedings of the 5th International Conference on Applied Engineering and Natural Sciences, Konya, Turkey*, pages 10–12, 2023.

K. Berg, H. Finne-Soveri, L. Gray, J. C. Henrard, J. Hirdes, N. Ikegami, G. Ljunggren, J. N. Morris, L. Paquay, L. Resnik, et al. Relationship between interrai hc and the icf: opportunity for operationalizing the icf. *BMC Health Services Research*, 9(1):47, 2009.

L. Billiet, R. M. van Nispen, S. De Baets, R. de Vries, D. Van de Velde, and H. P. van der Aa. The first step in developing an international classification of functioning, disability and health core set for vision loss: A systematic review. *Ophthalmic and Physiological Optics*, 44(2):413–425, 2024.

S. Blee, B. Rosenberg, J. M. Switchenko, R. Hianik, M. C. Thomson, M. Dixon, M. A. Bilen, and R. D. Pentz. Understanding immunotherapy terminology: an analysis of provider-patient conversations. *ImmunoMedicine*, 1(2):e1028, 2021.

H. Bredin. pyannote.audio 2.1 speaker diarization pipeline: principle, benchmark, and recipe. In *Proc. INTERSPEECH 2023*, 2023.

L. Cai, X. Mao, Y. Zhou, Z. Long, C. Wu, and M. Lan. A survey on temporal knowledge graph: Representation learning and applications. *arXiv preprint arXiv:2403.04782*, 2024.

R. M. Carvalho, A. S. Teixeira, and C. Pesquita. Building a temporal knowledge graph for electronic health records. In *CEUR Workshop Proceedings*, volume 3833. CEUR-WS, 2024.

S. Chen and P. Vossen. A cheap lunch: Synthetic annotation with reduced human effor for medical text mining. 2026.

A. K. Conduah, S. Ofoe, and D. Siaw-Marfo. Data privacy in healthcare: Global challenges and solutions. *Digital health*, 11:20552076251343959, 2025.

G. De Giacomo, M. Lenzerini, et al. Tbox and abox reasoning in expressive description logics. *KR*, 96(316-327):10, 1996.

R. Dendere, M. Hargrave, M. Lang, and S. B. Dekhil. Report: A survey and evaluation of comprehensive geriatric assessment systems for residential and community aged care. *Queensland, Australia: Centre for Health Services Research the University of Queensland*, 2023.

J. R. Dettori and D. C. Norvell. Kappa and beyond: is there agreement? *Global spine journal*, 10(4):499–501, 2020.

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, and T. Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL `https://aclanthology.org/N19-1423/`.

J. L. Fleiss, B. Levin, and M. C. Paik. *Statistical methods for rates and proportions*. john wiley & sons, 2013.

A. Fokkens, M. Van Erp, P. Vossen, S. Tonelli, W. R. Van Hage, L. Serafini, R. Sprugnoli, and J. Hoeksema. Gaf: A grounded annotation framework for events. In *Workshop on Events: Definition, Detection, Coreference, and Representation*, pages 11–20, 2013.

A. Fokkens, P. Vossen, M. Rospocher, R. Hoekstra, W. R. van Hage, and F. B. Kessler. Grasp: Grounded representation and source perspective. In *Proceedings of the Workshop Knowledge Resources for the Socio-Economic Sciences and Humanities associated with RANLP*, pages 19–25, 2017.

E. Grill, G. Stucki, C. Boldt, S. Joisten, and W. Swoboda. Identification of relevant icf categories by geriatric patients in an early post-acute rehabilitation facility. *Disability and rehabilitation*, 27(7-8):467–473, 2005.

J. P. Hirdes, B. E. Fries, J. N. Morris, K. Steel, V. Mor, D. Frijters, S. LaBine, C. Schalm, M. J. Stones, G. Teare, et al. Integrated health information systems based on the rai/mds series of instruments. In *Healthcare management forum*, volume 12, pages 30–40. SAGE Publications Sage CA: Los Angeles, CA, 1999.

J. P. Hirdes, B. E. Fries, J. N. Morris, N. Ikegami, D. Zimmerman, D. M. Dalby, P. Aliaga, S. Hammer, and R. Jones. Home care quality indicators (hcqis) based on the mds-hc. *The Gerontologist*, 44(5):665–679, 2004.

J. P. Hirdes, G. Ljunggren, J. N. Morris, D. H. Frijters, H. Finne Soveri, L. Gray, M. Björkgren, and R. Gilgen. Reliability of the interrai suite of assessment instruments: a 12-country study of an integrated health information system. *BMC health services research*, 8(1):277, 2008a.

J. P. Hirdes, J. W. Poss, and N. Curtin-Telegdi. The method for assigning priority levels (maple): a new decision-support system for allocating home care resources. *BMC medicine*, 6(1):9, 2008b.

A. Hogan. Web of data. In *The Web of Data*, pages 15–57. Springer, 2020.

A. Hogan, E. Blomqvist, M. Cochez, C. d'Amato, G. D. Melo, C. Gutierrez, S. Kirrane, J. E. L. Gayo, R. Navigli, S. Neumaier, et al. Knowledge graphs. *ACM Computing Surveys (Csur)*, 54(4):1–37, 2021.

S. Hooshafza, F. Orlandi, R. Flynn, L. McQuaid, G. Stephens, and L. O'Connor. Modelling temporal data in knowledge graphs: a systematic review protocol. *HRB Open Research*, 4:101, 2022.

InterRAI network. Speaking one language for high quality care worldwide. URL `https://interrai.org/`. Accessed: 30-03-2026.

U. Jakubauskaitė. From nine to one: Combining icf functioning level classifiers in the a-proof project. Master's thesis, Vrije Universiteit Amsterdam, December 2025.

Y. M. Jaradat, M. A. Alia, M. Z. Masoud, A. A. Manasrah, I. A. Jannoud, and O. Alheyasat. Beyond one-size-fits-all: Comparing and selecting regression metrics for robust model assessment. In *2025 12th International Conference on Information Technology (ICIT)*, pages 416–422. IEEE, 2025.

S. Ji, S. Pan, E. Cambria, P. Marttinen, and P. S. Yu. A survey on knowledge graphs: Representation, acquisition, and applications. *IEEE transactions on neural networks and learning systems*, 33(2):494–514, 2021.

T. Kehinde, W. A. Khan, and S.-H. Chung. Financial market forecasting using rnn, lstm, bilstm, gru and transformer-based deep learning algorithms. In *Proceedings of the IEOM International Conference on Smart Mobility and Vehicle Electrification, Detroit, MI, USA*, pages 10–12, 2023.

J. Kim, S. Verkijk, E. Geleijn, M. van der Leeden, C. Meskers, C. Meskers, S. van der Veen, P. Vossen, and G. Widdershoven. Modeling Dutch medical texts for detecting functional categories and levels of COVID-19 patients. In *Proceedings of the 13th Language Resources and Evaluation Conference*, Marseille, France, 2022.

J. R. Landis and G. G. Koch. The measurement of observer agreement for categorical data. *biometrics*, pages 159–174, 1977.

D. Y. Leung, A. Y. Leung, and I. Chi. A psychometric evaluation of a negative mood scale in the mds-hc using a large sample of community-dwelling hong kong chinese older adults. *Age and Ageing*, 41(3):317–322, 2012.

Y. Luo, L. Zhou, A. Jhingoer, K. V. Jongbloed, C. Jordans, B. Werkhoven, T. Seinen, E. van Mulligen, C. Rokx, and Y. Li. Multilingual bert language model for medical tasks: Evaluation on domain-specific adaptation and cross-linguality. *arXiv preprint arXiv:2510.27552*, 2025.

K. Ma, J. Zhang, X. Huang, and M. Wang. Leveraging transformer models to predict cognitive impairment: accuracy, efficiency, and interpretability. *BMC Public Health*, 25(1):504, 2025.

J. D. A. Mello, K. Hermans, C. Van Audenhove, J. Macq, and A. Declercq. Evaluations of home care interventions for frail older persons using the interrai home care instrument: a systematic review of the literature. *Journal of the American Medical Directors Association*, 16(2):173–e1, 2015.

C. Miller, T. Portlock, D. M. Nyaga, and J. M. O'Sullivan. A review of model evaluation metrics for machine learning in genetics and genomics. *Frontiers in bioinformatics*, 4:1457619, 2024.

J. N. Morris, B. E. Fries, D. R. Mehr, C. Hawes, C. Phillips, V. Mor, and L. A. Lipsitz. Mds cognitive performance scale©. *Journal of gerontology*, 49(4):M174–M182, 1994.

J. N. Morris, I. Carpenter, K. Berg, and R. N. Jones. Outcome measures for use with home care clients. *Canadian Journal on Aging/La Revue canadienne du vieillissement*, 19(S2):87–105, 2000.

H. Muizelaar, M. Haas, K. Van Dortmont, P. Van Der Putten, and M. Spruit. Extracting patient lifestyle characteristics from dutch clinical text with bert models. *BMC medical informatics and decision making*, 24(1):151, 2024.

OpenAI. Gpt-4o system card. `https://openai.com/index/gpt-4o-system-card/`, 2024. Accessed: 16-05-2026.

A. Plaquet and H. Bredin. Powerset multi-class cross entropy loss for neural speaker diarization. In *Proc. INTERSPEECH 2023*, 2023.

A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023.

E. Rajabi and S. Kafaie. Knowledge graphs and explainable ai in healthcare. *Information*, 13(10):459, 2022.

M. T. Rietberg, V. B. Nguyen, J. Geerdink, O. Vijlbrief, and C. Seifert. Accurate and reliable classification of unstructured reports on their diagnostic goal using bert models. *Diagnostics*, 13(7):1251, 2023.

L. A. Saketkoo, R. Escorpizo, J. Varga, K. J. Keen, K. Fligelstone, S. S. Birring, H. Alexanderson, H. Pettersson, H. A. Chaudhry, J. L. Poole, et al. World health organization (who) international classification of functioning, disability and health (icf) core set development for interstitial lung disease. *Frontiers in pharmacology*, 13:979788, 2022.

M. S. Salahudeen and P. S. Nishtala. A systematic review evaluating the use of the interrai home care instrument in research for older people. *Clinical Gerontologist*, 42 (5):463–484, 2019.

M. Scheeres. A comparative study of large language model applications in dutch electronic health records for symptom identification. Master's thesis, 2025.

Y. Song, Y. Tian, N. Wang, and F. Xia. Summarizing medical conversations via identifying important utterances. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 717–729, 2020.

S. L. Spoorenberg, S. A. Reijneveld, R. J. Uittenbroek, H. P. Kremer, and K. Wynia. Health-related problems and changes after 1 year as assessed with the geriatric icf core set (geriatrics) in community-living older adults who are frail receiving person-centered and integrated care from embrace. *Archives of Physical Medicine and Rehabilitation*, 100(12):2334–2345, 2019.

H. G. Van der Roest, L. van Eenoo, L. I. van Lier, G. Onder, V. Garms-Homolová, J. H. Smit, H. Finne-Soveri, P. V. Jónsson, S. Draisma, A. Declercq, et al. Development of a novel benchmark method to identify and characterize best practices in home care across six european countries: design, baseline, and rationale of the ibenc project. *BMC Health Services Research*, 19(1):310, 2019.

W. R. Van Hage, V. Malaisé, R. Segers, L. Hollink, and G. Schreiber. Design and use of the simple event model (sem). *Journal of Web Semantics*, 9(2):128–136, 2011.

C. Van Son, T. Caselli, A. Fokkens, I. Maks, R. Morante, L. Aroyo, and P. Vossen. Grasp: A multilayered annotation scheme for perspectives. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, pages 1177–1184, 2016.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

S. Verkijk and P. Vossen. Medroberta. nl: a language model for dutch electronic health records. In *Computational Linguistics in the Netherlands*, volume 11, pages 141–159. Computational Linguistics in the Netherlands, 2021.

S. Verkijk and P. Vossen. Creating, anonymizing and evaluating the first medical language model pre-trained on dutch electronic health records: Medroberta. nl. *Artificial Intelligence in Medicine*, 167:103148, 2025.

World Health Organisation (WHO). International classification of functioning, disability and health (icf). URL `https://icd.who.int/browse/2026-01/icf/en`. Accessed: 13-04-2026.

World Health Organization et al. Towards a common language for functioning, disability, and health: Icf. *The international classification of functioning, disability and health*, 2002.

S. Yildirim. Fine-tuning transformer-based encoder for turkish language understanding tasks. *arXiv preprint arXiv:2401.17396*, 2024.

M. Zamini, H. Reza, and M. Rabiei. A review of knowledge graph completion. *Information*, 13(8):396, 2022.

M. Zhang, Y. Gao, J. Xue, K. Li, L. Zhang, J. Yu, T. Yan, and X. Hou. Development of the assessment standards of the international classification of functioning, disability, and health (icf) geriatric core set through a modified delphi method. *BMC geriatrics*, 24(1):239, 2024.

J. Zhu. Comparative study of sequence-to-sequence models: From rnns to transformers. *Appl Comput Eng*, 42(67):2755–2721, 2023.